C. A. NALLINO

Notes on al-Battānī's *Zīj*

from *Al-Battānī sive Albatenii Opus astronomicum* (Milan, 1899–1907)

translated from the Latin

Nallino's edition of al-Battānī's *Zīj* gives the Arabic text (Part III, 1899), a Latin translation of the canons with a commentary on them (Part I, 1903), and the tables in Latin with notes on them, a glossary and indexes (Part II, 1907). The canons themselves are translated into English from the Arabic in the English *Zīj* of this edition; this volume translates Nallino's own writing, which is in Latin and has not been translated before. It is made from the line-by-line transcription of his Latin in this edition (S03 and S04). Every sentence is translated, in Nallino's order; nothing is summarised or left out. His Arabic, Greek and Hebrew quotations stand in their scripts, as he prints them, with his translations of them in English. Proper names are given in a modern transliteration. In the headings of the commentary, «p.» and «l.» are the pages and lines of Nallino's Latin translation (Part I, pp. 1–150); «t. III» is the Arabic text, whose pages the English *Zīj* gives in its margin. Nallino's footnotes are numbered; the few notes of the translator are lettered. Nallino's own corrections in the «Addenda et emendanda» of Part II are followed and noted. The pages of Nallino's volumes are given in the margin.

The translation is in progress. This version holds the commentary on chapter 3, the commentary on chapter 4, the commentary on chapter 5, the commentary on chapter 6, the commentary on chapter 7, the commentary on chapter 8, the commentary on chapter 9, the commentary on chapter 10, the commentary on chapter 11, the commentary on chapter 12, the commentary on chapter 13, the commentary on chapter 14, the commentary on chapter 15, the commentary on chapter 16, the commentary on chapter 17, the commentary on chapter 18, the commentary on chapter 19, the commentary on chapter 20, the commentary on chapter 21, the commentary on chapter 22, the commentary on chapter 24, the commentary on chapter 25, the commentary on chapter 26, the commentary on chapter 27, the commentary on chapter 28, the commentary on chapter 29, the commentary on chapter 30. It is AI-translated (Claude), 2026; no person has proofread it. Nallino's work is in the public domain; the translation is dedicated to the public domain (CC0 1.0).

Commentary

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On chapter 3

a) — pp. 9–10.

All the propositions on *chords* that the author gives are proved in the *Almagest* I, 9 (ed. Halma, vol. I, pp. 26–37). If we denote the radius by the letter R, they are expressed in our algebraic notation as follows:

(α) $\operatorname{chord}\frac{360^\circ}{6}$ or $\operatorname{chord} 60^\circ = R$

(β) $\operatorname{chord}\frac{360^\circ}{3}$ or $\operatorname{chord} 120^\circ = \sqrt{4R^2 - \operatorname{chord}^2\frac{360^\circ}{6}}$

(γ) $\operatorname{chord}(180^\circ - a) = \sqrt{4R^2 - \operatorname{chord}^2 a}$

(δ) $\operatorname{chord}\frac{360^\circ}{4}$ or $\operatorname{chord} 90^\circ = \sqrt{2R^2}$

(ε)[a] $\operatorname{chord}\frac{360^\circ}{10}$ or $\operatorname{chord} 36^\circ = \sqrt{R^2 + \left(\frac{2R}{4}\right)^2} - \frac{2R}{4}$

(ζ) $\operatorname{chord}\frac{360^\circ}{5}$ or $\operatorname{chord} 72^\circ = \sqrt{\operatorname{chord}^2\frac{360^\circ}{10} + R^2}$

(η) $\operatorname{chord}(a-b) = \frac{\operatorname{chord} a \operatorname{chord}(180^\circ - b) - \operatorname{chord} b \operatorname{chord}(180^\circ - a)}{2R}$

(ϑ) $\operatorname{chord}\frac{a}{2} = \sqrt{\frac{2R - \operatorname{chord}(180^\circ - a)}{2} \times 2R}$

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To find the chord of the sum of two arcs, Ptolemy (ed. Halma, vol. I, pp. 32–33), whom our author follows, has nothing like the rule (η), but goes by way of the chord of the supplement[1]; hence

(ι)[b] $\operatorname{chord}(180^\circ - [a+b]) = \frac{\operatorname{chord}(180^\circ - a)\operatorname{chord}(180^\circ - b) - \operatorname{chord} a \operatorname{chord} b}{2R}$

(κ) $\operatorname{chord}(a+b) = \sqrt{4R^2 - \operatorname{chord}^2(180^\circ - [a+b])}$

Almost all the historians of mathematics describe the way by which Ptolemy came to compute the chord of 1°; of the most recent, one may consult M. Cantor, *Vorlesungen über Geschichte der Mathematik*, Leipzig 1880 ff., vol. I, pp. 353–355 (or 390–392 of the second edition, of 1894); R. Wolf, *Handbuch der Astronomie, ihrer Geschichte und Litteratur*, Zürich 1890–93, vol. I, pp. 164–165, § 61*b*; A. von Braunmühl, *Vorlesungen über Geschichte der Trigonometrie*, Leipzig 1900 ff., vol. I, p. 21. The error, which F. Woepcke determined with Lagrange's interpolation formula (*Discussion de deux méthodes arabes pour déterminer une valeur rapprochée de sin 1°*, in the Journal de Mathématiques pures et appliquées published by J. Liouville, vol. XIX, 1854, p. 165), is 0.00000009, that is, less than 1/10,000,000; it therefore becomes noticeable only in the thirds,

[1] Delambre, *Astron. ancienne*, vol. II, p. 39, is mistaken (as Schiaparelli points out to me) in ascribing to Ptolemy the rule

$$\operatorname{crd}(p+q) = \frac{\operatorname{crd} p \operatorname{crd}(180^\circ - q) + \operatorname{crd} q \operatorname{crd}(180^\circ - p)}{2R}.$$

[*Translator:* The print has – between the two products; Nallino's Addenda et emendanda (Part II, p. XXI) correct it to +, which gives the chord of the sum.]

which Ptolemy leaves out of his table of chords. — A value of the sine of 1° nearer the truth was found by Abū l-Wafāʾ (940–998), whose error shows only in the fifths, that is, in the ninth decimal place (sin 30′ = $0^p\,31'\,24''\,55'''\,54^{IV}\,55^{V}$ instead of $54^{V}\,47^{VI}\,16^{VII}$); see Woepcke, *Recherches sur l'histoire des sciences mathématiques chez les Orientaux*, in the Journal Asiatique, 5th series, vol. XV, 1860, pp. 286–303; Cantor, I, 641–642 (= 703 of the second edition); von Braunmühl, I, 56–57. — According to what Mīram Čelebī (d. 931 H = 1524–25) has handed down to us in his commentary on Ulugh Beg (in Sédillot's French translation, pp. 69–77, note), other, early Arabs, whose names are not known, carried Ptolemy's method through and found a value of the sine of 1° that exceeds the truth by less than 1/100,000,000; for the error is 37/416 (or about 1/11) of Ptolemy's error, since they make sin 1° = $1^p\,2'\,49''\,43'''\,13^{IV}\,5^{V}\,30^{VI}$ instead of $11^{IV}\,14^{V}\,44^{VI}$. But had they not deliberately neglected certain small fractions in their computations, they could have found a far better number, whose error would have been 1/64 of Ptolemy's. See Woepcke, *Discussion de deux méthodes* etc., pp. 155–157, 159–167. — Lastly, Ghiyāth al-Dīn[1] Jamshīd al-Kāshī, in whom other Arabs had probably gone before him, computed sin 1° = $1^p\,2'\,49''\,43'''\,11^{IV}$ correctly and most elegantly, by solving the equation of the trisection of the angle approximately, that is, by expanding the equation of the third degree in a series; his method, handed down to us by Mīram Čelebī, loc. cit., pp. 77–83, was explained by Woepcke, *Discussion*, pp. 167–176, 301–303. See also H. Hankel, *Zur Geschichte der Mathematik im Alterthum und Mittelalter*, Leipzig 1874, pp. 290–293; Cantor, I, 671–672 (= 737 of the second edition); von Braunmühl, I, 73–75.

b) — pp. 10–11.

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The technical word *jayb*, by which the Arabs denote the sine, al-Battānī never mentions, although it was already in common use at the beginning of the ninth century, as appears from the unpublished Latin translation of the astronomical work of Muḥammad ibn Mūsā al-Khwārizmī, made by Adelard of Bath in the twelfth century. For there the Arabic name *elgeib* is kept (with the Arabic article *al* or *el* put before it), and «*elgeib elmustewi* or plane» [*al-jayb al-mustawī*], that is, the right sine, and «*elgeib elmacus* or diminished»[2], that is, the versed sine, are named. Likewise Ḥabash, a contemporary of al-Battānī, always uses the word *jayb*. But our author would not take up the word, which is of foreign origin and wholly alien to Greek usage. Not a few historians of mathematics, then, and among the best, are wrong when they ascribe the use of the word *jayb* to our author and think that Plato of Tivoli brought the word *sinus* to Europe with his translations; of the more recent I shall cite only Hankel, p. 280; Wolf, *Handb. der Astron.*, I, 166, § 62*a*; Cantor, I, 632 (= 693 of the second edition), II, 101 and 249. The old Kaestner, however, had already corrected the error, *Geschichte der Mathematik*, Göttingen 1796, vol. I, p. 522; see now M. Koppe, *Die Behandlung der Logarithmen und der Sinus*, Programm des Andreas-Realgymnasium, Berlin 1893, p. 32; von Braunmühl, I, 50. — The word *sinus* occurs only once in Plato of Tivoli's translation: «si autem per has cordas [*sic*] versas arcus scire desideras, si chorda quam habueris minus 60 fuerit, eam de 60 minue, et residui arcum scito, quem de 90 minues, et quod remanserit, erit arcus *sinus* versi» («but if you wish to know the arcs from these versed chords: if the chord you have is less than 60, take it from 60 and find the arc of the remainder; take that from 90, and what remains will be the arc of the versed *sinus*»; fol. 7v of the Nuremberg edition; the passage corresponds to p. 11, ll. 28–30 of my translation). But, as von Braunmühl, I, 50, note 3, rightly judges, the word must be taken for a gloss of Regiomontanus that has crept into the text by mistake.

[1] Some indeed call him, wrongly, ʿAtāb al-Dīn, reading عتاب ʿAtāb for غياث Ghiyāth. He died about the year 840 H, 1436–37, as H. Suter, *Die Mathematiker und Astronomen der Araber*, Leipzig 1900, pp. 173–4, no. 429, rightly established; to his arguments one may add that the little work on finding the sine of 1° (which still survives in the Cairo Library) is already praised by Qāḍīzāde al-Rūmī, who died between 840 and 850 H, as Mīram Čelebī relates.

[2] Adelard's error; *maʿkūs* = versed, inverted. The passage was cited by Chasles, *Recherches sur l'astronomie ancienne*, in the Comptes rendus de l'Académie des Sciences, vol. XXIII, session of 2 November 1846, p. 850.

Since chords are hardly ever used in trigonometric computations, it is no wonder that al-Battānī called the halves of the chords of the double arcs by the name of chords; a similar usage is found among the Indians as early as the fourth century, of whom one reads the following in the notes of Burgess and Whitney on the book entitled *Sūrya-siddhānta*: «In this passage, the sine is called *jyārdha*, 'half-chord'; hereafter, however, that term does not once occur, but *jyā* 'chord' (literally 'bowstring') is itself employed, as are also its synonyms *jīvā*, *māurvikā*, to denote the sine. The usage of Albategnius is the same» (*Translation of the Sūrya-Siddhānta, a text-book of Hindu astronomy, by* E. Burgess, note on ch. II, verse 28, in the Journal of the American Oriental Society, vol. VI, New Haven 1860, p. 201). — Nor should it be passed over that among the Arabs the name of the whole *chord* too comes from the usage of the Indians, as the passage just quoted shows; among the Greeks the chord was called ὑποτείνουσα, «subtending», ἡ ἐν τῷ κύκλῳ εὐθεῖα, «the straight line in the circle», εὐθεῖα ἐν κύκλῳ μὴ διὰ τοῦ κέντρου οὖσα, «a straight line in the circle not passing through the centre», εὐθεῖα ἐλάττων τοῦ διαμέτρου, «a straight line smaller than the diameter».

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The origin of the technical word *jayb* was explained satisfactorily by Munk, whom F. Woepcke, *Essai sur la propagation des chiffres indiens*, in the Journal Asiatique, 6th series, vol. I, 1863, p. 478, note, and almost all later writers have followed. From the Indian word *jīvā*, chord and sine, which I mentioned a little above, the Arabs, who have no letter *v*, made *jīb* جيب; for the Sanskrit *j* is pronounced like the Arabic *j*. But since *jīb* has no meaning in Arabic, while the letters جيب can also be read *jayb* without any change to the writing (for in ordinary writing the small signs that show the true pronunciation of the vowels are almost always left out), the Arabs preferred this form of the name, which in their language means not only «the round opening of a garment through which a man's neck and head pass», but also «the fold or pocket made in a garment for carrying small things»; occasionally, but rarely, it also means a man's bosom itself. We have already seen that Adelard of Bath kept the Arabic name *elgeib* in his translations; but his contemporary Gerard of Cremona (1114–1187), in translating the astronomy written by Jābir ibn Aflaḥ, used the word *sinus*, which came into our languages by this route, as Koppe, loc. cit., rightly observed[1]. J. Ruska, *Zur Geschichte des «Sinus»* (in the Zeitschrift für Mathematik und Physik, vol. XL, Leipzig 1895, historisch-literarische Abtheilung, p. 128), cites two Syriac writers of the thirteenth century[2] in whom the technical word *jayb* is always rendered by the Syriac *ʿūbā* ܥܘܒܐ, which means a man's bosom; but he is wrong when he writes: «Die Uebersetzung durch *sinus* ist demnach als richtig durch die syrische Belege gesichert. Die verständigere Uebersetzung aber, welche einen Mann voraussetzt, dem der mathematische Inhalt und dessen klarer Ausdruck wichtiger ist als pedantische Wörtlichkeit, lautet natürlich *chorda* . . . . . Beide Uebersetzungen finden sich vor: die zweite, wie Koppe nachgewiesen hat, bei Plato von Tivoli» («The translation by *sinus* is thus confirmed as correct by the Syriac evidence. The more sensible translation, however, which presupposes a man to whom the mathematical content and its clear expression matter more than pedantic literalness, is of course *chorda* . . . . . Both translations occur: the second, as Koppe has shown, in Plato of Tivoli»). For Plato of Tivoli writes *chorda* not from any striving after a more rational translation (how foreign that was to him and to the other translators of his age, their translations show plainly enough everywhere), but because al-Battānī uses this very word.

That the Arab mathematicians learned from the Indians the terms unknown to the Greeks — the chord, the sine, the right sine (Sanskrit *kramajyā*), the versed sine (Sanskrit *utkramajyā*), the sagitta (Sanskrit *śara* or *ishu*) — can hardly be doubted. To this one may add what the Arab writers relate of the Indian book entitled *al-Sindhind* (that is, the Sanskrit *Brahma-sphuṭa-siddhānta*), which was brought to the caliph at Baghdad in the year 156 H, 773. When all this is weighed, the

[1] But from the fact that it was Gerard of Cremona, not Plato of Tivoli, who spread the use of this word among us, I do not see how Koppe can call into doubt the origin of the Arabic *jayb* that Munk explained.

[2] They belong, then, to a late age not only of Syriac but also of Arabic letters, and not, as von Braunmühl, I, 50, note 1, writes, «in den ersten Zeiten des Mittelalters», to the first times of the Middle Ages.

history of the trigonometric functions can be told as follows. The Indians, moved above all by their use of orthogonal projections (von Braunmühl, I, 31–32), used right and versed sines instead of the chords of the Greeks; they computed them for the twenty-fourth parts of the quadrant, that is, for every 3° 45′, taking a radius of 3438 minutes. Once they had received the first notion of sines from the Indians, and had come to know Ptolemy's books at about the same time, the Arabs rightly understood that it was more convenient and more exact to construct new sines by halving Ptolemy's chords (which are computed for every half degree and a radius of sixty parts). This had already been done before al-Battānī.

c) — p. 11.

The *interpolation* for finding the sines of arcs that fall between two consecutive arcs of the table is the same as the one found in Ptolemy for computing chords: the change of the sines and of the chords is taken to be proportional to the change of the arcs within the interval of the table, half a degree. What Cantor, I, 355 (= 392 of the second edition), writes of Ptolemy, «ein Beweis für diese angenommene Proportionalität in engem Bereiche ist dagegen nicht vorhanden» («a proof of this proportionality, assumed within a narrow range, is however not given»), may rightly be said also of al-Battānī, of Ḥabash (fol. 18v) and of the other Arabs. More exact interpolations they used only in constructing tables. — Tables of sines for every minute of the quadrant, computed by Ibn Yūnus, survive in the Arabic codex 5752 of the Royal Library of Berlin; Ulugh Beg also computed others.

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Versed sines, whose use the Indians knew before the Arabs, the author defines:

$$\text{versed sine } a = R - \cos a.$$

He thought it superfluous to construct a table of them, which on the other hand we find in Ḥabash. At the foot of the page two formulas are given; one is

(α) $\quad \cos a = \sqrt{R^2 - \sin^2 a},$

which follows from the theorem of Pythagoras and, with $R = 1$, corresponds to our familiar rule

$$\cos a = \sqrt{1 - \sin^2 a}.$$

The other,

(β) $\quad \sin \dfrac{90^\circ - a}{2} = \sqrt{30^p\,(R - \sin a)},$

comes from the rule of al-Battānī and of Ptolemy mentioned in note *a* (p. 153),

$$\text{chord } \frac{a}{2} = \sqrt{\frac{2R - \text{chord}\,(180^\circ - a)}{2} \times 2R},$$

which, when sines are put for chords and $R = 60^p$ is taken, becomes

$$\sin \frac{a}{2} = \sqrt{30^p\,(R - \cos a)}.$$

Why al-Battānī took the complement of *a* in place of *a*, I do not know.

On chapter 4

a) — p. 12.

Of the Arabs who measured the *obliquity of the ecliptic* before the year 1000, Ibn Yūnus relates the following in the eleventh chapter of his work[1]: «Between Ptolemy and the authors of the *Tested Tables* (*al-mumtaḥan*) I know of no observations of the obliquity except one made a little after[2] the year 160 of the Hijra [which began on 18 October 776], whose author gives the obliquity as 23° 31′. As for the value found by the authors of the *Tested Tables*[3] in observations close to our own age: Yaḥyā ibn Abī Manṣūr, Sanad ibn ʿAlī, al-ʿAbbās ibn Saʿīd al-Jawharī and a group of learned and eminent men (وطايفة من اهل العلم والفضل) investigated it in [the quarter called] al-Shammāsiyya of the city of Baghdad, in the time of [the caliph] al-Maʾmūn, and found 23° 33′, as Muḥammad ibn Mūsā al-Khwārizmī in his astronomical work (في زيجه) and Muḥammad ibn Kathīr al-Farghānī in his book on the making of the astrolabe report on their authority; and the matter is well known[4]. Those who observed at Damascus after the death of Yaḥyā ibn Abī Manṣūr, with the instrument that al-Maʾmūn had ordered to be made when he set out for the land of the Byzantines (بالالة التي امر باتخاذها المامون حين توجـه الى بـلاد الروم) — they are Khālid ibn ʿAbd al-Malik al-Marwarrūdhī, Abū l-Ṭayyib Sanad ibn ʿAlī, ʿAlī ibn ʿĪsā[a] al-Asṭurlābī and others — report that they found the greatest declination of the ecliptic to be 23° 33′ 52″; and the measurement made with this instrument was in the year 201 of Yazdajird» [which began on 27 April 832][5]. I 158

«Abū l-Ḥasan Thābit ibn Qurra says: I found ancient operations from before Ptolemy which show (تدل) that the greatest declination is 23° 35′[6]. Muḥammad ibn Jābir ibn Sinān al-Battānī relates that he found 23° 35′ by his observations. The noble and eminent Abū l-Qāsim ʿAlī ibn al-Ḥusayn ibn Muḥammad ibn ʿĪsā, known as Ibn al-Aʿlam, a man of learning and of outstanding intelligence, says that he measured the greatest declination and, taking the utmost care in the measurement, found 23° 34′ 2″. Abū l-Ḥasan al-Ṣūfī ʿAbd al-Raḥmān ibn ʿUmar relates that he measured the greatest declination and, having carried out the measurement with the greatest accuracy, found 23° 33′ 45″[7]; that man held an eminent place in the science of geometrical proofs. Help comes from God.»

Ibn Yūnus, ch. IV (in Caussin, p. 120), says that the value 23° 35′ was chosen in the *Tested Tables*; and al-Farghānī reports the same in ch. V (p. 20 of Golius's edition, fol. 121r of the Paris codex 2504). But Ibn Rusta, who copies the whole of this chapter of his contemporary al-Farghānī, has

[1] P. 222 of the Leiden codex (of which the Paris codex no. 2495 is a most careful copy, keeping the same division of the pages). I compared the passage with the Paris codex 2496 (which gives selected chapters of Ibn Yūnus's work and is written more carelessly), fol. 25r.

[2] What I have rendered «a little after» is in Arabic نيف, that is, «more than two but fewer than ten» [years after].

[3] This is the astronomical work that the astronomers of the caliph al-Maʾmūn (reigned 813–833) compiled from their observations under the direction of Yaḥyā ibn Abī Manṣūr.

[4] Ibn Yūnus himself confirms this in the introduction to his work (in Caussin, p. 56), saying that in the year 214 H (which began on 10 March 829), 198 of Yazdajird (which began on 28 April 829), Yaḥyā ibn Abī Manṣūr, al-ʿAbbās ibn Saʿīd al-Jawharī, Sanad ibn ʿAlī and others found 23° 33′ in the city of Baghdad. — Al-Zarqālī, in the unpublished Latin translation of the *Canons* or *Toledan Tables*, and the Jew Abraham ibn Ezra (in the second half of the twelfth century) say that Yaḥyā ibn Abī Manṣūr found 23° 33′ 30″; see Delambre, p. 177 (whose codex of the Toledan Tables gave the name in a corrupt form, «Almeon fils d'Albumazar»), and M. Steinschneider, *Études sur Zarkali*, in the Bullettino di bibliografia e di storia delle scienze matematiche e fisiche, vol. XX, 1887–88, p. 87 (where a codex of the Tables is cited that has the name «Jahiben filius Albumasaris»).

[5] He relates the same in the introduction (Caussin, 56); the year there is 217 H (which began on 7 February 832), 201 of Yazdajird.

[6] But in his own works Thābit chose 23° 33′, as appears from Ḥabash, fol. 90v, and Delambre, p. 73.

[7] The Paris codex 2496 wrongly writes 23. 35. 45. 38.

[a] The print has ʿAlī ibn Mūsā; Nallino's Addenda et emendanda (Part II, p. XXI) correct it to ʿAlī ibn ʿĪsā.

23° 33′ for 23° 35′[1]; and 23° 33′ is also read in the Latin translation of al-Farghānī made by John of Seville and printed at Nuremberg in 1537 (fol. 5r). This seems the more probable when one compares what Ibn Yūnus himself relates of the measurement made by the authors of the *Tested Tables*, and what Rabiçag of Toledo writes in his *Libro de las armellas*[2], which is drawn wholly from Arabic sources: «Et sabet que en este nuestro tiempo segund el nostro rectificar. bien se paresció por los rectificamentos antigos et los nuevos. que lo que ha entre ellos [that is, the interval between the pole of the ecliptic and the pole of the equator] non es uno todauia. Et esto es porque Abrachis [= Hipparchus] falló la declinacion que era .XXIII. grados et .LI. menudos. et falláronla los rectificadores [that is, the authors of the Tested Tables] en el tiempo de Memun Abdalla. fiio de Haron Arraxit .XXIII. grados et XXXIII. menudos. Et fallóla Mohamat. fiio de Aben el Bateni [= Muḥammad ibn Sinān al-Battānī] .XXIII. grados et .XXXV. menudos» («And know that in this our time, by our rectifying, it has appeared clearly from the ancient and the new rectifications that what lies between them is not always one. And this is because Abrachis found the declination to be 23 degrees and 51 minutes; and the rectifiers found it in the time of Memun Abdalla, son of Haron Arraxit, to be 23 degrees and 33 minutes; and Mohamat, son of Aben el Bateni, found it to be 23 degrees and 35 minutes»). — Thābit ibn Qurra says the same in the book *On the motion of the eighth sphere, or of access and recess*, which survives in two different Latin translations in the Vatican Library; one (cod. Lat. 4275, f. 84v, col. I) says: «Et eius [that is, of the whole obliquity] summa est 24 graduum secundum quod translatum est ab Indis, et similiter reperit Ptholomeus 23 g. et 51 minut., et consideratores [that is, اصحاب الممتحن, the authors of the Tested Tables] invenerunt in tempore Maymonis 23 g. et 33 m.» («and its amount is 24 degrees according to what has been handed down from the Indians, and likewise Ptolemy found 23 degrees and 51 minutes, and the observers found in the time of al-Maʾmūn 23 degrees and 33 minutes»); the other (cod. Lat. 4083, f. 56v, col. II): «pervenit talis declinatio ad 24 gradus secundum quod ab Indis traditum fuit. Ptolomeus quoque invenit eam 23 graduum et 51 mt. Alii autem inspectores post Ptolomeum in partem Mamonis eam 23 graduum 33 minutorum invenerunt»[a] («this declination reached 24 degrees according to what was handed down by the Indians. Ptolemy too found it to be 23 degrees and 51 minutes. But other observers after Ptolemy, on the side of al-Maʾmūn, found it to be 23 degrees 33 minutes»). — Lastly, it should not be passed over that Thābit ibn Qurra, as we saw in note 1, chose 23° 33′; and so did Qusṭā ibn Lūqā, who died about 910 or 912, in his book on the construction and use of the celestial globe[3]. — They would hardly have chosen 23° 33′ had they not found this value in the *Tested Tables*.

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Of those who flourished after the authors of the *Tested Tables*, down to the last years of al-Battānī, we know that the family of the Banū Mūsā[4], al-Nayrīzī[5] and the family of the Banū Amājūr[6] found the obliquity to be 23° 35′ by observation. So did, among later astronomers, Abū l-Wafāʾ, in the city of Baghdad, in the year 987 (on the evidence of Delambre, p. 156), and Ibn Yūnus in the year 391 H (which began on 30 November 1000; ch. XI, in Caussin, p. 238; more exactly he had found 23° 34′ 52″). — Many others adopted this obliquity of 23° 35′ in their tables, such as Ḥabash (a

[1]Ibn Rusta, *Kitâb al-Aʿlâk an-Nafisa VII*, edited by M. J. de Goeje, Leiden 1889, p. 16 [= Bibliotheca Geographorum Arabicorum, vol. VII].

[2]Book I, ch. IV = *Libros del saber de astronomia del Rey D. Alfonso X de Castilla, copilados, anotados y comentados por D. Manuel Rico y Sinobas*, Madrid 1863–67, vol. II, p. 6. It belongs to the second half of the thirteenth century.

[3]Ch. XXXIV of the Spanish translation (*Libros del saber de astronomia*, vol. I, p. 187), ch. XXX of the unpublished Hebrew translation (in Baldi, *Vite di matematici arabi con note di* M. Steinschneider, Rome 1874, p. 70, note).

[4]Ibn Yūnus, ch. V, in Caussin, p. 152. — One of these Banū Mūsā died in the month of Rabīʿ I 259 H = January 873; on them see Suter, *Die Mathematiker und Astronomen der Araber und ihre Werke*, Leipzig 1900, pp. 20–21, no. 43 (= Abhandlungen zur Geschichte der mathematischen Wissenschaften, part X; supplement to the 45th year of the Zeitschrift für Mathematik und Physik).

[5]Ibn Yūnus, ch. IV, in Caussin, p. 120. — He died about the year 310 H (922–23); see Suter, p. 45, no. 88.

[6]Ibn Yūnus, ch. V, in Caussin, p. 152. — They observed in the years 272–321 H (885–933); see Suter, pp. 49–50, no. 99.

[a]The print has «quos» for «quoque» and «tempore» for «partem»; Nallino's Addenda et emendanda (Part I, p. LXXII) correct both.

contemporary of al-Battānī), al-Kharaqī in his *Muntahā al-idrāk*[1], the Hebrew-Spanish Abraham bar Ḥiyya, writing in 1135[2], and his contemporary ʿAbd al-Raḥmān al-Khāzinī[3]; likewise Naṣīr al-Dīn al-Ṭūsī in the books he wrote before the *Īlkhānī Tables*[4], the Greek Chrysococces in his *Persian Tables*[5], and lastly, at the beginning of the fourteenth century, al-Jaghmīnī (as appears from pp. 264 and 265). Barhebraeus, or Abū l-Faraj, in his Syriac Astronomy written in 1279, uses round numbers, and so in places he gives the obliquity as 24°[6]; but where he wants more exact numbers he takes it as 23° 35′, as can easily be inferred from his computations of the right and the oblique ascensions (pp. 165–167 of the Syriac text).

Abū ʿAlī al-Ḥasan (that is, Aboul Hhassan), ch. XXIII, pp. 175 and 176, says that the obliquity varies between the limits 23° 33′ and 23° 53′; but in ch. XXIV, pp. 180–187, and in his tables, he adopted 23° 35′.

It should be noted that an error of two minutes in fixing the obliquity is of little consequence in the computations derived from it; for in the right ascensions it can produce a discrepancy only in the seconds of arc[7]; in the oblique ascensions the greatest difference never reaches three minutes of arc (especially since the latitudes that the Arabs had to use were always smaller than 45°); and lastly, in computing the greatest half arc of the day of the sun in the lands under Arab rule, the difference can never exceed ten seconds of an hour. Such differences could be neglected in those times.

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So that one may judge with what care the Arabs observed the greatest declination of the ecliptic, I have made a small table of the chief observations, using in my computations Bessel's formula, which for t years before 1750 is

$$\varepsilon = 23^\circ\,28'\,18.0'' + 0.48368''\,t - 0.00000272295''\,t^2.$$

With an asterisk I mark the years of the observations that are certain; where the year is not known, I have put the epoch of the author's tables, or the year that can be assumed approximately. I have given the errors to the second; but everyone knows that these numbers cannot always be taken absolutely, since sometimes the year of the observation is not known for certain, or the Arab observer deliberately neglected the seconds[8].

[1] Laurentian codex (Orient.-Palat. 110), f. 40r (maqāla I, bāb XIV, faṣl I).

[2] *Sphaera mundi auctore Rabbi Abrahamo Hispano filio R. Haijae. Osw. Schreckenfuchsius vertit in linguam Latinam, Seb. Munsterus illustravit annotationibus.* Basel 1546, ch. X.

[3] Vatican Arabic codex 761, f. 10v and in many places.

[4] As in the *Tadhkira*, Laurentian codex (Palat.-Orient. 271), f. 38v (bāb II, faṣl IV).

[5] In Ismael Bullialdus, *Astronomia Philolaica*, Paris 1545, part II, p. 224. — Bullialdus thinks that he flourished in the first half of the thirteenth century (loc. cit., p. 213); but Krumbacher, *Geschichte der Byzantinischen Litteratur*, 2nd ed., Munich 1897, says that he lived about the year 1346.

[6] This value was known to the Greeks before Eratosthenes; the Indians always used it. — See also below, p. 161, note 3.

[7] [The greatest difference is in the octants, that is, at the right ascensions 45°, 135°, 225°, 315°, where an error of one minute in fixing the obliquity produces an error of 0.217′ (or very nearly 13″) in the right ascension. — Schiaparelli]

[8] The small tables that, after Lalande (I think), circulate among our writers (the most careful Houzeau and Wolf not excepted) contain errors whose origin I do not know; for they ascribe to Eratosthenes 23° 45′ 7″, to al-Battānī 23° 35′ 41″, to Ibn Yūnus 23° 34′ 26″, to Ulugh Beg 23° 31′ 48″, to Abū ʿAlī al-Ḥasan (Aboul Hhassan) 23° 33′ 45″!

Observers	*Place of the observation*	*Year*	*Obliquity observed*	*True (mean) obliquity*	*Error*
Eratosthenes	Alexandria	230 BC	23° 51′ 20″	23° 43′ 45.01″	+ 7′ 35″
Hipparchus[1]	Rhodes	130 BC	23 51 20	23 42 56.62	+ 8 23
The authors of the Tested Tables[2]	Baghdad	* 829	23 33	23 35 41.16	− 2 41
Other astronomers of the caliph al-Maʾmūn	Damascus	* 832	23 33 52	23 35 39.71	− 1 48
Banū Mūsā	Baghdad	860	23 35	23 35 26.32	− 0 26
al-Battānī	al-Raqqa	880	23 35	23 35 16.74	− 0 17
Ibn al-Aʿlam	Baghdad?	960	23 34 2	23 34 37.92	− 0 36
ʿAbd al-Raḥmān al-Ṣūfī	Baghdad?	965	23 33 45	23 34 35.50	− 0 50
Abū l-Wafāʾ	Baghdad	* 987	23 35	23 34 25.46	+ 0 35
Wayjan ibn Rustam al-Kūhī[3]	Baghdad	* 988	23 51 1	23 34 24.98	+ 16 36
Ibn Yūnus	Cairo	* 1001	23 34 52	23 34 18.74	+ 0 33
Alfonsine Tables[4]	Toledo	1250	23 32 29	23 32 19.16	+ 0 10
Naṣīr al-Dīn al-Ṭūsī[5]	Marāgha	1270	23 30	23 32 9.49	− 2 9
Ibn al-Shāṭir	Damascus	1363	23 31	23 31 24.78	− 0 24
Ulugh Beg	Samarqand	1437	23 30 17	23 30 49.13	− 0 32

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The author's words, p. 12, ll. 14–24, fully confirm what I had already said in note 9, p. 5: that al-Battānī could by no means have inferred a *continuous* decrease of the obliquity of the ecliptic from the uncertain observations of the Greeks. How could he have argued it, when from Eratosthenes to Theon, that is, over an interval of about 600 years, all the Greeks always say they found 23° 51′ 20″?

I 162

To the later Arabs, who could compare many good observations with one another, the hypothesis of a continuous decrease was not unknown, although the small discrepancies could have been put down to errors of the observers and of the instruments. Indeed, we read the following in the astronomical book entitled the Sanjarī (*al-zīj al-sanjarī*), which Abū Manṣūr ʿAbd al-Raḥmān al-Khāzinī wrote about the year 520 H (1126)[6]: «In [fixing] the obliquity of the ecliptic there are many discrepancies, about whose cause opinions differ. The most likely of all is that they arise only from the refraction of the rays (وغالب الظنّ ان سببه انعطاف الشعاع دون سايره), so that, when the latitude of two observers is one and the same, the same obliquity appears to both. The greatest declination is 24° 0′ with the Indians, 23° 51′ with Ptolemy and Hipparchus, 23° 35′ with the authors of the Tested [Tables] and al-Battānī, 23° 33′ with some of the moderns (بعض المحدثين). Because of these discrepancies, and because of the irregularity in the motion of the fixed stars, some thought fit (ظهر لبعضهم) to posit a motion in the space between the two spheres, that of the universe and that of the fixed stars, which would complete the whole circle every 1285 (١٢٨٥) solar years, and whose poles, like the poles of the fixed stars too, would not lie on the axis of the universe; so that the poles of the signs of the zodiac would approach the equators or recede from them, and the motion of the fixed stars would be quickened or slowed, in a way that it

[1] Ptolemy, in the year 140, keeps the same value; his error is therefore + 10′ 10″. The error of Theon, who wrote about the year 360, is + 12′ 1″.

[2] If 23° 33′ 30″ is admitted, as some say (see p. 157, note 4), the error becomes − 2′ 11″. [*Translator:* The print refers to «p. 6, note 3»; Nallino's Addenda et emendanda (Part II, p. XXI) correct the reference to p. 157, note 4.]

[3] According to the passage of Ibn al-Qifṭī that Casiri, *Bibliotheca Arabico-hispana Escurialensis*, Madrid 1760–70, vol. I, p. 441, and Sédillot, in the preface to the text of Oloug Beg, pp. LII–LIII, cite. The codex of Ibn al-Qifṭī kept at Rome in the Library of the Accademia dei Lincei gives the same reading.

[4] So in many places in the *Libro de las taulas alfonsies* (*Libros del saber de astronomia*, vol. IV) and in the *Libro del quadrante para rectificar* (ibid., vol. III, pp. 296–297).

[5] According to Sédillot's note on the text of Oloug Beg, p. 63; the passage he cites is found on f. 105r of the Florence codex of these same *Īlkhānī Tables*. The first declinations (ff. 120r–121r) and the second (ff. 121v–122v) are computed with this value.

[6] Vatican Arabic codex no. 761, in the third chapter (*bāb*) of the preface, f. 10v.

would take too long to set out now. But since the obliquity is 23° 35′ with most [authors], we shall use this value in our reckonings, as one rightly accepted (استعملناه في الاعتبار مُسلا)». — Naṣīr al-Dīn al-Ṭūsī, in his Memorandum of what is chiefly to be remembered in astronomy[1], relates the following: «The total declination found by the ancient and by the more recent observations is not one and the same; the value the ancients found is greater than the one the more recent have measured. What a more recent observer found is sometimes taken to be less than what an earlier one found, the greatest value found not reaching 24°[2], while the least does not go below 23° 1/2 + 1/(2 × 10) [= 23° 33′][3]. Most (الجمهور) set it at 23° 1/4 + 1/3 [= 23° 35′]. Because of these discrepancies some have thought that the ecliptic moves in latitude and approaches the equator; if this were true, another sphere would have to be posited that carries the zodiac with such a motion. If the ecliptic moved, it could complete the circle; or it could not complete it, but reach a certain limit from which it would afterwards turn back. Any such limit could lie after the two falls of the ecliptic onto the equator, or at the time of the second fall, or between the two falls; and this last [in three ways, namely] after half the circle has been completed, or exactly at the half circle, or before the semicircle is completed. Lastly, the ecliptic could also not reach the space between the two falls, if it is thought to turn back either at the time of its first fall onto the equator or before it. There are, then, eight hypotheses. By the first five, the two halves of the zodiac, the northern and the southern, would exchange places, together with their properties; and by three of these [five], each half of the zodiac would [at times] fall onto each half of the equator. By the first seven hypotheses, a half of the ecliptic would coincide with the half of the equator nearest to it; whenever this happened, days and nights would be equal in all places of the earth, and the different seasons of the year would fail. By the last [that is, the eighth] hypothesis[4], however, there would follow only an increase or a decrease in the altitudes [of the sun] and in the lengths of the days and the nights».

Quṭb al-Dīn Maḥmūd al-Shīrāzī, who died in the month of Ramaḍān 710 (February 1311), says in his *al-Tuḥfa al-shāhiyya* (ff. 5v–6r) that no definite rate of decrease is found among the different values of the obliquity measured at different times.

Those who have written the history of astronomy say that in the second half of the sixteenth century some first recognized the continuous decrease of the obliquity, such as Ignazio Danti in his *Trattato dell'Astrolabio*, Florence 1569, though many others of that age called it into doubt; that Tycho Brahe (1546–1601) inferred it from the changed latitudes of the stars; and that, lastly, Euler (1707–1783) proved it by the laws of celestial mechanics.

b) — p. 12, ll. 27–32.

[1] *Al-Tadhkira fī ʿilm al-hayʾa*, in the fourth chapter (*faṣl*) of the second section (*bāb*), ff. 38r–39r of the codex of the Laurentian Library, which also contains the commentary of Qāḍīzāde. — A French summary of the passage is given by Carra de Vaux, *Les sphères célestes selon Nasîr-eddîn Attûsî*, at the end of P. Tannery's *Recherches sur l'histoire de l'astronomie ancienne*, Paris 1893, p. 346.

[2] The Indian astronomers always have 24°. — Qāḍīzāde remarks on this passage: «Some think that the sages of the Indians found it [24°], and that this same value was very well known in the time of Euclid, which is why [Euclid] in his book, after computing the side of the hexagon inscribed in a circle, derived the side of the pentadecagon inscribed in a circle, since 24 is the fifteenth part of the circumference». — So too Quṭb al-Dīn Maḥmūd al-Shīrāzī, *al-Tuḥfa al-shāhiyya*, f. 5v: «Some think that all the Indians agree on this value [24°], and that 24° was the obliquity in the time of Euclid; which is why he derived the side of the pentadecagon in his book, since 24 is the fifteenth part of the circumference». — Strange enough that some saw the obliquity of the ecliptic foreshadowed in the fifteenth and sixteenth propositions of the fourth book of Euclid, on inscribing regular hexagons and pentadecagons in a circle! [*Translator:* Nallino's Addenda et emendanda (Part II, pp. XXI–XXII) trace this story to Heron of Alexandria's commentary on Euclid, as quoted by al-Nayrīzī, and cite Proclus and, after Eudemus, Theon of Smyrna, who give the distance between the two poles as the side of the fifteen-sided polygon, 24°; the Addenda are translated with Part II.]

[3] Qāḍīzāde: «So at the time this book was written; but afterwards the author himself, with the new observations that he set up [in the city of] Marāgha, found 23° 30′, which is the least value found so far». And in fact Naṣīr al-Dīn adopted the obliquity 23° 30′ in his *Īlkhānī* tables.

[4] By which the ecliptic is thought never to coincide with the equator.

The rule for finding the **declination** of the *sun*, or of any degree of the ecliptic, follows from Ptolemy's construction, *Almag.* I, 12 (ed. Halma, vol. I, pp. 56–58). Let ε be the obliquity of the ecliptic, δ the declination of the given degree, L the longitude of the sun or of the given degree; by the well-known theorem of Menelaus Ptolemy proves

$$\frac{\text{chord}\,180^\circ}{\text{chord}\,2\varepsilon} = \frac{\text{chord}\,180^\circ}{\text{chord}\,2\delta} \times \frac{\text{chord}\,2L}{\text{chord}\,180^\circ}.$$

Since chord 180° is the diameter (2R), if sines are used instead of chords, there results

$$\frac{2R}{2\sin\varepsilon} = \frac{2R}{2\sin\delta} \times \frac{2\sin L}{2R},$$

whence al-Battānī's formula[1]

$$\sin\delta = \frac{\sin L\ \sin\varepsilon}{R}.$$

c) — p. 13, ll. 15–18.

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Nothing like this is read in Ptolemy, but it is found in Theon's *Canon*, ed. Halma, vol. I, p. 54: «For when the sun is between 1° and 90° we shall say that it is descending towards the north [*sic*, βοῤῥᾶν κατιέναι]; when between 91° and 180°, that it is descending to the south; between 181° and 270°, that it is ascending towards the south [*sic*, νότον ἀνιέναι]; between 271° and 360°, that it is ascending to the north». Likewise in the tables of the declinations of the sun, under the columns of the degrees of the ecliptic, Theon, I, 144, writes:

From 0° to 90°:	Βοῤῥαῖος καταβαθμός, northern descent.
From 90° to 180°:	Νότιος καταβαθμός, southern descent.
From 180° to 270°:	Νότιος ἀναβαθμός, southern ascent.
From 270° to 360°:	Βοῤῥαῖος ἀναβαθμός, northern ascent.

As in al-Battānī, we read in the *Libro de las taulas alfonsies*, ch. XX (*Libros del saber de astronomia*, vol. IV, p. 136): «et si el logar fuer de un grado fata tres signos. el sol es suuiente en septentrion. et si fuer de tres signos fata .VI. es decendiente en septentrion. et si fuer de .VI. signos fata .IX. es decendiente en mediodía. et si fuer de .IX. signos fata .XII. es suuiente en mediodía» («and if the place is from one degree to three signs, the sun is ascending in the north; and if it is from three signs to six, it is descending in the north; and if it is from six signs to nine, it is descending in the south; and if it is from nine signs to twelve, it is ascending in the south»).[a]

[1] With Ptolemy and al-Battānī the radius is of sixty parts; in our trigonometric formulas, on the other hand, $R = 1$ is taken, so that in our usage the author's rule becomes $\sin\delta = \sin L \sin\varepsilon$.

[a] Nallino's Addenda et emendanda (Part I, p. LXXII) add to this note that Theon's strange definitions are found also in the Astrologia of Vettius Valens, and that Julianus of Laodicea agrees with al-Battānī's usage; the Addenda are translated with Part I's front matter.

On chapter 5

a) — pp. 13–14.

The computation of the *right ascension.* Let δ be the declination of the given degree of the ecliptic, α its right ascension, ε the obliquity of the ecliptic; Ptolemy, *Almag.* I, 13 (ed. Halma, vol. I, pp. 60–61), proves by the theorem of Menelaus

$$\frac{\text{chord}\, 2\alpha}{\text{chord}\, 180^\circ} = \frac{\text{chord}\, 2\delta}{\text{chord}\, (180^\circ - 2\delta)} \times \frac{\text{chord}\, (180^\circ - 2\varepsilon)}{\text{chord}\, 2\varepsilon}.$$

Using sines instead of chords, one derives

$$\frac{2\sin\alpha}{2R} = \frac{2\sin\delta}{2\cos\delta} \times \frac{2\cos\varepsilon}{2\sin\varepsilon},$$

whence al-Battānī's formula

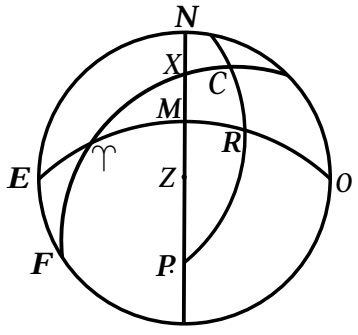
$$\sin\alpha = R\,\frac{\sin\delta\,\cos\varepsilon}{\cos\delta\,\sin\varepsilon}.$$

Ḥabash, who lived in the same age as al-Battānī, had constructed tables of tangents and cotangents (taking a radius of 60 parts), and so could teach in his work (fol. 88r), more briefly[a],

$$\sin\alpha = \frac{\text{tg}\,\delta\,\,\text{cotg}\,\varepsilon}{R}.$$

b) — p. 14, ll. 10–12.

[Now that the computation of the right ascensions of any point of the ecliptic is known, the author wants to show how the time of the culmination of those points is found, that is, the degree of the celestial equator that rises from the horizon of the given city at the time of the culmination.



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Let the celestial sphere be projected onto the plane of the horizon FENO of the given city; let P be the pole, PN the meridian, EMO the equinoctial circle, ♈ the beginning of Aries, F♈C the ecliptic, F the ascendant, C the beginning of Capricorn, PC the solstitial colure. Necessarily ♈R, that is, the arc of the celestial equator cut off between the point ♈ of the vernal equinox and the point R of the solstitial colure, is of 90 degrees; likewise the arc EM, cut off by the horizon and the meridian, must always be 90°. Clearly, then, at any time E♈ = MR. — Let X be the point of the ecliptic whose time of culmination we want to know; XC will be its longitude counted from

[a]The print has sin α = R tg δ cotg ε; Nallino's Addenda et emendanda (Part II, p. XXIV) correct it to tg δ cotg ε : R, which is right when the sine and the tangents are all taken to the radius R.

the beginning of Capricorn, MR its right ascension from the winter solstitial colure. Since we have seen that MR equals E♈ (which is the oblique ascension corresponding to the arc ♈F of the ecliptic), the number of degrees contained in the arc MR will equal the degree of the equator then rising from the horizon; and this in all places of the earth. The computation of the arc MR, since XC is given, reduces very easily to the computation of the arc M♈ when X♈ is given, which is the problem al-Battānī solved a little before. So the author's «one and the same number» means that at any time and in any place of the earth there holds the relation

right ascension of the culminating degree X of the ecliptic, counted from the colure of Capricorn = oblique ascension of the rising degree F of the ecliptic,

that is,

90° + right ascension of the culminating degree, counted from the beginning of Aries = oblique ascension of the rising degree.

Al-Battānī's tables of right ascensions, fol. 179v–181r, therefore count the ascensions from the colure of Capricorn, not, as is the common usage of the Greeks and our own, from the beginning of Aries. In this way, to give an example, one learns at once from the tables that the end of the tenth degree of Pisces (that is, 340° from the beginning of Aries) culminates at the same time as the point of the equinoctial circle that is 71° 34′ from the vernal equinox rises. — Schiaparelli.]

Ḥabash always counts the right ascensions from the beginning of Aries, as the other Arab astronomers published so far usually do; Abū ʿAlī al-Ḥasan, however (Aboul Hhassan, ch. XXXII, pp. 218–227 and in the tables), follows al-Battānī's usage. Ibn Yūnus (on the evidence of Delambre, p. 122) made a double table of right ascensions, counting them both from the beginning of Aries and from the solstitial colure of Capricorn; and so did Ulugh Beg (Oloug Beg, p. 67 of the text, 93 of the translation, in ch. VII of the second part), who afterwards uses the ascensions counted from the winter solstitial colure in the astrological computations of the direction of the significator (p. 161 of the text, 214 of the translation) and of the projection of the rays (p. 158 of the text, 210 of the translation), in which it is more convenient to take the distances from the solstitial colure than from the equinoctial one. Tables of right ascensions beginning from the winter solstitial colure, for every minute of the ecliptic, were also constructed by ʿAbd al-Qādir ibn Muḥammad al-Fayyūmī, who died in 1022 H (1613–14); they are kept in the Cairo Library; see Suter, *Die Mathematiker und Astronomen der Araber*, p. 193, no. 479.

The oblique ascensions, on the other hand, are always counted from the beginning of Aries.[a]

On chapter 6

a) — p. 17, ll. 10–11.

I 165

Plato wrongly has 3500. — Those ancients, then, reckoned the degree of the earth's circumference at 75 miles; the 65 (in Plato 85) that are read at the end of the chapter seem to be an error, as I said on p. 19, note 8. Such a degree is of Graeco-Syrian origin. The Roman mile (1479.5 metres) contains seven *philetaerian* stadia (213.048 m), which were in very wide use in Egypt and Syria

[a] Nallino's Addenda et emendanda add to this note further tables of right ascensions counted from the beginning of Capricorn (Part I, p. LXXII: al-Khāzinī, Kūshyār ibn Labbān, al-Zarqālī, Ibn al-Musharraf, al-Akhṣāṣī; Part II, p. XVII: Theon's tables, and later Arabic manuscripts that call them «the ascensions of the culmination» or «the ascensions of the Cupola»); the Addenda are translated with the front matter of the two parts.

in the decline of the Roman empire. The Graeco-Syrian writers believed that the stadia of which Ptolemy gives 180,000 to the circumference of the earth were philetaerian; they therefore held that the equator of the earth contains 27,000 miles, and the degree 75 miles. Some Arabs, however, confusing the Roman miles with the Arabic ones (1973 m), took over the numbers of the Syrian writers without change. Compare what I discussed in *Il valore metrico del grado di meridiano secondo i geografi arabi*, in Guido Cora's *Cosmos*, vol. XI, Turin 1892, pp. 50–53 (= pp. 9–11 of the offprint); add that Jacob of Edessa, who died in 708 or 710, writes in his Syriac *Hexaemeron* that some make the degree 90 miles, others 75 (A. Hjelt, *Études sur l'Hexaméron de Jacques d'Édesse*, Helsingfors 1892, p. 20).

Al-Battānī does not mention the measurement of a degree of the meridian that Arab astronomers carried out in the reign of al-Ma'mūn (813–833) between the cities of Palmyra (Tadmur) and al-Raqqa, and in the plain of Sinjār (in Mesopotamia). They had found that, between about 35° and 36° of latitude, the degree is 56 2/3 Arabic miles, that is, 111,814.67 m; their error was therefore only 877 metres. See what I wrote, loc. cit., pp. 53–63, 105–121[a] (= pp. 12–39 of the offprint).

b) — p. 17, l. 22.

The Indians held that the island of *Laṅkā* (Taprobane, today Ceylon), placed on the equator, occupied the middle place in the longitude of the earth; the Arabs, not rightly understanding the Indian tradition, held that the central meridian of the earth passed through an island known by no name and lying on the equator between India and Ethiopia. The place where the central meridian cuts the equator they called *qubbat al-arḍ*, «the dome of the earth», as the author says, or simply *al-qubba*, «the dome, the cupola»; whence the corrupt word *Catha* in the old Latin translation of the astrology of ʿAlī ibn Abī l-Rijāl[1].

It was also known by the name *qubbat Uzayn* (or *Azīn* ازين); for the Indians held that the first meridian of the island of Laṅkā passed through the city of *Ujjayinī*, very famous among them (Ptolemy's Ὀζήνη, today Ujjain اجين in the province of Mālwa in central India, 75° 51′ 45″ E of Greenwich, 23° 11′ 10″ N), which flourished greatly in trade in the first three centuries of the Christian era and was perhaps the chief seat of astronomical studies in India. The Arabs mostly confused *Uzayn* or *Azīn* (for, with the signs of the vowels left out as usual, ازين can be read either way) with the island of Laṅkā itself, and put it on the equator; later writers, leaving out a dot in the writing, pronounced it *Arīn* (ارين), whence *Arin*, *Arim*, *Arym* in our writers of the Middle Ages. Read the excellent discussion of Reinaud, *Introduction générale à la Géographie d'Aboulféda*, Paris 1848, pp. CCXXXVI–CCLIV; of the Arabs, al-Bīrūnī above all should be consulted (*India*, pp. 158–159 of the text, vol. I, pp. 306–308 of the English translation).

c) — p. 17, ll. 22–27.

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I noted already in the translation that lines 7–25 are read in the same words in al-Masʿūdī, *Prairies*, vol. I, pp. 179–181, who ascribes them to «the sages» (*al-ḥukamā'*); al-Battānī, on the other hand, to anonymous «ancients» (see l. 3). We shall see in note *d* that the first source of this sixth chapter is Graeco-Syrian. Our author reports what he has received, neither approving nor disapproving it, as in ch. L (pp. 120–123) on the distances and the sizes of the celestial spheres and bodies; it is strange, all the same, that he did not reject the two gross errors of these lines. [For the area of a spherical quadrilateral bounded by two parallels and two meridians is not found by multiplying

[1] *Albohazen Haly filii Abenragel, De iudiciis astrorum libri octo, doctorum aliquot virorum opera in Latinum sermonem conversi, postremo a barbarie vindicati et puritati linguae donati, per Antonium Stupam.* Basel 1571, book VIII, ch. 8, p. 374.

[a] The print has 195–121.

the angular extent in latitude by the angular extent in longitude, since the degrees of one parallel, in linear measure, necessarily differ from the degrees of another. But even if one could compute areas as the author says, the computation would be wrong: half the circumference ought to be multiplied not by a sixth of the whole 360°, but by a third of the 180° that the latitude spans from pole to pole, so that the area found would be not 1/12 but 1/6 of the earth[1]. — The area of the spherical zone between the equator and the parallel of 60° is to the whole sphere as the sine of 60° is to the diameter; so half that zone will be to the whole sphere as the sine of 60° (that is, ½√3, or 0.8660) to twice the diameter (that is, 4, the radius being taken as 1). Instead of 0.8660 : 4, or 0.2165 : 1, those ancients found 1 : 12, or 0.0833 : 1, which is nearly three times less than the truth. — Schiaparelli.]

d) — pp. 17–19.

The description of the seas and the lands. — In the translation I have already pointed out, in their places, the Arab authors who drew from the same unknown source from which al-Battānī took his information; and I said that our author's passage had already been edited and translated into French by Reinaud. It remains to inquire what this common source is.

The oldest of all the Arab writers who tell such things about the seas in their books still extant today are Ibn Rusta, who wrote between the years 285 and 300 H (898–913), and Qudāma ibn Jaʿfar, who died in the year 337 H (which began on 10 July 948); al-Kharaqī, on the other hand, as we shall see below, says that he borrowed his information from al-Jayhānī (who wrote a little after 301 H, 913–14) «and other sages». All of them, therefore, belong to the age of our author. But a passage of al-Masʿūdī, *Tanbīh*, p. 51, sheds more light for us; for of the Indian sea he tells the following: «According to those who have applied themselves to measuring and depicting the earth by its places in astronomical longitudes and latitudes, the length of this sea from east to west comprises 8000 miles; its breadth to the north 2700, and [also] 1900 is said. So Ptolemy reports, and others who lived before and after him; the last who has attained these things in Islam is Yaʿqūb ibn Isḥāq al-Kindī, in his epistle on the seas and their ebb and flow, and his pupil Aḥmad ibn al-Ṭayyib in his epistle on the uses of the seas, the mountains and the rivers». Al-Masʿūdī mentions the same men in the *Prairies*, vol. I, p. 276, where he speaks of the breadth of the sea of the Byzantines[2]. In this way we reach a time before the age of al-Battānī; for al-Kindī, who had ordered a translation of Ptolemy's Geography to be made for himself[3], died about the year 260 H (which began on 26 October 873); his pupil Aḥmad ibn al-Ṭayyib al-Sarakhsī in the year 286 (which began on 16 January 899).

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Consider now that the degree of 75 miles comes from the Graeco-Syrian writers, as I showed in note *a*; add what I shall prove in the second part of this work[4], namely that unmistakable Syriac traces are to be recognized in al-Battānī's tables of the longitudes and latitudes of cities: it will not be a rash conjecture that the early Arabs received that geographical information and those measures of the seas from the Syrians. And the conjecture becomes quite certain, since in fact all these measures are found already in the famous Syriac writer Jacob of Edessa[5].

[1] The error is read again on p. 19, l. 21.

[2] Elsewhere (*Prairies*, I, p. 256), speaking again of the sea of the Byzantines, he cites al-Battānī: «Many authors of astronomical tables, among them Muḥammad ibn Jābir al-Battānī and others, say in their astronomical works that its length is 5000 miles, and that its breadth varies from 800 to 700 and 600 miles».

[3] See my *al-Ḫuwârizmî e il suo rifacimento della Geografia di Tolomeo*, p. 6, note 10 (Memorie della R. Accademia dei Lincei, Classe di scienze morali, 5th series, vol. II, 1894).

[4] I have already set this out in my little work *Le tabelle geografiche d'al-Battânî tradotte ed annotate*, Turin 1898, p. 3 (= Guido Cora's *Cosmos*, 2nd series, vol. XII, p. 163).

[5] He was born about 633 and died in 708 or 710. The geographical part is read in Syriac and in Latin in A. Hjelt, *Études sur l'Hexaméron de Jacques d'Édesse, notamment sur ses notions géographiques contenues dans le 3me Traité*, Helsingfors 1892.

If we now ask whence the Graeco-Syrians collected all this, we can only agree with the words of al-Masʿūdī quoted above, that it takes its first origin from Ptolemy's *Geography*. Let the miles be turned into degrees, taking the degree as 75 miles and making no distinction between the degree of the meridian and the degrees of the parallels (an error which, as we saw in note *c*, those authors had already made): everything will be found to agree perfectly with Ptolemy's map. It is surprising that Reinaud[1] and Lelewel[2] not only did not recognize this but at times denied it. For the former, considering the reading of the codex that gives the Indian sea 1900 (instead of 900) miles of breadth beyond the equator, writes on p. CCLXXXIV: «Mille et neuf cents milles, à raison de soixante et quinze milles par degré, donnaient un peu plus de vingt-cinq degrés. Voilà un des motifs qui me font croire qu'Albateny n'a pas admis le système de Ptolémée, relativement à la prolongation du continent africain du côté de l'est» («Nineteen hundred miles, at seventy-five miles to the degree, gave a little more than twenty-five degrees. This is one of the reasons that make me believe that Albateny did not accept Ptolemy's system as regards the extension of the African continent towards the east»). So, in drawing the picture of the world according to al-Battānī (or rather according to the ancient Greeks and Syrians whom al-Battānī copied), he makes the African coast, immediately south of the equator, turn west from the farthest peninsula of the Somalis in an almost straight line and join the coast of the Atlantic Ocean. Lelewel, vol. I, p. 37, note 86, attacks this, rightly observing, among other considerations, that to determine the breadth of the Indian Ocean is the same as to put land to the south of it; but afterwards he adds: «Mais nourrir l'idée d'une mer interne Hind, n'est pas suivre le système de Ptolémée ou copier sa carte, et je désire disculper Albateni d'une semblable accusation» («But to entertain the idea of an inner sea of India is not to follow Ptolemy's system or to copy his map, and I wish to clear Albateni of such an accusation»). And on p. 37, note 83, after reporting some of al-Battānī's measures of the seas, he remarks quite wrongly: «Comptant 75 milles au degré, ces nombres paraissent trop forts même pour le système de Ptolémée. La longueur des deux mers intérieurs [that is, the sea of the Byzantines and the Indian sea], 5000 et 8000, ferait évaluer la longueur de l'habitable sur le grand cercle à 185 1/3 degrés [*sic*, but 13000 : 75 = 173 1/3], à quoi il faudrait ajouter quelques degrés de plus. Pour adapter ces nombres à la carte ptoléméenne, il faudrait compter au moins 80 milles au degré du grand cercle» («Counting 75 miles to the degree, these numbers seem too large even for Ptolemy's system. The length of the two inner seas, 5000 and 8000, would put the length of the inhabited land on the great circle at 185 1/3 degrees, to which some degrees more would have to be added. To fit these numbers to Ptolemy's map one would have to count at least 80 miles to the degree of the great circle»). An error of arithmetical division, then, was perhaps the chief reason why Lelewel denied the Ptolemaic origin of this chapter of al-Battānī; a lesser reason was that at that time the Arabic and Syriac writers who give the same measures were not known.

To show better all I have said of the common, and Ptolemaic, origin of the descriptions of the seas, and so as to be able to correct our author's numbers here and there, I shall now give a survey of the measures reported by al-Battānī (with whom Ibn Rusta agrees), by al-Jayhānī (according to al-Kharaqī) and by Jacob of Edessa, comparing the other Arab writers too[3]. I 168 I 169

[1] In the book I cited on p. 17, note 1.

[2] J. Lelewel, *Géographie du moyen âge*, Breslau 1852, vol. I: Cartes de géographes, pp. 36–38.

[3] The fragments read in Qudāma, pp. 230–231, correspond to p. 17, ll. 29–33 (as far as the number 200), p. 18, ll. 9–16 and 20–24 (leaving out the miles of the circuit of the islands), and p. 19, ll. 2–4, of my translation of al-Battānī.

	al-Battānī and Ibn Rusta	*al-Ǧayhānī in al-Kharaqī*	*Ǧacob of Edessa*
The inhabited land:			
length	13,500 [= 180°]	13,500	13,500
breadth	. . . . [= 60°]	. . . .	. . . . [= 60°]
The sea of the Indians[1]:			
length	8000 [= 106° 2/3]	8000	8000
whole breadth	2700 [= 36°]	2700	2700
of which south of the equator	1900 [= 25° 1/3]	1700	900 [= 12°]
The Barbarian Gulf[2]:			
length	500 [= 6° 2/3]	500	. . . .
breadth	100 [= 1° 1/3]	100	. . . .
The sea of al-Qulzum[3]:			
length	1400 [= 18° 2/3]	1400	1400
breadth at the end	200 [= 2° 2/3]	200	200
breadth at the beginning	700 [= 9° 1/3]	700	400 [= 5° 1/3]
The sea of al-Baṣra[4]:			
length	1400 [= 18° 2/3]	1400	1400
breadth at the beginning	500 [= 6° 2/3]	500	700 [= 9° 1/3] (both)
breadth on the direct course	150 [= 2°]	150	
The land between the sea of al-Qulzum and the Persian Gulf:			
length	1500 [= 20°]	1500	. . . .
The Green Gulf:			
length	1500 [= 20°]	1500	. . . .
The sea of the Byzantines and of Egypt[5]:			
length	5000 [= 66° 2/3]	5000	5000
breadth	600 [= 8°]	600	
	700 [= 9° 1/3]	700	400 [= 5° 1/3] (all three)
	800 [= 10° 2/3]	800	
The Adriatic Gulf:			
length	500 [= 6° 2/3]	500	. . . .
The Gulf of Narbonne:			
length	200 [= 2° 2/3]	200	. . . .
The Pontus[6]:			
length	1060 [= 14° 2/15]	1300	1300 [= 17° 1/3]
breadth	300 [= 4°]	300	300
The Maeotic Lake[7]:			
length	300 [= 4°]	300	a little smaller
breadth	100 [= 1° 1/3]	100	than the Pontus (both)
The strait of Constantinople:			
breadth	3	3	. . . .
The sea of Jurǧān[8]:			
length	800 [= 10° 2/3]	800	1800 [= 24°]
breadth	600 [= 8°]	600	600

Of the two islands of the sea of Jurjān (the Caspian), unknown to Ptolemy, Jacob of Edessa, p. LXV, has the following: «Quo in mari, quia identidem multitudine aquarum, quae perpetuo infunduntur, nimis augetur, fieri non potuit ut insulae exstiterint. Vix enim ab antiquissimis temporibus usque ad hoc tempus duae solae insulae parvae ibi conspectae sunt, quae ad habitandum parum sunt aptae» («In this sea, because it is again and again greatly increased by the mass of the waters that pour into it continually, it has not been possible for islands to arise. For from the most ancient times down to the present hardly two small islands have been seen there, and they are little fit for habitation»). See what al-Masʿūdī, *Prairies*, II, 26; *Tanbīh*, p. 60; Aboulféda, pp. 47 ff. (text), etc., relate of volcanic islands in this sea.

	al-Battānī and Ibn Rusta	*al-Ǧayhānī in al-Kharaqī*	*Ǧacob of Edessa*
In the **Indian sea** there are islands[9]	1370	1370	1378
of which around Taprobane	59[10a]	19	19
The circuit of **Taprobane**	3000 miles	3000	4000

The circuits of the islands of Cyrnos, Sardinia, Cyprus, Sicily and Crete are read as in al-Battānī in the Astronomy of Barhebraeus, p. 135 of the text, except only that the 350 miles of Cyprus are wrongly given to Cyrnos, and the 200 of Cyrnos to Cyprus.

I shall now give the whole unpublished passage of al-Kharaqī that describes the seas[11].[b]

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[1]Ibn Rusta, Qudāma and al-Masʿūdī, as I said on p. 17, note 4, confirm al-Battānī's numbers; the 1700 of al-Kharaqī is a copyist's error (سبع 7 for تسع 9) that is very frequent in Arabic manuscripts. The Arabic source (al-Kindī?), then, already had 1900; but the true and oldest reading is 900, as appears from Ptolemy's map and from Jacob of Edessa («in latitudinem circa 900 milliaria a loco aequinoctiali meridiem versus irrumpit usque ad terram a terminis eius australibus versus meridiem sitam, quae ab hominibus incognita appellatur et prorsus inhabitabilis est», «in breadth it breaks in about 900 miles from the equinoctial towards the south, as far as the land that lies to the south of its southern borders, which is called unknown to men and is wholly uninhabitable»; Hjelt, *L'Examéron*, p. LXVI). See the passage of al-Masʿūdī, *Tanbīh*, p. 51, quoted above (p. 166).

[2]It is Ptolemy's Βαρβαρικὸν πέλαγος, but with these Arabs it seems to denote the Gulf of Aden. Ptolemy calls Βαρβαρία the coasts that stretch from Cape Guardafui to the island of Zanzibar. — The same numbers in al-Masʿūdī, *Prairies*, I, 231, and in Barhebraeus, p. 135 of the text.

[3]The breadth of 700 miles is also in Ibn Rusta and in al-Masʿūdī, *Prairies*, I, 237; it is therefore the reading of the ancient Arabic source. But Jacob of Edessa and Barhebraeus (loc. cit.) have 400, more correctly. — On this sea see note *e*.

[4]Ibn Rusta has the same numbers; and al-Masʿūdī, *Prairies*, I, 238, writes: «the length of this gulf is 1400 miles; its breadth at the beginning 500, but at times the breadth of its two extremities becomes 150 miles». But «of its two extremities» (طرفيه) is plainly a scribal error for the original «of its direct course» (طريقه) that is read in al-Battānī. The same error occurs in another passage of al-Masʿūdī; see note *e*.

[5]The same numbers in Ibn Rusta; al-Masʿūdī, *Tanbīh*, pp. 56–57; Barhebraeus, p. 135 of the text. In the *Prairies*, I, 256, we read: «Many authors of astronomical tables, among them Muḥammad ibn Jābir al-Battānī and others, say in their books on the stars that its length is 5000 miles, and that its breadth varies from 800 to 700 and 600». Al-Masʿūdī himself, on the other hand, *Prairies*, I, 276 (see *Tanbīh*, p. 56), reports the same numbers as Jacob of Edessa, on the authority of al-Kindī and of Aḥmad ibn al-Ṭayyib al-Sarakhsī.

[6]The length 1060 is an error; Ibn Rusta, al-Masʿūdī (*Tanbīh*, p. 66) and Barhebraeus (p. 137 of the text) have 1300, like al-Kharaqī and Jacob. In al-Masʿūdī, *Prairies*, I, 260, the length 1100 is an evident scribal error for 1300. — See p. 18, note 12.

[7]The same numbers in al-Masʿūdī, *Tanbīh*, p. 67.

[8]The same numbers in al-Masʿūdī, *Prairies*, I, 263, and *Tanbīh*, p. 60.

[9]Ibn Rusta and Barhebraeus (p. 136 of the text) also have 1370; but Jacob of Edessa's 1378 is Ptolemy's number (*Geography*, VII, 4, 11).

[10]The true reading is 19, as I said on p. 18, note 5.

[11]It is ch. II (*bāb*) of the second treatise (*maqāla*); Paris codex ff. 74r–76r, Laurentian codex ff. 57v–58v. — The author died in the city of Merv in the year 533 H (which began on 7 September 1138).

[a]The print has no call for Nallino's note 5; it belongs to the 59 of al-Battānī.

[b]Nallino's Addenda et emendanda (Part II, pp. XXII–XXIII) add to pp. 167–175 the like descriptions of the seas by Niẓām al-Dīn al-Naysābūrī and Fakhr al-Dīn al-Rāzī; they are translated with Part II.

«Chapter II: on the description of the seas and their position in the inhabited lands. Al-Jayhānī and other sages [versed] in the knowledge of the position and the extent of the seas relate that there are five known seas: the first, *the sea of the Indians*, also called *the sea of China*; the second, *the sea of the West* (*baḥr al-maghrib*); the third, *the sea of Syria, of the Byzantines and of Egypt*; the fourth, *the Pontus* (*bunṭus*); the fifth, *the sea of Jurjān* (that is, *the Caspian*). — *The length of the sea of the Indians* stretches from west to east, from the farthest part of *Ethiopia* (*arḍ al-ḥabasha*) to the farthest land of the *Indians* and the *Chinese*, and comprises 8000 miles; its breadth is 2700 miles, and it passes the line of the equator by 1700 miles. A gulf goes out of it near *Ethiopia* and stretches towards *al-Barbar*; it is called *the Barbarian Gulf*; its length is 500 miles, the breadth of its extremity[1] 100. — Out of it [the sea of the Indians] another gulf too goes out towards *Ayla*, namely *the sea of al-Qulzum*, whose length comprises 1400 miles, its breadth at its beginning 700, and its end towards the nearest extremity, which is called *the Green Sea*, 200[2]; at its almost pointed extremity is [the city of] *al-Qulzum*, from which the sea took its name; to the east of it is the land of *al-Yaman* and *ʿAdan*, to the south *Ethiopia*. — A gulf also goes out of it [the Indian sea] towards the land of *Persia* (*fāris*), called *the Persian Gulf*, or *the sea of al-Baṣra and of Persia*, on whose eastern [shore] are *Tīz* and [the province of] *Mukrān*, on the western ….. [3] *ʿUmān*; its length is 1400 miles, its breadth at its beginning 500, on the direct course 150. Between these two gulfs, that is, of *Ayla* and of *Persia*, are the land of *al-Ḥijāz*, *al-Yaman* and the other regions of the *Arabs*, 1500 miles in extent. — Another gulf too goes out towards the farthest lands of the *Indians*, which near its end is called *the Red Gulf*[4], and whose length is 1500 miles. — In this *sea of the Indians* there are 1370 islands, inhabited or deserted, among which is a great island at the far end of the sea, facing the land of the *Indians* from the east, near the region of *China*, called *Taprobane* or *Sarandīb*; it has a circuit of 3000 miles; in it are high mountains and many rivers; from it[5] come rubies and sapphires. Around it are 19 inhabited islands, rich in cities and villages; among the islands of this sea are the islands of *al-Zābij*[6], the islands of *al-Dībaḥāt*[7] and *Qumayr*[8], the island of *Kalah*[9], from which tin (الرصاص القلعي) is brought, and the island of *Sarbuza*[10], from which camphor comes. I 171

[1]The manuscripts have طرفه, but without doubt طريقه, «of its direct course», should be read, as in al-Battānī.

[2]On this passage see note *e*, pp. 175–177.

[3]I do not know how to correct the corrupt words of the manuscripts.

[4]So the manuscripts, with which one might compare the Greek name of the Indian Ocean (Ἐρυθρὰ θάλασσα, the Red Sea); but why a small gulf should have taken the name of the whole sea is not understood. When al-Battānī (p. 18, l. 4 and note 2) and Ibn Rusta are compared, there can hardly be a doubt that الاحمر, «red», is an error of the manuscripts, or of the author himself, for الاخضر, «green». [*Translator:* Nallino's Addenda et emendanda (Part II, p. XXIII) note that Fakhr al-Dīn al-Rāzī rightly calls this gulf the Green Gulf, al-khalīj al-akhḍar.]

[5]So the manuscripts, ومنها يخرج, rightly; al-Battānī leaves out the و, so that the pronoun must refer to the mountains and rivers.

[6]It is the island we call *Java* (in Malay *Jāwa* جاو, among the Arabs of the thirteenth century *Jāwah*); see Van der Lith's discussion in *Le livre des merveilles de l'Inde par le capitaine Bozorg fils de Chahriyâr, texte arabe et traduction française par P. A. Van der Lith et M. Devic*, Leiden 1883, pp. 231–33.

[7]The manuscripts are without dots; others read *al-Dībajāt* (ج j for ح ḥ). The name arose from the Indian *dvīpa*, that is, island; it denotes the islands we call today the *Laccadives* and the *Maldives* (*Dībat al-maḥal* among the Arabs of the fourteenth century); see Van der Lith, 216–17.

[8]I keep the reading of manuscript L (قمير); P has قمر. The name is also found written so in al-Bīrūnī, *India*, p. 103 of the text (= I, 210 of the translation), where without doubt it denotes the same region that the other Arabs call قمار *Qumār* (that is, *Cambodia*, whose ancient inhabitants were the *Khmēr*). The passage of al-Bīrūnī, which Sachau did not quite rightly understand, was better interpreted by de Goeje in his translation of Ibn Khurradādhbih, p. 50, note 1.

[9]It is the port called today *Kēdā* (*Quedah*), in the Malay peninsula (Malāka ملاك), as Van der Lith has shown, pp. 255–264; see also Ibn Rusta, p. 88, and al-Masʿūdī, *Tanbīh*, p. 61.

[10]It was a village or region in the island of Sumatra, on the banks of the river *Mūsī*, near the place where the city of *Palembang* (فلمبغ) now stands; in the Chinese annals it is called *Sam-bo-tsai*. See Van der Lith, pp. 247–253 (and 102, 105, 111, 124, 158–160, 165, 176, 190); al-Masʿūdī, *Tanbīh*, p. 61; *Prairies*, I, 243 (where the name is corrupted to *Serīra* سريرة for سريزة); Yāqūt, I, 61, III, 68 and 354; Aboulféda, pp. 22, 368, 375 (= II, 26, III, 126 and 132 of the translation), where it is always *Serīra*; al-Dimashqī, pp. 22, 199, 204 (*Serīra*), etc.

«*The sea of the West*, which is also called *the Encircling sea* and by the Greeks *Oceanus*, is joined to the *sea of the Indians*, and nothing of it is known except the part which, lying in the west and the north, faces the land of the *Russians* (*al-rūs*) and of the *Slavs* (*al-ṣaqāliba*)[1]. Beginning from its last limit in the south, it faces the *land of the Blacks* (*arḍ al-sūdān*) and passes along the borders of *Farthest al-Sūs*[2], *Tangier* (*Ṭanja*), *Tāhart*[3], *Spain*, *al-Ǧalāliqa*[4] and the *Slavs*[5] as far as *Britain* (*al-Barṭāniya*); then from there it stretches, beyond impassable mountains and uninhabited lands, towards the *sea of the East*. This sea, *Oceanus*, is not furrowed by ships and is sailed only near its shores; in it are six islands facing the land of the *Ethiopians*, which are called the *Eternal* (*al-khālidāt*) or the *Fortunate* (*al-suʿadāʾ*). Another island, called *Gadira*, lies opposite *Spain*, near the *gulf* that goes out of this sea; the gulf is 7 miles wide where it goes out, lies between *Spain* and *Tangier*, and was [anciently] called the *Ford of Hercules* (*maʿbarat Hiraqlas*); today it is called *al-Zuqāq* [that is, the narrow lane]; this gulf flows into the *sea of the Byzantines and of Syria*. In the northern stretch of *Oceanus* are the 12 islands of *Britain*. Out of it too, in the north of the *Slavs*, arises a great gulf, which is also called the *sea of the Varangians* (*warank*), after the people that inhabits its shores[6]; it stretches to the land of the Muslim *Bulgars* (*bulghār*), and extends 300 miles in length from east to west and 100 in breadth.

«*The sea of the Byzantines, of Africa, of Egypt and of Syria*, proceeding eastwards from the gulf that goes out of the *sea of the West* as far as *Tyre* and *Sidon*, has a length of 5000 miles, a breadth in one place of 600, in another of 700, in another of 800. A gulf goes out of it which, going northwards near *Rome*, extends 500 miles in length and is called *Adrias*[7]. Another gulf goes out towards the land of *Narbonne*; it has a length of 200 miles. In this sea there are 162 inhabited islands, of which 5[8] are large, namely *Rhodes*, *Cyprus*, *Samos*, *Sicily* and *Crete*.

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«*The Pontus* is called the land-locked (that is, inner) *sea*, and [also] *the sea of Trebizond*, after a port on its shore; it stretches from *Lādhiqa* beyond *Constantinople*, into the land of the *Russians* and the *Slavs*; it has a length of 1300 miles and a breadth of 300. Into this sea flows the river called *Tanais*, which runs from the northern side with respect to the *Maeotic sea*, which is the *sea of the Varangians* (*warank*)[9]. From the *Pontus*, near the walls of *Constantinople*, a gulf goes out like a river[10] and flows down into the *sea of Egypt*; near *Constantinople*, which is on its western

[1]By this name the Arab writers denote the Slavs and many Germanic tribes; see G. Jacob, *Studien in arabischen Geographen*, Berlin 1891–92, part I, p. 17.

[2]*Farthest al-Sūs* is the southern province of the empire of Morocco, whose chief city today is *Tārūdānt*.

[3]It was the name of two great cities five Arabic miles (about ten km) apart, of which one had the surname «the old» (*al-qadīma*), the other «the new» (*ḥadītha*). The former is called *Tiharet* (*Tiyāret*) on our maps, the other *Tāqdemt*; both are in western Algeria. Read de Goeje's discussion, *Specimen e litteris orientalibus exhibens descriptionem al-Magribi sumtam e Libro Regionum al-Ǧaʿqubii*, Leiden 1860, pp. 103 ff., and later in *Description de l'Afrique et de l'Espagne par Edrisi*, Leiden 1866, p. 100; see also R. Basset, *Documents géographiques sur l'Afrique septentrionale traduits de l'arabe*, Paris 1898, p. 23. [*Translator:* Nallino's Addenda et emendanda (Part I, p. LXXIII) add J. Canal, *Tiaret, monographie ancienne et moderne*, in the Bulletin trimestriel de géographie et d'archéologie d'Oran, vol. XX, 1900, pp. 1–44.]

[4]They are the *Callaeci* of the Romans, the *Gallaeci* of the Middle Ages, whose name survives in the present province of *Galicia* (in northern Spain).

[5]See note 1.

[6]It is the Baltic Sea, whose eastern shores were inhabited in the ninth century by the *Varangians* (German *Wäringer* or *Waräger*), a Norman people. The name is uncommon in the Arab writers; Aboulféda, p. 35 (= II, 42 of the translation), testifies that he found it written only in al-Bīrūnī and in Naṣīr al-Dīn's *Tadhkira*. — What follows in al-Kharaqī, however, strangely confuses, as we shall also see below, the Baltic Sea with the Maeotic Lake (the Sea of Azov), near which the Muslim Bulgars lived, and to which the measures given in this passage without doubt belong. [*Translator:* Nallino's Addenda et emendanda (Part II, p. XXIII) take back «uncommon»: the name is found also in al-Naysābūrī, al-Qazwīnī and al-Dimashqī, and C. M. Frähn, *Ibn Foszlan's und anderer Araber Berichte über die Russen älterer Zeit* (St Petersburg 1823), pp. 177–204, gathered the Arabic and Persian witnesses of this sea.]

[7]With the Syrians *Adrias* (ܐܕܪܝܐܣ *adryūs*) is the Mediterranean Sea; see F. Nau, *Notice sur le livre des Trésors de Jacques de Bartela*, in the Journal Asiatique, 9th series, vol. VII, 1896, pp. 309–312 (= 26–29 of the offprint).

[8]Manuscript P has خمسون, 50; but the sense and the grammatical construction (for جزائر عظام is read afterwards) demand خمس, 5, as manuscript L, al-Battānī and Ibn Rusta have.

[9]He confuses the Baltic Sea with the Maeotic Lake, as I said on p. 171, note 6.

[10]The manuscripts have the corrupt بحرى فيه مايه نهر, which I restore with al-Battānī and Ibn Rusta to يجري كانه نهر.

shore, it has a breadth of 3 miles, and grows ever narrower until it flows into the *sea of Syria and of Egypt*. On the southern shore of this sea [*of Egypt and of the Byzantines*] are the lands of *al-Maghrib*[1] and of *Africa* as far as *Alexandria* and *Egypt*; opposite, on the northern side, lie the lands of *Spain*, of the *Byzantines* and of the *Franks* (*faranja*) as far as *Antioch*, which *Syria* and *Palestine* follow.

«*The sea of Jurjān*, or of the [*Caucasian*] *Gate*, extends 800 miles in length from west to east and 600 in breadth; it contains two islands, which were inhabited in times past. This sea is also called *the sea of Ābaskūn*, after one of its ports[2]; it then reaches to [the provinces of] *Ṭabaristān*, *al-Daylam* and *Shirwān*, [the city of] *Bāb al-Abwāb*[3], and the region of the *Alans* (*al-Lān*)[4]; then to the *Khazars* (*al-khazar*) and the mouth of the river *Itil*[5]; lastly to the regions of the *Persians*, until it reaches *Ābaskūn* again. It is called by the name of each region that it washes; in ancient times it was called *the sea of Jurjān*, today it is called *the sea of Ābaskūn* and *the sea of the Khazars*; it does not flow into any other sea.

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«These are the great seas; the rest are lakes and marshes, such as the lake of *Khwārizm* (that is, of *Chorasmia*, today the *Aral*), the lake of *Tiberias* in *Syria*, and *Īssik-kūl* near *Barsukhān*[6]. *Aristotle*[7] relates that *Oceanus* surrounds [the whole] earth like a crown; that out of it arises a gulf on the western side, at the place called the *Ford of Hercules*[8], which, entering the inhabited land, proceeds eastwards and is the *sea of Egypt, of the Byzantines, of Africa and of Syria*. On the eastern side, on the other hand, another gulf comes out of it [*Oceanus*], entering the inhabited land, which stretches westwards and is the *sea of China, of the Indians, of Persia, of al-Qulzum, of the Barbarians* (*al-Barbar*). This is what we know of the seas».

I give the Arabic text of the whole passage of al-Kharaqī after the Paris (P) and the Laurentian (L) manuscripts:

﴿الباب الثاني﴾ في ذكر البحور وكيفيّة وَضْعها من المساكن. قد حكى الجَيْهانيُّ[1] وغيره من العلماء بمواقع البحور وامتداداتها ان البحور المعروفة خمسة احدها بحر الهند وهو الذي يقال له ايضًا بحـر الصين والثاني بحر المغرب والثالث بحر الشام والروم ومصر والرابع بحر بنطس[2] والخامس بحر جرجان فاما بحر الهند فانه يمتـدّ طوله من المغرب الى المشرق

[1]The *Mauretania* of the ancients; today the empire of Morocco, Algeria and the province of Tunis.

[2]Abuskūn, Abiskūn or Aboskūn was a great sea port of the region and the city of Jurjān (whose ruins stand near Pisarak پسرك). There are two opinions on the site of the place: B. Dorn, *Caspia, Ueber die Einfälle der alten Russen in Tabaristan*, St Petersburg 1875, pp. 5, 6, 7, puts it at the mouth of the river Gurgān گرگان, not far from the village of *Gümüsh-tepeh* گومش تپه, where Mouravieff (in the first quarter of the nineteenth century) found many ruins; Melgunof, *Das südliche Ufer des kaspischen Meeres*, Leipzig 1868, pp. 62, 63, 112, 119–120, around the present mausoleum (imām-zādeh) called *Kharāb-i-shehr*, 10 km from the shore and 6 east of the town of Gez جز (in the district of Anezān of the province of Asterābād). But it is always near the place where the eastern shore of the sea turns westwards.

[3]Today *Derbent*.

[4]They are the people called today *Ossetes*.

[5]What the ancients called the *Rha*, we call the *Volga*.

[6]The names are corrupt in the manuscripts. The city is known to the older Arab geographers, such as Qudāma, p. 203 (where it is corrupted to نوشجان Nūshajān for برسخان), and al-Muqaddasī, pp. 48, 263, 275, 340; the lake, on the other hand, is unknown to them, except to Qudāma, who mentions it without a name, loc. cit. Yet the lake, which lies in eastern Turkestan, is already called *Īssik-kūl* (ايسكول, or corruptly السكول) by a Persian book of an unknown author, written in the year 372 H (which began on 26 June 982) and entitled *Ḥudūd al-ʿālam*, «the Limits of the world»; see V. Barthold, *Otčet pojezdkije v srednjuju Aziju s naučnoju cjelju, 1893–94* (Report on a journey in Central Asia undertaken for a scientific purpose in the years 1893–94), St Petersburg 1897, pp. 31–32 (= Mémoires de l'Académie Impériale des Sciences de St.-Pétersbourg, 8th series, Classe historico-philologique, vol. I, no. 4). The Turkish name means «warm lake»; it is written in various ways: *īssīq*, *issīq*, *īssīgh*, *issīgh*. Abū Saʿīd ʿAbd al-Ḥayy Gardīzī (گردیزي) describes the city of Barsukhān and the lake (*īssigh-kūl*) at length, about the year 1050, in his book *Zayn al-akhbār*; his words are given in Persian in Barthold, pp. 89–90, and in Russian translation, pp. 113–114. The name *īssīgh-kūl* is written by Muḥammad Ḥaydar in his Persian «Rashīdī History» (*Taʾrīkh-i rashīdī*), in the sixteenth century; see Barthold, p. 50.

[7]Quṭb al-Dīn Maḥmūd al-Shīrāzī also gives this whole passage of Aristotle, *al-Tuḥfa al-shāhiyya*, f. 158r.

[8]See above, p. 171, l. 12, and below, on the Arabic text, pp. 174 and 175.

من اقصى ارض الحبشة الى اقصى ارض الهند والصين يكون مقدار ذلك ثمانية آلاف ميل وعرضه الفان وسبعمائة ميــل ويجاوز خطّ الاستواء الف وسبعمائة ميل يخرج منه خليج عند ارض الحبشة ويمتدّ الى ناحيــة البربر يُسمَّي ذلك الخليج البربريّ يكون طوله مقدار خمسمائة ميل وعرض طرفه مائة ميل ويخرج منه ايضًا خليج آخَرُ نحو أَيْلةَ وهو بحر القلزم طوله الف واربعمائة ميل وعرضه في الاصل سبعمائة ميل ومنتهاه الى طرف[3] الأَذْنَى الذي يُسمَّى البحر الاخضر مقدار مائتَيْ ميل وعلى طرفه المستدقّ القلزم فلذلك سُمِّيَ بــه على شرقيّه ارض اليمــن وعدن وعلى جنوبيّه ارض الحبشة ويخرج منه خليج آخَرُ نحــو ارض فارس يُسمَّى الخليج الفارسيّ وهو بحــر البصرة وفارس الذي على شرقيّه تِيز ومُكران وعلى غربيّه *وبحاله ى حمه[4] عمان طوله الف واربعمائة ميل وعرضه في الاصل خمسمائة ميل وعرض طريقه مائة وخمسون ميــلًا وبين هذَيْن الخليجَيْن اعني خليج ايلة وخليج فارس ارض الحجاز واليمن وسائر بــلاد العرب فيما بين مسافة الف وخمسمائة ميل ويخرج منه خليج آخَرُ الى اقصى ارض الهند عند تمامــه يُسمَّى الخليج الاحمر وطوله الف وخمسمائة ميل وفي بحر الهند هذا من الجزائر العامرة وغــير العامرة الف وثلثمائة وسبعون جزيرة

1) P الجبهانى, L الجهيانى — 2) P نبطس — 3) So the manuscripts, for طرفه or الطرف — 4) So P; L وبحالهِ ى خنبه

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منها جزيرة ضَخمة في اقصى البحر مقابل ارض الهند في ناحية المشرق عند بلد الصين تُسمَّى طبروباني[1] وهي سَرَنديب يحيط بها ثلثة آلاف ميل فيها جِبال عظيمة وانهار كثيرة ومنها يخرج الياقوت الاحمر ولون السماء وحَوْلَ هذه الجزيرة تسع عشرة جزيرة عامرة فيها مدائن وقُرَى كثيرة ومن جزائر هــذا البحر جزائر الزابج وجزائر الدِّيبَجات وقُمَيْر[2] ومنها جزيرة كَلَه التي يُجْلَب منها الرَّصاص القَلْعِيّ وجزيرة سَرْبُزَة[3] التي يُجْلَب منها الكافور واما بحر المغرب فهو الذي يُسمَّى المحيط وتُسَمّيه اليونانيّون اوقيانوس ويتَّصِل به بحر الهند ولا يُعْرَف منه إلّا في ناحية المغرب والشمال عند مُحاذاة ارض الروس والصقالبة فيأخذ من اقصى المُنتَهَى في الجنوب محاذيًا لارض السودان مارًّا على حدود السوس الأَقْصَى وطنجة وتاهَرْت[4] ثمّ الاندلس والجلالقة والصقالبة الى البرطانية ثمّ يمتــدّ من هناك وراء الجبال[5] المسلوكة والاراضي غير المسكونة نحو بحر المشرق وهذا البحر اعني اوقيانوس لا يجري فيه السُّفُن وانّما يُسْلَك بالقُرْب من ساحله وفيه ستّ جزائر مقابل ارض الحبشة تُسمَّى جزائر الخالدات وتُسمَّى ايضًا جزائر السَّعَداء وجزيرة اخرى تُسمَّى غديرة[6] مقابل الاندلس عند خليج يخرج من هذا البحر وعرضه حيث يخرج سبعة اميال وهــو بين الاندلس وطنجة كان يُعْرَف بمَعْبَرَة هِرَقْلَس[7] والآن يُعْرَف بالزُّقاق وهذا الخليج ينصبّ في بحر الروم والشام وفيه اعني بحر اوقيانوس من ناحية الشمال جزائر برطانية وهي اثنتا عشرة جزيرة ويخرج منه ايضًا خليج عظيم في شمال الصقالبة يُسمَّى ايضًا بحر وَرَنْك[8] وهم امّــة على ساحله ويمتدّ هــذا الخليج الى ارض بلغــار المسلمين طوله من المشرق الى المغرب ثلثمائة ميــل وعرضه مائة ميل واما بحر الروم وافريقية ومصر والشام طوله من عنــد الخليج الذي يخرج من بحر المغرب الى ناحية المشرق ويمتدّ الى صور وصَيْداء يكون ذلك مقدار خمسة الاف ميل وعرضه في مكان ستّمائة ميل وفي مكان سبعمائة ميل وفي مكان ثمانمائة ميل ويخرج منه خليج الى ناحية الشمال قريب من الرومية يكون طوله خمسمائة ميل يُسمَّى اذريس[9] ويخرج منه خليج آخَرُ الى ارض تربُونة[10] يكون طوله مائتَيْ ميل وفي هذا البحر مائة واثنتان وستّون جزيرةً عامرةً منها خمس[11] جزائر عظام وهي جزيرة

1) P طروبانى, L طرفانى — 2) P قمر, L قمير — 3) P سريره, L سربره — 4) P وماهرت, L وباهرت — 5) So the manuscripts, but without doubt غير should be added — 6) P عديره, L عدره — 7) P بمغره مرقلس, L بمعره هرقلس. See also p. 175, note 11; Yāqūt, I, 61, l. 19, معبرة هيرَقْلَس; Qāḍīzāde, in his commentary on the first chapter of the third section of the *Tadhkira*, f. 120v, معره هرقل; Quṭb al-Dīn Maḥmūd al-Shīrāzī, *al-Tuḥfa al-shāhiyya*, f. 158r, معبره هرقلس — 8) P وربله, L وربك — 9) The manuscripts خمسون, but afterwards جزاير — 10) P ادرس, L بربريه — 11) P بربره

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رودص وقبرس وسامس وسقلية واقريطش وامّا بحر بنطس الارضيّ فإنّه يُسمَّى طرابزند[1] لأنّها فُرْضة عليه ويمتدّ من

لاذقة[2] الى خَلْفِ قسطنطينيّة في ارض الروس والصقالبة طوله مقدار الف وثلثمائة ميل وعرضه ثَلثمائة ميل ويدخل فيه النهر الذي يُسَمّي طانيس وهو يجري من ناحية الشمال من البحر الذي يُسمَّى مايطس وهو بحر وَرَنْك وينفجر من بحر بنطس عند سُوَر قسطنطينيّة خليج يجري فيه مائة[3] نهر وينصبّ في بحر مصر وعرضه عند قسطنطينيّة قدر ثلثة اميال وقسطنطينيّة هي على شاطئه الغربيّ ولا يزال يتضايق حتّى يقع في بحر الشام ومصر الذي على جنوبيّهِ بلادُ المغرب وافريقيّة الى الاسكندريّة ومصر وبحِذائه في الشمال ارض الاندلس والروم وفَرَنْجة[4] الى انطاكِيَة ويليها بلاد الشام وفِلَسْطِين وامّا بحر جُرْجان وهو بحر الباب[5] طوله من المغرب الى المشرق ثمانمائة ميل وعرضه سّتّمائة ميل وفيه جزيرتان كانتا عامرتَيْن فيما مضى من الزمان ويُعْرَف هذا البحر ببحر آبَسْكُون[6] لأنّها على فُرْضته ثمّ يمتدّ الى طبرستان والدَّيْلَم وشِرْوان وباب الابواب وناحية اللّان ثمّ الخزر على مَصَبّ نهر إتِل ثمّ ديار الفُرْس الى ان يعود الى آبسكون[7] ويُسمَّى باسم كلّ بُقْعة حاذاها كان في القديم يُعْرَف ببحر جرجان والآن ببحر آبَسْكُون[8] وببحر الخزر وليس يتَّصِل ببحر آخَرَ وهذه هي البحور العظام فامّا غيرها بُحَيْرات وبطائح كبُحَيْرة خوارزم وبحيرة طَبَرِيَّة بأرض الشام وإِسِكّول[10] بالقرب من بَرْسُخَان[9] وقد حكى ارسطاطاليس انّ بحر اوقيانوس محيط الارض بمَنْزِلة إكْليل لها وانّه ينفتح منه خليج في ناحية المغرب من الموضع المسمّى بمَعْبَرَة هِرَقْلَس[11] فيدخل الى المعمورة ماراًّ نحو المشرق وهو بحر مصر والروم وافريقيّة والشام وينفتح منه ايضًا في ناحية المشرق خليج فيدخل الى المعمورة ماراًّ نحو المغرب وهـو بحر الصين والهند وفارس والقلزم والبربر فهذا ما عرفناه من امر البحور

a

1) P طراوند, L طراوبد — 2) P لادقيه, L لاذقيه — 3) So the manuscripts; but see p. 172, note 2. — 4) P فرخه, L فرحه — 5) The manuscripts الراب — 6) P السلون, L انسكون — 7) P السكون — 8) P السلون — 9) The manuscripts والسلول[b] — 10) P برسحان, L برشخان — 11) P بمعره مرفلس, L بمعره هرقلس; see above, p. 174, note 7.

e) — p. 17, l. 33.

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On this *Green Sea* opinions have differed. Reinaud, loc. cit., p. CCLXXXV, note 2, writes: «En grec, mer πρασώδης, du mot πράσον, signifiant *poireau*» («In Greek, the πρασώδης sea, from the word πράσον, meaning *leek*»); clearly, then, he took it to be the *Indian Ocean*, a part of which, washing the coasts of Africa from the island of Menuthias (Zanzibar?) to the farthest east, is called ἡ Πρασώδης θάλασσα by Ptolemy (see p. 18, note 2). But «the direct course, called the Green Sea» cannot lie outside the sea of al-Qulzum itself; moreover, the author afterwards denotes the part of the sea of al-Qulzum that adjoins the Indian Ocean by the words «at the beginning», to which, following Ptolemy, he gives the enormous breadth of 700 miles[1].

The parallel passage in Ibn Rusta, p. 84, is: «and the breadth near its end in the west, that is, at its nearest extremity, which is called the *Last sea*, 200» (وعند منتهاه في المغرب اعني طرفه الادنى الذي يسمى بالبحر الاخير مائتا ميل). With a small change of letters, «its extremity» (طرفه) answers to «its direct course» (طريقه), which al-Battānī has; likewise, in Arabic writing, الاخير, «Last», differs little from الاخضر, «Green».

Qudāma, p. 230, writes: «and the breadth near its end in the west, and at the place adjoining (or: joined to) the Green Sea, 200» (وعند منتهاه في المغرب والموضع المتصل بالبحر الاخضر مائتا ميل). In the French translation of Qudāma's book, p. 174, note 2, de Goeje explains this: «Reinaud, Aboulféda,

[1]Or rather 400; see p. 168, note 3.

[a]Nallino's Addenda et emendanda (Part I, p. LXXIII) exchange the note numbers 9 and 10 in the thirteenth line of this page: 9 belongs to وإِسِكّول, 10 to بَرْسُخَان, as the readings of the two notes show.

[b]The print has واسلول; Nallino's Addenda et emendanda (Part I, p. LXXIII) correct it to والسلول.

p. 211 (comp. aussi Ibn Khaldoun *Prolég.* I, 96) dit que ce nom appartient à l'Océan Indien. Mais on l'emploie souvent aussi pour désigner l'Atlantique, et même la Méditerranée porte parfois ce nom. On pourrait croire que Kodâma a supposé avec Bâtiny [*sic*, for al-Battānī] que l'Afrique, au sud de l'Abyssinie, tournait à l'ouest et se dirigeait vers l'océan Atlantique (comp. Reinaud, Introd., p. CCLXXXVI, note 2)[1], mais la comparaison du texte avec celui d'Ibn Rosteh prouve que Kodâma n'a pas compris sa source et que c'est en réalité à l'Atlantique que le nom de mer Verte appartient ici» («Reinaud, Aboulféda, p. 211 (compare also Ibn Khaldūn, *Prolégomènes* I, 96), says that this name belongs to the Indian Ocean. But it is often used also for the Atlantic, and even the Mediterranean sometimes bears this name. One might think that Qudāma supposed with al-Battānī that Africa, south of Abyssinia, turned west and ran towards the Atlantic Ocean, but a comparison of the text with that of Ibn Rusta proves that Qudāma did not understand his source, and that it is in reality to the Atlantic that the name of Green Sea belongs here»). But how can the Atlantic Ocean be called joined or adjoining to the sea of al-Qulzum, which we today call the Red Sea? Anyone can see that de Goeje's opinion too must be wholly rejected.

Let us now consider similar passages in other writers. Al-Masʿūdī, *Prairies*, I, 237, has the following: «its length is 1400 miles; the breadth of its two extremities, which is the nearest place in its breadth, 200; the breadth in the middle, lastly, 700, which is its greatest breadth» (طوله الف واربعمائة ميل وعرض طرفيه مائتا ميل وهو اقرب المواضع في عرضه وعرضه في الوسط سبعمائة ميل وهو اكبر العرض فيه). The words «of its two extremities» (طرفيه) are a small scribal error[2] for طريقه, «of its direct course»; so al-Masʿūdī's words agree with al-Battānī. — Al-Kharaqī, fol. 74v of manuscript P = 57v of manuscript L: «its end, towards the nearest extremity, which is called the *Green Sea*, is 200 miles» (الادنى الذي يسمّى البحر الاخضر مقدار مائتي ميل (codd. طرف) ومنتهاه الى طرفه).

Quṭb al-Dīn Maḥmūd al-Shīrāzī, *al-Tuḥfa al-shāhiyya*, f. 155v: «Two hundred parasangs [= 600 miles] is the breadth of the beginning of the second gulf, which is called the Red Gulf, and whose extremity is called the Green Sea; its length towards the north is 460 parasangs [= 1380 miles]; where it becomes narrower, and that is its end, it comprises 60 parasangs [= 180 miles]» وهي عرض بالبحر الاخضر وطوله في الشمال اربعمائة وستّون فرسخًا [add. يُعْرَف] اصل الخليج الثاني ويُسمَّى الخليج الاحمر وطَرَفه وحيث يستدقّ وهو منتهاه ستّون فرسخًا. Qāḍīzāde, in his commentary on ch. I (*faṣl*) of the third section (*bāb*) of the book *al-Tadhkira*, f. 120v, says that the Red Gulf comprises 200 parasangs [= 600 miles] near its beginning (عند اصله); «afterwards it becomes narrower, until its end comes to 60 parasangs [= 180 miles]» ثمّ انّه يستدقّ الى أن يصير منتهاه ستّين فرسخًا.

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«The nearest extremity», «the nearest direct course», etc., of almost all the Arab writers can only be explained as «nearest to us»; and since both the Arab authors and the Syrians (from whom the Arabs received those notions about the seas) lived in Syria and in Mesopotamia, the part of the sea of al-Qulzum nearest to them must plainly be the northern one. Nor can «the end» be understood otherwise, since it is opposed to the beginning, that is, to the strait we call Bāb el-Mandeb. Jacob of Edessa confirms it most certainly; for he writes in the *Hexaemeron* (p. LXVI of Hjelt's translation): «in latitudinem extrema parte australi 400 fere milliaria, septentrionali vero ad ducenta aut fortasse minus patet» («in breadth it extends about 400 miles at its southern end, but at the northern up to two hundred or perhaps less»). This being so, one might think that the true reading is the one Qudāma gives, and that in this passage the name *Green Sea* belongs to the Mediterranean (as the Arabs sometimes, though rarely, say), which can rightly be called adjoining the northern end of the sea of al-Qulzum. There are, however, difficulties: it seems hardly probable that Qudāma alone kept the good reading and that all the others, even those older than Qudāma, corrupted the text; nor could one explain why the name *Green Sea* is never mentioned in the passages of these descriptions that deal with the Mediterranean.

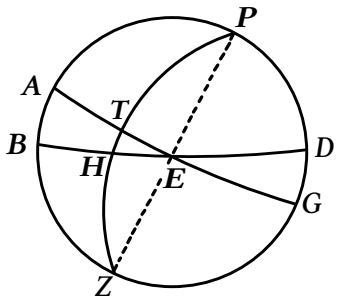
[1] But we have already seen (note *d*, p. 167) that this conjecture of Reinaud is wholly false.

[2] The same error in the description of the sea of al-Baṣra; see note *d*, p. 168, note 4.

The true interpretation can, I think, be drawn from a passage of the Syriac astronomy of Barhebraeus, edited very recently by Nau. The author, who wrote in 1279, used the oldest Syriac authors and the Arab writers alike in describing the seas, lakes and rivers; of the Red Sea (*yamā sūmāqā*) he has the following (p. 135 of the text): «The second [gulf that arises from the Indian Ocean] is the Red Sea, which is small and extends 400 miles in length; the breadth of its extremity, which is the *yamā dh-sūph*, comprises 200 miles». This *yamā dh-sūph* ܝܡܐ ܕܣܘܦ, which is read already in the oldest Syriac translations of the Old and the New Testament, is the Hebrew name *yam sūph* adapted to the nature of the Syriac language; it means «the sea of weeds» and denotes the stretch of the Red Sea in which Pharaoh, pursuing the Israelites, perished most miserably with his men. The first Syrian who compiled that description of the seas from Ptolemy's Geography would not leave out a name most famous in the sacred traditions; but the Arab who translated the Syriac, not knowing well what plant the word *sūph* answered to, rightly rendered it, for the sense, as «*Green Sea*». — So al-Battānī's reading is the best; Qudāma's المتصل, «joined, adjoining», is an error or a false correction for المسمى, «called».[a]

On chapter 7

a) — p. 20, ll. 22–30.



To compute the *solstitial amplitude*[1], given the greatest half arc of the day (P) and the obliquity of the ecliptic (ε).

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Let ABGD be the meridian circle, AEG the equinoctial circle (whose poles are P and Z), BED the horizon, H the solstitial point of the ecliptic; then TH = ε, AT = P, HE = the solstitial amplitude. With this construction Ptolemy, *Almag.* II, 2 (ed. Halma, vol. I, pp. 67–69), proves

$$\frac{\text{chord}\,2AT}{\text{chord}\,2AE} = \frac{\text{chord}\,2TZ}{\text{chord}\,2ZH} \times \frac{\text{chord}\,2HB}{\text{chord}\,2BE},$$

that is,

$$\frac{\text{chord}\,2P}{\text{chord}\,180^\circ} = \frac{\text{chord}\,180^\circ}{\text{chord}\,(180^\circ - 2\varepsilon)} \times \frac{\text{chord}\,(180^\circ - 2\ \text{sol. ampl.})}{\text{chord}\,180^\circ}.$$

[1] It is the arc of the horizon between the point of the east or the west and the place of the horizon from which the solstitial point of the ecliptic rises; if it is counted from the east point it is called the *rising amplitude* (*sa'at al-mashriq*), if from the west point, the *setting amplitude* (*sa'at al-maghrib*). The author can also call the amplitudes *azimuth*, since with the Arabs the beginning of the azimuths is taken from the east or the west point, not, as always with us, from the north or the south point.

[a] Nallino's Addenda et emendanda (Part II, pp. XXIII–XXIV) give a letter of de Goeje on this note, who finds Nallino's solution very ingenious but uncertain, and holds «طريقه» a corruption of «طرفه»; they are translated with Part II.

Using sines instead of chords, one finds

$$\frac{\sin P}{R} = \frac{R \cos(\text{sol. ampl.})}{\cos\varepsilon \times R},$$

whence al-Battānī's formula:

(α) $\quad \cos(\text{sol. ampl.}) = \dfrac{\cos\varepsilon \ \sin P}{R}.$

b) — p. 20, ll. 34–41.

To compute the *amplitude* of any point of the ecliptic that is not the solstitial point, given the half arc of the day of that point and the obliquity of the ecliptic.

In the formula of the preceding problem it is enough to call P the half arc of the day of the given point of the ecliptic, and to use δ, the declination of the same given point, instead of the whole obliquity. Al-Battānī's formula follows at once:

(α) $\quad \cos(\text{ampl.}) = \dfrac{\cos\delta \ \sin P}{R}.$

c) — p. 21, ll. 1–6.

Given the latitude φ of the city and the declination δ of the proposed point of the ecliptic, to compute the *amplitude*.

In the preceding figure BZ is plainly the altitude of the pole above the horizon of the given city, that is, the latitude φ. Ptolemy, *Almag.* II, 3 (ed. Halma, vol. I, pp. 71–72), proves geometrically

$$\frac{\text{chord}\, 2ZA}{\text{chord}\, 2AB} = \frac{\text{chord}\, 2ZT}{\text{chord}\, 2TH} \times \frac{\text{chord}\, 2HE}{\text{chord}\, 2EB},$$

that is[a],

$$\frac{\text{chord}\, 180^\circ}{\text{chord}\, (180^\circ - 2\varphi)} = \frac{\text{chord}\, 180^\circ}{\text{chord}\, 2\varepsilon} \times \frac{\text{chord 2 sol. ampl.}}{\text{chord}\, 180^\circ}.$$

With sines put for chords, the amplitude of any point of the ecliptic for that of the solstitial point, and so the declination δ of the given point for the whole obliquity ε, one finds

$$\frac{R}{\cos\varphi} = \frac{\sin(\text{ampl.})}{\sin\delta},$$

whence at once al-Battānī's formula:

(α) $\quad \sin(\text{ampl.}) = R\,\dfrac{\sin\delta}{\cos\varphi}.$

Ḥabash, a contemporary of al-Battānī, fol. 92r, has three other formulas besides this one. The first (since $\text{cosec}\ x = R^2/\sin x$, and Ḥabash, ff. 85r–87v, gives a table of the cosecants divided by *R*, that is, cosec/R = R/sin) is

$$\sin(\text{ampl.}) = \frac{\text{cosec}(90^\circ - \varphi)}{R} \ \sin\delta.$$

The second, since $\sin\delta = \sin\varepsilon \sin L/R$ (where L is the longitude of the point of the ecliptic), is

$$\sin(\text{ampl.}) = \frac{\sin\varepsilon \ \sin L}{\cos\varphi}.$$

The third was plainly found by orthographic projection:

$$\sin(\text{ampl.}) = \sqrt{(\text{tg}\,\varphi \ \sin\delta)^2 + \sin^2\delta}.$$

[a] In the numerator of the third fraction the print has «chord sol. ampl.» for the chord of the double arc, «chord 2 sol. ampl.», as in the formula before it; the 2 is set here.

On chapter 8

a) — p. 21.

To compute the *altitude* φ *of the pole*, given the difference between the equinoctial day (of 12 hours) and the longest day, that is, given the difference between the equinoctial half arc of the day (of 90 degrees) and the greatest half arc of the day.

First al-Battānī teaches how to find «the distance of the rising of the beginning of Cancer from the north point», which in the following figure is plainly represented by the arc HD, the same as the arc HE (that is, the solstitial amplitude) increased by ninety degrees; so the sine of that distance is the cosine of the solstitial amplitude, and al-Battānī's rule follows at once from the formula of note *a* on ch. VII:

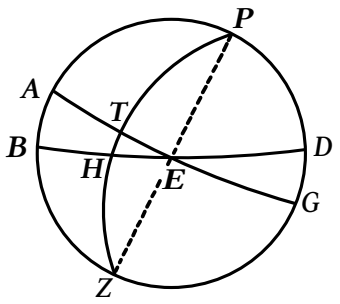
$$(\alpha) \qquad \sin(\text{dist. of the rising of } 0^\circ \text{ Cancer from the north point}) = \frac{\cos\varepsilon\,\sin P}{R}.$$

The arc EH is what the author now calls «the distance of the rising of the beginning of Cancer from the point of the rising of the equinox», that is, what he had called «the solstitial amplitude» in the preceding chapter. This premised, al-Battānī gives the formula

$$(\beta) \qquad \sin\varphi = \frac{\sin(P-90^\circ)\times\sin(\text{dist. of } 0^\circ \text{ Cancer from the north point})}{\sin(\text{dist. of Cancer from the equinoctial rising})}\times\frac{R}{\sin P},$$

which, since $\sin(P-90^\circ) = -\cos P$, may be written[1]

$$\sin\varphi = -\frac{\cos P}{\sin P}\times\frac{\cos(\text{sol. ampl.})}{\sin(\text{sol. ampl.})}\times R.$$



The formula comes from Ptolemy's construction, *Almag.* II, 3 (ed. Halma, vol. I, pp. 69–70). Let the figure be as in the notes on the preceding chapter; Ptolemy proves

$$\frac{\text{chord } 2ET}{\text{chord } 2AT} = \frac{\text{chord } 2EH}{\text{chord } 2HB}\times\frac{\text{chord } 2BZ}{\text{chord } 2ZA},$$

that is,

$$\frac{\text{chord}\,(180^\circ-2P)}{\text{chord } 2P} = \frac{\text{chord 2 sol. ampl.}}{\text{chord}\,(180^\circ-\text{2 sol. ampl.})}\times\frac{\text{chord } 2\varphi}{\text{chord } 180^\circ}.$$

With sines put for chords,

$$\frac{\cos P}{\sin P} = \frac{\sin(\text{sol. ampl.})}{\cos(\text{sol. ampl.})}\times\frac{\sin\varphi}{R},$$

[1] It is easily reduced to our $\operatorname{tg}\varphi = -\cos P\operatorname{cotg}\varepsilon$ if $\cos\varepsilon\sin P/R$ is put for cos sol. ampl., and $R\sin\varepsilon/\cos\varphi$ for sin sol. ampl., as the author taught in ch. VII.

whence

$$\sin\varphi = \frac{\cos P}{\sin P} \times \frac{\cos(\text{sol. ampl.})}{\sin(\text{sol. ampl.})} \times R,$$

where cos P is positive because Ptolemy considered the difference between the least half arc of the day and the equinoctial half arc of the day, that is, 90° – P instead of P – 90°.

On chapter 9

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a) — pp. 21–22.

Given the altitude φ of the pole, to find the *greatest* or the *least half arc of the day of the sun.*

With the preceding figure Ptolemy, *Almag.* II, 3 (ed. Halma, vol. I, pp. 70–71), proves

$$\frac{\text{chord}\, 2ZB}{\text{chord}\, 2BA} = \frac{\text{chord}\, 2ZH}{\text{chord}\, 2HT} \times \frac{\text{chord}\, 2TE}{\text{chord}\, 2EA},$$

that is,

$$\frac{\text{chord}\, 2\varphi}{\text{chord}\,(180^\circ - 2\varphi)} = \frac{\text{chord}\,(180^\circ - 2\varepsilon)}{\text{chord}\, 2\varepsilon} \times \frac{\text{chord}\,(180^\circ - 2P)}{\text{chord}\, 180^\circ}.$$

With sines put for chords,

$$\frac{\sin\varphi}{\cos\varphi} = \frac{\cos\varepsilon\ \sin(90^\circ - P)}{\sin\varepsilon \times R},$$

whence

$$\sin(90^\circ - P) = R\,\frac{\sin\varphi\ \sin\varepsilon}{\cos\varphi\ \cos\varepsilon}.$$

Ptolemy had considered the shortfall of the shortest day with respect to the equinoctial day, that is, 90° – *P*; al-Battānī, on the other hand, the excess of the longest day, that is, *P* – 90°, so that his formula is

(α) $\quad \sin(P - 90^\circ) = -\cos P = R\,\dfrac{\sin\varphi\ \sin\varepsilon}{\cos\varphi\ \cos\varepsilon}.$

If another arc is sought instead of the greatest half arc of the day, let the declination (δ) of the point of the ecliptic that the sun occupies on the given day be put in place of the obliquity ε; al-Battānī's formula becomes[1]

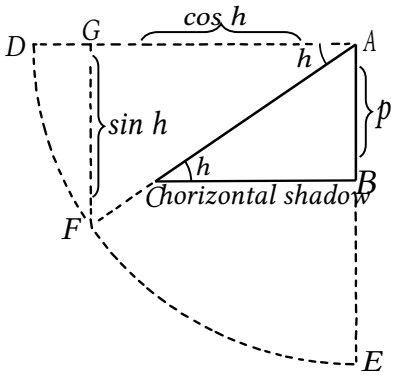
(β) $\quad -\cos P = R\,\dfrac{\sin\varphi\ \sin\delta}{\cos\varphi\ \cos\delta}.$

[1] We, using tangents and taking the radius $R = 1$, write more briefly $-\cos P = \text{tg}\,\varphi\ \text{tg}\,\delta$.

On chapter 10

a) — p. 22.

To compute the *altitude h of the sun*, or the *zenith distance* ζ (= 90° − *h*), from the shadow of the gnomon, divided into 12 parts.



I. From the *horizontal shadow*. Let the parts of the gnomon (that is, of the side AB) be called *p*; the author teaches, ll. 13–15, that

(α) horizontal shadow $= p\,\dfrac{\cos h}{\sin h}$.

The proposition can easily be derived in this way: with the radius AD of sixty parts (to which the author refers the trigonometric functions) let the quadrant of a circle EFD be described; from the similarity of the right-angled triangles ABC and AGF one gathers at once

$$CB : AB = GA : FG,$$

$$\text{horizontal shadow} : p = \cos h : \sin h.$$

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It could have been derived more easily from the theorem that in any plane triangle the sides are to one another as the sines of the angles opposite them. But this proposition seems to have been unknown to the Arabs of that age[1]; it is read later in Naṣīr al-Dīn al-Ṭūsī, who died in July 1274, as von Braunmühl, I, pp. 66–67, rightly says[a].

The author then teaches how to find «the hypotenuse[2] of the shadow triangle», that is, the side AC; plainly

(β) hypotenuse of the shadow triangle $= \sqrt{\text{horizontal shadow}^2 + p^2}$.

And since $\sin h = R\,AB/AC$, we shall also have

[1]The proposition «in any spherical triangle the sines of the angles are directly proportional to the sines of the opposite sides», on the other hand, as far as right-angled triangles are concerned, seems from a passage of the book on the quadrilateral written by Naṣīr al-Dīn al-Ṭūsī to have been known already before al-Battānī to the mathematician Thābit ibn Qurra, who died in the month of Ṣafar 288 H (February 901); see H. Suter, *Zur Geschichte der Trigonometrie*, in Bibliotheca Mathematica, edited by G. Enestrom, new series, vol. VII, Stockholm 1893, p. 7.

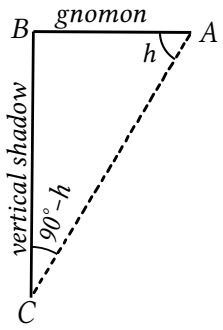
[2]Or «the diameter»; in Arabic *quṭr* means both things, for the hypotenuse is always the diameter of the circle circumscribed about a right-angled triangle.

[a]Nallino's Addenda et emendanda (Part I, p. LXXIII) add that al-Bīrūnī already mentions the theorem as a thing known to all (*Chronology*, p. 184 of the text, p. 166 of the translation).

(γ) $\quad \sin h = \dfrac{Rp}{\sqrt{\text{horizontal shadow}^2 + p^2}}.$

II. From the *vertical shadow*. The author, ll. 30–32, states it as

(δ) $\quad \text{vertical shadow} = p\,\dfrac{\sin h}{\cos h};$



then he gives two formulas:

(ε) $\quad \sin\zeta = \cos h = \dfrac{Rp}{\sqrt{\text{vertical shadow}^2 + p^2}},$

which is derived very easily, like (γ);

(ζ) $\quad \sin h = \dfrac{R \times \text{vertical shadow}}{\sqrt{\text{vertical shadow}^2 + p^2}},$

which only represents the well-known definition $\sin h = R\,BC/AC$.

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Had the author divided the gnomon into 60 parts (as he had already taken a radius of 60 parts, to which he refers the sine, the versed sine and the cosine), these shadows would plainly have become true trigonometric functions, and the formulas just found could have been written

(α′) $\quad \operatorname{cotg} h = R\,\dfrac{\cos h}{\sin h}$ $\qquad$ (β′) $\quad \operatorname{cosec} h = \sqrt{R^2 + \operatorname{cotg}^2 h}$

(γ′) $\quad \sin h = \dfrac{R^2}{\operatorname{cosec} h}$ $\qquad$ (δ′) $\quad \operatorname{tg} h = R\,\dfrac{\sin h}{\cos h}$

(ε′) $\quad \cos h = \dfrac{R^2}{\sec h}$ $\qquad$ (ζ′) $\quad \sin h = R\,\dfrac{\operatorname{tg} h}{\sec h};$

which, when $R = 1$ is taken, as is the present custom, correspond entirely to our trigonometric formulas. It is worth quoting here the words of Delambre, pp. 16–17: «Les Arabes avaient donc, au tems d'Albategni, des tables des quantités, ombre $= \sigma = \nu\,\frac{\cos h}{\sin h} = 12\cot h = 12\operatorname{tang} N$, et par conséquent des tables de tangentes, mais calculées pour le rayon 12, tandis que les sinus étaient calculés pour le rayon 60; mais en multipliant ces tangentes par 5, ou 10/2, on aurait eu les tangentes pour le rayon 60. ….. Voilà bien la première idée des tangentes; voilà des tables qui donnent les tangentes de tous les arcs, et par lesquelles tout arc peut être connu d'après sa tangente. Le malheur est ce choix du rayon de douze parties, qui étaient encore une imitation d'une pratique grecque[1]. Albategni ne sentit pas l'utilité de ces tangentes pour les calculs trigonometriques, et la preuve, c'est qu'il n'a pas changé le rayon» («So in al-Battānī's time the Arabs had tables of the quantities shadow = σ = ν cos h/sin h = 12 cot h = 12 tan N, and consequently tables of tangents, but computed for the radius 12, while the sines were computed for the

[1]More correctly von Braunmühl, I, p. 51: «Unzweifelhaft folgt er hierin indischen Vorbildern, denn, wie wir schon sahen (S. 39), teilten auch die Inder das Gnomon in 12 Finger, während Ptolemaeus die Einteilung in 60^p hat [*Almag.* II, 5, ed. Halma, vol. I, p. 75]» («Without doubt he follows Indian models in this, for, as we have already seen (p. 39), the Indians too divided the gnomon into 12 fingers, while Ptolemy has the division into 60 parts»).

radius 60; by multiplying these tangents by 5, or 10/2, one would have had the tangents for the radius 60. ….. Here indeed is the first idea of the tangents; here are tables that give the tangents of all arcs, and by which any arc can be known from its tangent. The misfortune is this choice of a radius of twelve parts, which was again an imitation of a Greek practice. Albategni did not feel the usefulness of these tangents for trigonometric computations, and the proof is that he did not change the radius»).

The historians of mathematics among us agree with one voice that Abū l-Wafāʾ (who died in the month of Rajab 388 = July 998) was the first to use the shadows as trigonometric functions and to make tables of cotangents for the radius 1 and for the radius 60^p. This universal opinion, however, does not seem right to me, since already in the astronomical work of Ḥabash, al-Battānī's contemporary, the shadows are referred to the radius 60^p, like the sines (fol. 97r–v); indeed, ff. 85r–87v give tables which, taking the radius as 60^p, contain for every degree of the quadrant the tangents (*ẓill maʿkūs*, «the inverted shadow»), the cotangents (*ẓill mustawī*, «the straight shadow»)[1], the tangents multiplied by $0^p\ 26'\ 11''\ 34'''$, and lastly the cosecants (*quṭr al-ẓill*, «the hypotenuse of the shadow») divided by the radius, that is, cosec/R = R/sin. And he uses these trigonometric functions, computed to the seconds, here and there in his formulas.

What al-Battānī calls *ẓill mabsūṭ*, «the extended or spread shadow», other Arabs call *ẓill thānī*, «the second shadow», or *ẓill mustawī*, «the straight shadow». The tangent, on the other hand, which al-Battānī calls *ẓill qāʾim*, «the standing shadow», or *ẓill muntaṣab*, «the erect shadow», others mostly call *ẓill awwal*, «the first shadow», or *ẓill maʿkūs*, «the inverted shadow».

With the Arabs a threefold division of the gnomon is found. The first is the one the Indians always use, which al-Battānī adopted as the most usual of all with the calculators; it comprises 12 parts, called digits (*aṣābiʿ*), and can be used only in dialling. Another, also set up for dialling but rarer than the first[2], counts 7, or 6 2/3, or 6 1/2 parts, called feet (*aqdām*). The third, lastly, which serves dialling and trigonometry at once, gives a division into sixtieths (*ajzāʾ*). — Oloug Beg, p. 61 of the text, 85 of the translation, says that the calculators always refer the first shadow (the tangent) to a gnomon of 60 parts, and the second shadow (the cotangent) indifferently to a gnomon of 60, 12 or 7 parts.

The gnomon shows the shadow of the upper limb of the solar disc, not of the centre of the sun, so that the altitude found with the gnomon exceeds the true altitude by about 16 minutes. Neither the Greeks nor al-Battānī seem to have noticed this; only later Ibn Yūnus, ch. XI, as Delambre, p. 102, relates, «dans un discours sur l'ombre, il démontre géometriquement que l'ombre du gnomon est celle du bord supérieur; il ne dit rien de la pénombre» («in a discourse on the shadow, proves geometrically that the shadow of the gnomon is that of the upper limb; he says nothing of the penumbra»).

On chapter 11

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a) — p. 23.

To find the *azimuth of the altitude*, that is, from the given altitude *h* of the sun to find *the azimuth of this circle of altitude* (that is, *the azimuth of the sun*) on the horizon of a place on the earth of known latitude φ.

[1] M. Curtze (in *Bibliotheca Mathematica herausgegeben von G. Enestrom*, new series, vol. X, 1896, p. 71) is mistaken when he says of the words umbra recta and umbra versa: «vor Abû 'l-Wafâ, der 998 starb, waren sie überhaupt als *termini technici* nicht bekannt» («before Abū l-Wafāʾ, who died in 998, they were not known at all as technical terms»).

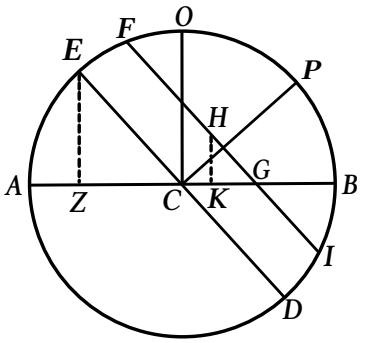
[2] And later; for, as far as the published Arab writers go, it is not found before the thirteenth century; see Aboul Hhassan, ch. XV, p. 152 (where 6 2/3); Barhebraeus, p. 178 of the text (where 7); al-Jaghmīnī, p. 270 (where 7 and 6 1/2); Oloug Beg, p. 61 of the text, 85 of the translation (where 7).

Leaving out the proofs, the author teaches the formulas:

(α) sine of the difference of the horizon $= \dfrac{\sin h \, \sin \varphi}{\cos \varphi}$

(β) $\sin(\text{azimuth}) = R \dfrac{\sin(\text{rising ampl.}) \pm \sin(\text{diff. of the horizon})}{\cos h}$.

Both were plainly derived by the orthographic projection of the sphere. The name *sine of the difference* (or *of the variation*) *of the horizon*, together with al-Battānī's formula (α), is read also in Ibn Yūnus, ch. XIX (in Delambre, p. 114); Abū l-Wafā'[1], Aboul Hhassan, ch. LXII, p. 308, and Oloug Beg, p. 78 of the text, 106–107 of the translation, on the other hand, call it the argument of the azimuth (*ḥiṣṣat al-samt*). Geometrically, as Abū l-Wafā' and Mīram Čelebī (in his commentary on Ulugh Beg) say, it is the line that joins the foot of the perpendicular drawn from the star to the plane of the horizon, that is, the foot of the sine of the altitude, with the line of intersection of the horizon and the parallel of the star.



Let the horizon ACB, the celestial equator ECD and the parallel circle FI, which the sun describes in its daily motion, be projected orthogonally onto the plane of the meridian circle AOBD. Suppose the sun to have reached H; HK will then be the projection of the altitude *h*. Let EZ be drawn perpendicular to AB, and so parallel to the line HK; since P is the pole of the equator, we shall have

$$KG = \text{sine of the difference of the horizon}, \qquad BP = EO = \varphi,$$

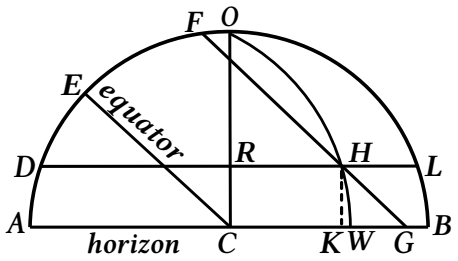
$$ZC = \sin \varphi, \qquad HK = \sin h, \qquad EZ = \cos \varphi.$$

From the similarity of the right-angled triangles EZC and HKG it follows at once that

$$EZ : HK = ZC : KG, \quad \text{that is,} \quad \cos \varphi : \sin h = \sin \varphi : \sin(\text{diff. horiz.}),$$

whence al-Battānī's rule

(α) $\sin(\text{diff. horiz.}) = \dfrac{\sin h \, \sin \varphi}{\cos \varphi}$.



It remains for us now to prove (β), by the same way that al-Battānī probably followed. Let the figure drawn here be the orthographic, or orthogonal, projection of the celestial sphere as seen from the east point; let AOB be the meridian, ACB the horizon, OC the vertical circle perpendicular to the meridian, EC the celestial equator, FG the parallel described by the sun, which we put at H. Through H let the circle of altitude OHW be drawn, and the circle DRL parallel to the horizon. Lastly, let the line HK be drawn perpendicular to the plane of the horizon.

The arcs CW and RH, cut off by the same vertical circles OC and OW, will contain the same number of degrees; so their sines, which the straight lines CW, RH represent in the figure, will be proportional to the radii CB, RL with which those arcs are described. Since LB = HW is the altitude h of the sun, and its sine is HK = RC, while the arc OH = OL is the complement of the altitude and has the sine RL, by what was said above we shall plainly have

$$CB : RL = CW : RH,$$

that is,

$$R : \cos h = \sin(\text{azimuth}) : RH,$$

whence

$$\sin(\text{azimuth}) = R\,\frac{RH}{\cos h}.$$

By the constructions made, RH = CK = CG – KG. But CG is plainly the sine of the rising amplitude, that is, the sine of the arc of the horizon between the east point C and the point from which the sun rises on the given day; KG is what the author calls the sine of the difference of the horizon; hence al-Battānī's formula

$$(\beta) \qquad \sin(\text{azimuth}) = R\,\frac{\sin(\text{rising ampl.}) - \sin(\text{diff. horiz.})}{\cos h},$$

which holds when the azimuth, the amplitude and the declination are counted positive towards the north. If the difference, which is the numerator of the fraction, is negative[1], the azimuth will be towards the south.

The same formula (β) is found in Ibn Yūnus, ch. XX (Delambre, p. 116), in Aboul Hhassan, ch. LXII, p. 308, and in Oloug Beg, p. 78 of the text, 106–107 of the translation. Ḥabash, f. 92v, puts for the sine of the difference of the horizon its value by rule (α), and so writes

$$\sin(\text{azimuth}) = \frac{R}{\cos h}\left(\frac{\sin\varphi}{\cos\varphi}\,\sin h - \sin(\text{rising ampl.})\right).$$

Let us put into formula (β) the value of the sine of the difference of the horizon found before (that is, $\sin h \sin\varphi/\cos\varphi$), and the value of the sine of the rising amplitude found in ch. VII (that is, $R\sin\delta/\cos\varphi$); there results

$$\sin(\text{azimuth}) = R\,\frac{R\sin\delta - \sin h\,\sin\varphi}{\cos h\,\cos\varphi}.$$

But since al-Battānī, like almost all the Arabs, counts the azimuth from the line of the rising and setting, while we proceed from the north point towards the east, the sine of the azimuth of the Arabs becomes with us $\sin(90^\circ - \text{azimuth}) = \cos(\text{azimuth})$. Lastly, taking the radius as one part, as is our custom, we find at once the present-day formula

$$\cos(\text{azimuth}) = \frac{\sin\delta - \sin h\,\sin\varphi}{\cos h\,\cos\varphi},$$

[1] Carra de Vaux, *L'Almageste d'Aboû 'l-Wéfa al-Bûzdjâni*, in the Journal Asiatique, 8th series, vol. 19, 1892, p. 429.

[1] When the sine of the difference of the horizon is greater than the sine of the rising amplitude. — When the sun is on the southern side of the equator, not the difference but the sum of the sines must be used.

which is an immediate consequence of the fundamental rule of spherical trigonometry (called by the Germans the *Cosinussatz*)

$$\cos a = \cos b \cos c + \sin b \sin c \cos A,$$

if one puts $a = 90^\circ - \delta$, $b = 90^\circ - h$, $c = 90^\circ - \varphi$, $A =$ azimuth.

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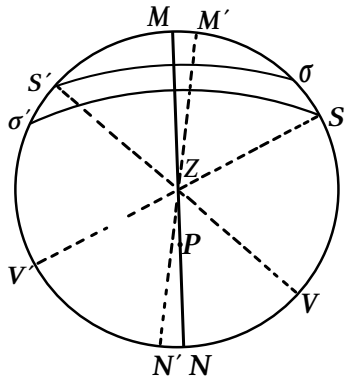
What Delambre writes on p. 17, mixing his algebraic explanations with al-Battānī's rules, was not rightly interpreted by Hankel, *Zur Gesch. der Mathem.*, p. 281, by Cantor, I, 633 (= 694 of the second edition), and by Suter, *Zur Geschichte der Trigonometrie* (in Bibliotheca Mathematica, ed. G. Enestrom, new series, vol. VII, Stockholm 1893, pp. 6 and 8), who say that al-Battānī knew the proposition $\cos a = \cos b \cos c + \sin b \sin c \cos A$. We can indeed easily derive it from (β); but the author neither made such a reduction nor wished to make it; rather, the very form of rule (β) shows that al-Battānī by no means regarded it as a general proposition for solving spherical triangles. Von Braunmühl, I, p. 53, judged better, but not absolutely enough.

Regiomontanus, in a long note on this chapter, which the printed editions wrongly give in the middle of chapter XII, rightly guessed the way that al-Battānī (or the Arabs before him) probably followed to find his two formulas[1]. For he writes: «Albategnius in demonstrationibus suis multis ad lineas rectas refugit, et ex similitudine triangulorum multa conclusit. Modus iste intellectui facilis videretur, nisi sectiones superficierum obscuritatem afferrent» («In many of his proofs Albategnius takes refuge in straight lines, and he concluded much from the similarity of triangles. This method would seem easy to the understanding, were it not that the sections of the surfaces bring obscurity»). He adds a proof that differs little from mine, but is perhaps more obscure.

On chapter 12

a) — p. 24, ll. 26–31.

[«*The sun moves slowly in declination*». For near the solstices the declination of the sun changes little in a day, so that the sun can then be said, to a close approximation, to run along a true parallel; which, on the other hand, would be false near the equinoxes, for between its rising and its setting the sun can vary by more than 11 minutes in declination.



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[1]Because of the form of the rule, and because of what al-Battānī proves geometrically elsewhere, the conjecture was not too difficult for Regiomontanus, who had pondered al-Battānī's work so assiduously. So I do not think one should agree with von Braunmühl, I, 52, note 1, when he writes: «Es scheint demnach, das Regiomontanus die Beweise des Albategnius, die in der Druckausgabe nich angeführt sind, kannte» («It seems, then, that Regiomontanus knew the proofs of Albategnius, which are not given in the printed edition»).

In what follows, the Arabic text ought to be translated literally: «It is also manifest that the time of midday cannot be determined truly, because of the hasty passage of the sun through (or in) the meridian circle .....». Plato: «Manifestum etenim est quod medii diei hora propter festinum Solis motum in medii diei circulo veraciter non terminatur, quae tamen vel ei propior apprehenditur» («For it is manifest that the hour of midday, because of the hasty motion of the sun in the circle of midday, is not truly determined, though it, or a time nearer to it, is grasped»). — But the interpretation we have adopted seems more probable, and rests on these considerations. When the sun rises at the point S′ of the horizon, its setting, if its declination were constant, would have to take place at the point σ of the same celestial parallel. But because of the daily motion in declination, the sun passes gradually from that parallel to another, σ′S, and sets at S. At the time of its rising it casts the shadow along the direction ZV, at the time of its setting along ZV′, which makes with the meridian line MN an angle different from the angle contained by the meridian itself and the shadow ZV. So by bisecting the angle V′ZV one finds not the meridian line MN but the line M′N′; whence it is clear that at the moment of time equally distant from the rising and from the setting («at the time of midday») the sun is not on the meridian circle, but in the vertical plane that bisects the angle V′ZV. In the regions of the Arabs, from the equator to the parallel of 45°, the small angle MZM′ can rise to 8′ or 9′; whence we understand why al-Battānī says that this drawing of the meridian line should be undertaken about the times of the solstices. — Schiaparelli.]

b) — p. 25, ll. 16–22, and p. 25, l. 37 – p. 26.

Given the rising amplitude and the meridian altitude, to compute *the altitude of the sun that has no azimuth*, that is, *the altitude on the prime vertical circle* (which passes through the east and the west points, from which the azimuths of the Arabs are counted).

So far the author had stated his formulas without proofs; now, on the other hand, he shows geometrically how the rule is found. Why he did so is not certainly known. It seems probable to me that he left out the earlier proofs as being already known from other authors, whereas this solution of the problem was new, and for this reason al-Battānī wished to prove it. For we have seen that the propositions of spherical astronomy contained in the earlier chapters either can easily be derived from Ptolemy's constructions (with sines put for chords) or are read also in Ḥabash[1], our author's contemporary; the problem of this chapter, on the other hand, I find solved in this way only in later writers. Likewise in chapter XXVI we read the proof of another trigonometric formula which, there can be no doubt, was first found by al-Battānī.

Al-Battānī's proof rests on the analemma, or the orthographic projection of the sphere, whose use is frequent with the Indians and more frequent with the Arabs[2]. — The last words of the proof (p. 26, ll. 27–29), which Plato leaves out, were spoiled by some unskilled reader, who disfigured the text in the same way in chapter XXVI (p. 38, ll. 27–28). [Al-Battānī meant: so if the line GF is diminished in the ratio of CM to CF, the line EM remains; likewise, if GF is diminished in the ratio of EC to CG, the line EM remains. That is, since $EC : CG = EM : GF = CM : CF$, plainly $EM = \frac{GF \times CM}{CF} = \frac{GF \times EC}{CG}$. — Schiaparelli.]

Since EC = the sine of the rising amplitude of the sun on the given day, GF = the sine of the meridian altitude of the sun, and CG = EC + EG = the sine of the rising amplitude + the cosine of the meridian altitude, al-Battānī's formula follows at once:

$$(\alpha) \qquad \sin(\text{alt. without azimuth}) = \frac{\sin(\text{rising ampl.}) \times \sin(\text{merid. alt.})}{\sin(\text{rising ampl.}) + \cos(\text{merid. alt.})},$$

which is read later, together with others, in Ibn Yūnus, ch. XVIII (Delambre, p. 114).

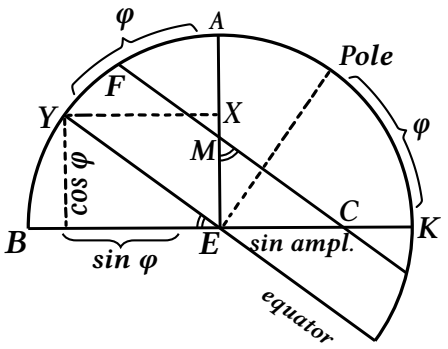
[1] Who, however, always leaves out the proofs.

[2] As von Braunmühl rightly understood, I, p. 38 (on the Indians) and in many places.

It is useless, and does not answer to the author's thought, to develop this formula as Delambre does, p. 19, putting for the sine of the rising amplitude its value $R \sin\delta / \cos\varphi$ found in chapter VII, for the sine of the meridian altitude the value $\cos(\varphi - \delta)$ that follows from chapter XV, and lastly $\sin(\varphi - \delta)$ for the cosine of the meridian altitude. One then obtains[1]

$$\sin(\text{alt. without azimuth}) = R\,\frac{\sin\delta}{\cos\varphi}\left(\frac{\cos(\varphi-\delta)}{\sin(\varphi-\delta) + R\,\frac{\sin\delta}{\cos\varphi}}\right) = \frac{R\sin\delta\,\cos(\varphi-\delta)}{\cos\varphi\,\sin(\varphi-\delta) + R\sin\delta}.$$

It should rather be noted that in place of al-Battānī's formula three others are found in Ḥabash, f. 93r:



I. $$\sin(\text{alt. without azimuth}) = \frac{\sin(\text{rising ampl.}) \times \cos\varphi}{\sin\varphi},$$

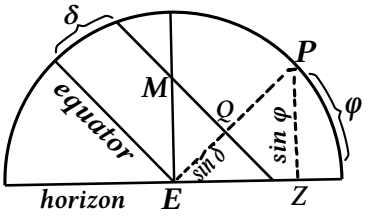
probably also found by orthographic projection; for from the similarity of the triangles MEC, YXE the proportion is easily gathered

$$EM : EC = EX : XY,$$

$$\sin(\text{alt. without azim.}) : \sin(\text{rising ampl.}) = \cos\varphi : \sin\varphi.$$

II. $$\sin(\text{alt. without azimuth}) = \frac{\sin(\text{rising ampl.})}{\frac{\sin\varphi}{\cos\varphi}},$$

which is only another way of writing formula I.



III. $$\sin(\text{alt. without azimuth}) = R\,\frac{\sin\delta}{\sin\varphi},$$

which can be derived algebraically from I, when its value $R \sin\delta / \cos\varphi$ is put for the sine of the rising amplitude; but it also follows directly in the orthographic projection, from the similarity of the triangles MEQ, EPZ:

$$EM : EQ = EP : PZ,$$

[1] Delambre puts $\sin(\varphi - \delta)$ in the numerator and $\cos(\varphi - \delta)$ in the denominator of the fraction; but it is a mere slip of the pen.

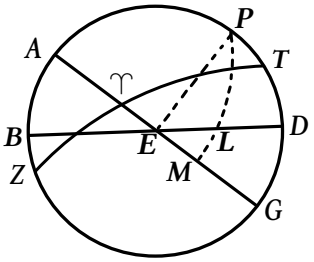
$$\sin(\text{alt. without azim.}) : \sin\delta = R : \sin\varphi.$$

Formula III is also in Aboul Hhassan, ch. LXI, p. 306.

On chapter 13

a) — p. 27, ll. 17–24.

From the right ascension of any degree of the ecliptic and the greatest length of the day in a given city, to find the *oblique ascension* of that degree of the ecliptic in that city.



The *oblique ascension*, which the Arabs call *the ascensions of the clime* or *in the clime*, is the arc of the equinoctial circle between the beginning of Aries and the point of the equinoctial circle that is on the horizon at the same time as the given star or degree of the ecliptic. Let P be the pole of the equator, AEG the equinoctial circle, ♈ the beginning of Aries, Z♈T the ecliptic, BED the horizon of the given place on the earth, L the given star, which rises from the horizon, PM the circle of declination passing through the star L; the whole arc ♈M will be the right ascension of the star, ♈E its oblique ascension, the difference EM the ascensional difference.

The problem plainly reduces to finding the ascensional difference EM, which al-Battānī calls *the argument of the difference of the day in the quadrant*[1]. The author gives two formulas. I 188

I. The author considers the oblique ascension of a given degree of the ecliptic, not of any star. Let *m* be the ascensional difference; P the greatest half arc of the day, or the greatest length of half the day, computed in degrees of the equator, in the given city, so that P – 90° is half the excess of the longest day over the equinoctial day (= 2 × 90° = 12^h); α the right ascension of the given degree of the ecliptic. The author's formula is

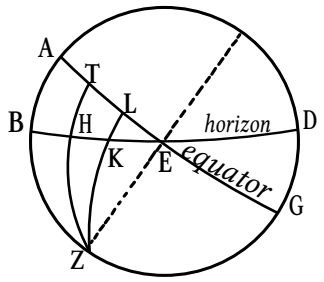
$$(\alpha) \qquad \sin m = \frac{\sin\alpha\ \sin(P - 90^\circ)}{R},$$

which we can write

$$\sin m = -\frac{\sin\alpha\ \cos P}{R}.$$

It is contained implicitly in Ptolemy's proof, *Almag.* II, 7 (ed. Halma, vol. I, pp. 97–98).

[1]The name plainly comes from the author's considering the ascension of a given degree of the ecliptic; so that, as appears from ch. XIX, the half arc of the day of that degree, that is, half the length of the day when the sun has reached that degree, is 90° ± the ascensional difference.



Let ABGD be the meridian circle; AEG the equator, whose southern pole is Z; BED the horizon; ZHT and LKZ two arcs of circles of declination; H the point common to the horizon and to the parallel that passes through the winter solstitial point; K the point common to the horizon and to the parallel that passes through the given place of the ecliptic. TEG is plainly the greatest half arc of the day in the place of the earth to which the celestial horizon BED corresponds. Ptolemy found

$$\frac{\text{chord}\,2TH}{\text{chord}\,2ZH} = \frac{\text{chord}\,2TE}{\text{chord}\,2EL} \times \frac{\text{chord}\,2KL}{\text{chord}\,2KZ},$$

that is,

$$\frac{\text{chord}\,2\varepsilon}{\text{chord}\,(180^\circ - 2\varepsilon)} = \frac{\text{chord}\,(2P - 180^\circ)}{\text{chord}\,2\text{ asc. diff.}} \times \frac{\text{chord}\,2\delta}{\text{chord}\,(180^\circ - 2\delta)},$$

and, with sines put for chords,

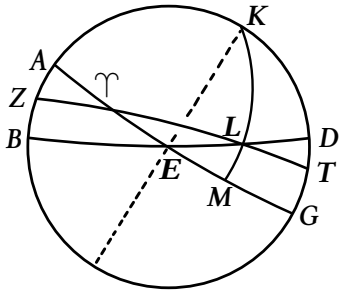
$$\frac{\sin\varepsilon}{\cos\varepsilon} = \frac{\sin(P - 90^\circ)}{\sin(\text{asc. diff.})} \times \frac{\sin\delta}{\cos\delta},$$

whence

$$\sin(\text{asc. diff.}) = -\cos P\,\frac{\sin\delta\,\cos\varepsilon}{\cos\delta\,\sin\varepsilon}.$$

But al-Battānī had taught in chapter V that $\sin\alpha = R\,\frac{\sin\delta\,\cos\varepsilon}{\cos\delta\,\sin\varepsilon}$, so that

$$\sin m = -\frac{\cos P\,\sin\alpha}{R}.$$



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II. The other formula is derived directly from Ptolemy's proof, *Almag.* II, 7 (ed. Halma, vol. I, pp. 92–94). Let ABGD be the meridian circle; BED the horizon; AEG the celestial equator; Z♈T the ecliptic, whose point ♈ is the beginning of Aries; K the northern pole of the equator; KM an arc of a circle of declination. ♈E will be the oblique ascension of the given point L of the ecliptic, ♈M its right ascension, EM the ascensional difference. Ptolemy proves

$$\frac{\text{chord}\,2KD}{\text{chord}\,2DG} = \frac{\text{chord}\,2KL}{\text{chord}\,2LM} \times \frac{\text{chord}\,2ME}{\text{chord}\,2EG},$$

that is,

$$\frac{\text{chord}\, 2\varphi}{\text{chord}\,(180^\circ - 2\varphi)} = \frac{\text{chord}\,(180^\circ - 2\delta)}{\text{chord}\, 2\delta} \times \frac{\text{chord}\, 2m}{\text{chord}\, 180^\circ}.$$

With sines put for chords, one obtains

$$\frac{\sin\varphi}{\cos\varphi} = \frac{\cos\delta}{\sin\delta} \times \frac{\sin m}{R},$$

whence al-Battānī's formula

(β) $$\sin m = R\,\frac{\sin\varphi\ \sin\delta}{\cos\varphi\ \cos\delta},$$

which Ḥabash, f. 104r, using trigonometric tangents, writes more briefly[a]

$$\sin m = \frac{\text{tg}\,\varphi\ \text{tg}\,\delta}{R},$$

where φ is the geographical latitude of the given city.

On chapter 14

a) — pp. 29–30.

To find the *geographical latitude* φ of a given place on the earth from the meridian altitude of the sun.

The rules are very well known; when the sun (⊙) is north of the celestial equator,

(α) $\varphi = 90^\circ - (\text{merid. alt. of } \odot - \text{declin. of } \odot)$;

when the sun is in the southern signs of the zodiac,

(β) $\varphi = 90^\circ - (\text{merid. alt. of } \odot + \text{declin. of } \odot)$.

We write them briefly

$$\varphi = \text{complement of the merid. alt.} \pm \text{declin.}$$

On chapter 15

a) — p. 30.

To find the *meridian altitude* of the sun from the geographical latitude of the given city and the declination of the sun.

Al-Battānī gives the very well known formulas in two ways; when the sun is in the northern signs of the zodiac,

(α) $\text{merid. alt.} = 90^\circ - (\text{latit.} - \text{declin.})$

(α′) $\text{merid. alt.} = \text{complement of the latit.} + \text{declin.}$;

when the sun is in the southern signs,

(β) $\text{merid. alt.} = 90^\circ - (\text{latit.} + \text{declin.})$

(β′) $\text{merid. alt.} = \text{complement of the latit.} - \text{declin.}$

[a] The print has sin m = R tg φ tg δ; Nallino's Addenda et emendanda (Part II, p. XXIV) correct it to tg φ tg δ : R.

On chapter 16

a) — p. 30.

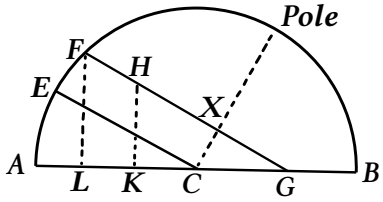
To find the *hours elapsed since the rising of the sun* at a given moment of time, from the altitude of the sun.

The problem reduces to finding the hour angle t (that is, the angle contained by the meridian and the circle of declination passing through the sun); it is added to the half arc of the day if the given time is after noon, or subtracted if the time is before noon. The sum or the remainder will be the arc of the revolution of the sphere, whose degrees are easily turned into equal or seasonal hours.

I 190

Let P be the half arc of the day, h the altitude of the sun at the given time; al-Battānī's formula is

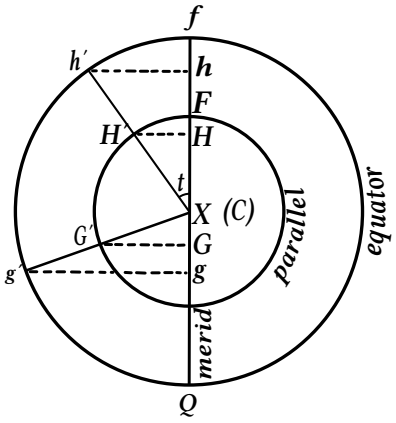
$$(\alpha) \qquad \operatorname{vers} t = \operatorname{vers} P - \frac{\sin h \operatorname{vers} P}{\sin(\text{merid. alt.})}.$$



This formula was plainly derived by the orthographic projection of the sphere. Onto the plane of the meridian circle AFPB let there be projected orthographically the horizon ACB, the celestial equator EC, and the parallel FG described by the sun, which is supposed to have reached H; the perpendicular FL will be the sine of the meridian altitude, HK the sine of the altitude at the given time. In the triangle FLG, since HK is parallel to the side FL, we have the proportion

$$FG : HG = FL : HK,$$

$$FG : HG = \sin(\text{merid. alt.}) : \sin h.$$



Now, to know what FG and HG are, let us imagine the orthographic projection of the figure onto the plane of the equinoctial circle, leaving out the horizon circle; FXQ will be the meridian, the

angle fXh' = t, the angle fXg' = P, hX = cos t, Xg = cos P. From the similarity of the triangles $h'h$X and H′HX one gathers

$$FH : fh = H'X : h'X,$$

that is,

$$FH : fh = FX : R;$$

but $fh = fX - hX = R - \cos t = \operatorname{vers} t$, therefore

$$FH = \operatorname{vers} t \, \frac{FX}{R}.$$

Similarly the proportion is derived

$$FG : fg = G'X : g'X,$$

or

$$FG : fg = FX : R;$$

but, since in the figure $P > 90°$,

$$fg = fX + Xg = R + \cos P = \operatorname{vers} P,$$

therefore $FG = \operatorname{vers} P \, \frac{FX}{R}$. Lastly

$$HG = FG - FH = \frac{FX}{R} \, (\operatorname{vers} P - \operatorname{vers} t).$$

The first proportion found therefore becomes

$$\frac{FX}{R} \operatorname{vers} P : \frac{FX}{R} \, (\operatorname{vers} P - \operatorname{vers} t) = \sin(\text{merid. alt.}) : \sin h,$$

whence $\operatorname{vers} P - \operatorname{vers} t = \frac{\sin h \, \operatorname{vers} P}{\sin(\text{merid. alt.})}$, and at last al-Battānī's formula:

$$(\alpha) \qquad \operatorname{vers} t = \operatorname{vers} P - \frac{\sin h \, \operatorname{vers} P}{\sin(\text{merid. alt.})}.$$

The same is read in Ḥabash, f. 99v (who calls vers P *jayb al-nahār*, «the sine of the day»), in Aboul Hhassan, ch. XLI, p. 261, and in Oloug Beg, p. 95 of the text, 122 of the translation. I 191

Algebraically it is not difficult to develop this formula further. By the rule of chapter IX

$$\operatorname{vers} P = R - \cos P = R + R \, \frac{\sin \varphi \, \sin \delta}{\cos \varphi \, \cos \delta};$$

from chapter XV one gathers

$$\sin(\text{merid. alt.}) = \sin(\text{compl. of } \varphi \pm \delta) = \frac{\cos \varphi \, \cos \delta \pm \sin \varphi \, \sin \delta}{R},$$

$$\sin(\text{merid. alt.}) = \cos(\varphi \mp \delta);$$

so that formula (α) can be written

$$\operatorname{vers} t = \left(R + R \, \frac{\sin \varphi \, \sin \delta}{\cos \varphi \, \cos \delta}\right) - \frac{\sin h \left(R + R \, \frac{\sin \varphi \, \sin \delta}{\cos \varphi \, \cos \delta}\right)}{\cos(\varphi \mp \delta)}$$

$$= \frac{R \, (\cos \varphi \, \cos \delta + \sin \varphi \, \sin \delta)}{\cos \varphi \, \cos \delta} - \frac{R \sin h \, (\cos \varphi \, \cos \delta + \sin \varphi \, \sin \delta)}{\cos \varphi \, \cos \delta \, \cos(\varphi \mp \delta)}.$$

But (as appears from chapter III) $\cos\varphi\cos\delta + \sin\varphi\sin\delta = R\cos(\varphi - \delta)$, therefore

$$\operatorname{vers} t = R^2\,\frac{\cos(\varphi-\delta)}{\cos\varphi\,\cos\delta} - R^2\,\frac{\sin h\,\cos(\varphi-\delta)}{\cos\varphi\,\cos\delta\,\cos(\varphi\mp\delta)},$$

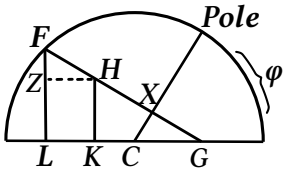
and at last

(I) $$\operatorname{vers} t = R^2\,\frac{\cos(\varphi-\delta)-\sin h}{\cos\varphi\,\cos\delta},$$

a formula which Delambre, p. 20, wrongly ascribes to al-Battānī himself (leaving out R^2, since we refer the trigonometric lines to $R = 1$). And since we derive the hour angle t directly from the well-known relation $\sin h = \sin\varphi\sin\delta + \cos\varphi\cos\delta\cos t$, that is,

$$\cos t = \frac{\sin h - \sin\varphi\,\sin\delta}{\cos\varphi\,\cos\delta},$$

where two multiplications have to be made instead of the single one (R^2 left out) found in (I), Delambre himself, p. 164, afterwards cites formula (I) as if it were the first and oldest example of *prosthaphaeresis*, the procedure by which our mathematicians, before logarithms were discovered, tried to put additions and subtractions in place of multiplications and divisions. Hankel, p. 281, and Cantor, I, p. 633 (= 694 of the second edition), repeat the same; but quite wrongly, for al-Battānī never converted formula (α), obtained by orthographic projection, into (I), as von Braunmühl, I, 53, note 2, very well perceived.



Formula (I), written

$$\operatorname{vers} t = R^2\,\frac{\sin(\text{merid. alt.})-\sin h}{\cos\varphi\,\cos\delta},$$

is in Oloug Beg, p. 96 of the text, 124 of the translation; who, however, without doubt derived it not algebraically from (α) but directly from the orthographic projection. In the first figure of this note let the perpendicular HZ be drawn; then ZL = HK = sin *h*, and FZ = sin merid. alt. – sin *h*. The angle XCG is the altitude of the pole above the horizon, that is, φ; therefore the angle FHZ = the angle XGC = 90° – φ. Plainly also CX = sin δ, FX = cos δ. Since the sides of any triangle are to one another as the sines of the opposite angles[1],

$$FZ : FH = \sin(\text{angle } H) : \sin(\text{angle } Z) = \cos\varphi : \sin 90^\circ = \cos\varphi : R;$$

therefore $FZ = FH\,\frac{\cos\varphi}{R}$. We saw above that $FH = \operatorname{vers} t\,\frac{FX}{R} = \operatorname{vers} t\,\frac{\cos\delta}{R}$; so

$$FZ = \frac{\operatorname{vers} t\,\cos\delta}{R}\times\frac{\cos\varphi}{R} = \frac{1}{R^2}\,\operatorname{vers} t\,\cos\delta\,\cos\varphi.$$

But $FZ = \sin(\text{merid. alt.}) - \sin h$; therefore

$$\sin(\text{merid. alt.}) - \sin h = \frac{1}{R^2}\,\operatorname{vers} t\,\cos\delta\,\cos\varphi,$$

whence

$$\operatorname{vers} t = R^2\,\frac{\sin(\text{merid. alt.})-\sin h}{\cos\delta\,\cos\varphi},$$

as was to be proved.

[1]Ulugh Beg was not unaware of the proposition; see p. 181, note 1.

On chapter 17

I 192

a) — p. 31.

To find the *altitude of the sun* from the hours of the day elapsed since its rising.

The problem is the converse of the preceding one; al-Battānī, then, inverting the formula of the preceding chapter, teaches

$$(\alpha) \qquad \sin h = \frac{(\operatorname{vers} P - \operatorname{vers} t)\ \sin(\text{merid. alt.})}{\operatorname{vers} P},$$

as we read also in Ḥabash, f. 100r, and in Aboul Hhassan, ch. XLIV, p. 266.

On chapter 18

a) — p. 31, ll. 30–32.

[«*But the stars on either side*» etc. — The whole sentence is very obscure. The declination is greater or smaller than the latitude in various ways in the different regions of the sky, and it is not understood how it could be useful to take the sum or the difference of two quantities which, as the author himself says just below, are counted on two different circles. Either it is a gloss of a very unskilled reader, or a fragment of a fuller passage that it is now impossible to restore. Perhaps the arc that we shall call β′ in what follows was defined in this place; but it is a mere conjecture. — Schiaparelli.]

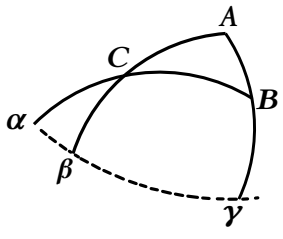
b) — pp. 31–32.

To compute the *declination* δ of a star whose longitude L and latitude β are known.

Let β′ be the latitude of the point of the celestial equator that has the same longitude as the star[1]; al-Battānī's rule is

$$(\alpha) \qquad \sin\delta = \frac{\sin(\beta \pm \beta')\ \cos\varepsilon}{\cos\beta'},$$

which Aboul Hhassan, ch. XXV, p. 189, also gives later, and, together with other formulas, Ibn Yūnus, ch. XXXIX (in Delambre, p. 141), and Oloug Beg, p. 64 of the text, 90 of the translation.



I 193

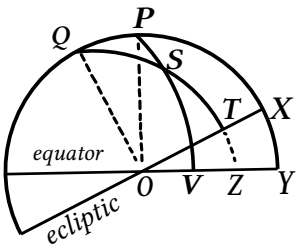
[This proposition was probably derived with the help of *the theorem of Menelaus*[2], which Ptolemy always uses to solve spherical triangles, and which can be stated in this way. Let ABC be a

[1]The quantity β′ al-Battānī calls «declination», that is, the second declination, as I have already said in the translation, p. 32, note 1.

[2]Menelaus, *Sphaerica* III, 2; it is also called *the rule of the six quantities.* The Arabs called its figure *al-shakl al-qaṭṭāʿ*, which the Latins of the Middle Ages translated *figura sector*, or, keeping the Arabic name, *figura alkata*; the Germans of today call it *Transversalenfigur*, and the proposition *Transversalensatz*.

spherical triangle whose sides are arcs of great circles; let a great circle αβγ cut the three sides of the triangle, produced as far as needed; among the six segments that can be conceived between each vertex of the triangle and the cutting arc αβγ there holds a relation which, putting sines for Menelaus' chords of the double arcs and using the present-day algebraic notation, is expressed in this way:

$$\sin A\beta \cdot \sin C\alpha \cdot \sin B\gamma = \sin C\beta \cdot \sin A\gamma \cdot \sin B\alpha.$$



Now let S be the given star, P the pole of the equator, Q the pole of the ecliptic; let the produced sides of the triangle QPS be cut by the equator OY; then

$$\sin QZ \cdot \sin SV \cdot \sin PY = \sin SZ \cdot \sin QY \cdot \sin PV.$$

But

$$QZ = QT + TZ = 90^\circ + \beta', \qquad QY = QX + XY = 90^\circ + \varepsilon,$$
$$SV = \delta, \qquad SZ = ST + TZ = \beta + \beta',$$
$$PY = 90^\circ, \qquad PV = 90^\circ.$$

So the equation found before becomes

$$\cos\beta' \ \sin\delta \times R = \sin(\beta + \beta') \ \cos\varepsilon \times R,$$

whence al-Battānī's rule. — Schiaparelli.]

But al-Battānī never teaches how β′ is computed; which is truly strange, unless a lacuna is supposed in the archetype of the Escorial codex and of Plato's translation; see note *a* on this very chapter (p. 192). In Ibn Yūnus, ch. XXXIX (Delambre, p. 141), in Aboul Hhassan, ch. XXIV, p. 181, and in Oloug Beg, pp. 63–64 of the text, 88 of the translation, the quantity β′ is found in two ways[a]:

$$\sin\beta' = \frac{\sin\delta'}{\cos d'}, \qquad \cos\beta' = \frac{\cos\varepsilon}{\cos d'},$$

where δ′ is the declination of the point *T* of the ecliptic (which is the longitude of the star), and *d′* the declination of the point of the ecliptic whose longitude equals the arc of the ecliptic between the point *T* and the nearest solstitial point. — But β′ can also be derived from the tables. Let the longitude *L* = *OT* be regarded as a right ascension; let it be increased by 90 degrees, since the author counts the right ascensions from the beginning of Capricorn, not of Aries, and let the sum be sought in the column of the degrees of the ascensions in the table of right ascensions (f. 179v ff.); the number corresponding to it in the column of the degrees of the ecliptic will be, after ninety degrees are taken off, the arc *OZ* of the equator, which we shall now regard as if it were an arc of the ecliptic. In the table of the declination of the sun (f. 177v f.) let us take the declination corresponding to the fictitious arc of the ecliptic *OZ*; the fictitious declination found will plainly be the arc *TZ*, that is, β′.

[a] The print has sin d′ in the first denominator. With sin δ′ = sin ε sin L and sin d′ = sin ε cos L, the second declination β′ (tg β′ = tg ε sin L) satisfies sin β′ = sin δ′ / cos d′, which is set here; the second formula is as printed.

In the spherical triangle *OTZ*, right-angled at *T*, plainly (since the angle *XOY* = *TOZ* = ε)

$$\sin(\text{angle } Z) = \frac{\cos\varepsilon}{\cos\beta'};$$

al-Battānī's formula can therefore be written

$$\sin\delta = \sin(\beta + \beta')\ \sin Z.$$

Delambre, pp. 289 and 290, seems to me to have had this in mind when he says that Regiomontanus, in his *Tabulae Directionum* (of 1475), started from al-Battānī's idea in constructing a table that gives, for all celestial longitudes, the *radix ascensionum* (that is, the arc *OZ*), the *arcus* (that is, β′, the second declination), sin *Z* and cotg *Z*. In this way the ecliptic coordinates (longitude and latitude) could easily be turned into equatorial ones (declination and right ascension), using the formulas[a]

$$\sin\delta = \sin(\beta + \beta')\ \sin Z,$$

$$\sin(OZ - \text{right asc.}) = \operatorname{tg}\delta\ \operatorname{cotg} Z.$$

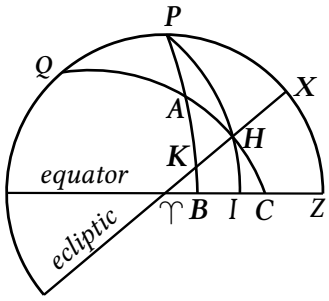
Nor should it be forgotten that the Arabs often use the second declination, and indeed constructed tables of this declination for every degree of the ecliptic.

c) — p. 32, ll. 13–17.

To derive *the degree of the ecliptic that culminates with a star* from the known latitude β, longitude *L* and declination δ of the star.

I 194

The author computes what he calls «the difference (or variation) of the passage of the star through the middle of the sky» (*ikhtilāf majāz al-kawkab fī wasaṭ al-samā'*); this quantity is added to the right ascension of the degree of the star's longitude, or subtracted from it, as the case requires; the sum or the remainder will be the right ascension of the degree that culminates with the star. The longitude of that degree is drawn from the tables of right ascensions.



[If the reading of the codex and of Plato were good, the author's rule would have to be expressed thus:

$$\sin(\text{diff. of passage}) = \frac{\sin\beta\ \sin\varepsilon\ \cos L}{R\ \cos\delta}.$$

Let us now establish what this difference of passage, or difference of culmination, is. Let P be the pole of the equator, Q the pole of the ecliptic; through the given star A let the circle of declination PAB and the circle of latitude QAC be drawn; through the point H of the ecliptic, which is the degree of the star's longitude, let the circle of declination PHI be described. Plainly the arc BI

[a]The print has cotg C for cotg Z: in the triangle SVZ, right-angled at V, sin VZ = tg δ cotg Z, with VZ = OZ – right ascension.

of the equator is the interval between the passage of the star A through the meridian and the passage of the point H of the ecliptic, which is the degree of the star; or the difference between the right ascension (ϒI) of the degree of the star and the right ascension (ϒB) of the star.

Since PB = PI = 90° by the constructions made, the arc BI (that is, the difference of passage sought) will be the measure of the angle APH. If we call D the declination of the point H, that is, the arc HI, we have

$$AH = \beta, \quad PH = 90^\circ - D, \quad PA = 90^\circ - \delta, \quad QA = 90^\circ - \beta, \quad QP = XZ = \varepsilon.$$

And since *QH* = *QX* = 90°, the arc *HX* (that is, 90° – *L*) will be the measure of the angle *PQA*. By the proportionality of the sines of the angles to the sines of the opposite sides, in the triangle AHP

$$\sin(\text{diff. of passage}) = \sin\beta\,\frac{\sin PAQ}{\cos D};$$

likewise in the triangle AQP

$$\sin PAQ = \sin\varepsilon\,\frac{\cos L}{\cos\delta};$$

therefore

$$(\alpha) \qquad \sin(\text{diff. of passage}) = \frac{\sin\beta\,\sin\varepsilon\,\cos L}{\cos D\,\cos\delta}.$$

It is necessary, then, to put cos D for the *R* that the Escorial codex and Plato have; otherwise the rule is false. As was noted on p. 48 of the Arabic text, the words «the chord of what remains of the declination of the degree in which the star is, to complete ninety» (that is, cos *D*), which perhaps filled a line, seem to have been left out by mistake by a copyist in the archetype; the passage so corrupted («divide the product by, and take the arc of what results») an unskilled reader, who spoiled other things too, thought to restore by adding «half the diameter» after «by».

I 195

Delambre, pp. 21–22, noticed no error in al-Battānī's rule, but interpreted it falsely. In our notation (for Delambre writes *B*, *D*, ω, λ, λ′ for our *H*, *B*, ε, β, β′), these are his words: «Cherchez cos *AC* et cos ϒ*H*, vous aurez cos *AB* cos *CB* = cos *AC*, ou cos $CB = \frac{\cos AC}{\cos AB}$, puis AR = ϒ*B* = ϒ*C* – *CB*; nous ferions tang *CB* = tang *AC* cos *C*, ou

$$\frac{\sin CB}{\cos CB} = \left(\frac{\sin AC}{\cos AC}\right)\cos C = \frac{\sin AC}{\cos AB\,\cos CB}\,\sin\varepsilon\,\cos\curlyvee H,$$

et

$$\sin CB = \frac{\sin AC\,\sin\varepsilon\,\cos\curlyvee H}{\cos AB} = \frac{\sin(\beta+\beta')\,\sin\varepsilon\,\cos L}{\cos\delta}.$$

….. On voit que les Arabes connaissaient les formules sin arc perp. = sin hypoténuse . sin angle à la base; cos hypoténuse = cos base . cos côté perpendiculaire»[a] («Find cos AC and cos ϒH; you will have cos AB cos CB = cos AC, or cos CB = cos AC / cos AB, then the right ascension = ϒB = ϒC – CB; we would put tan CB = tan AC cos C, or [the formulas above]. One sees that the Arabs knew the formulas: sine of the perpendicular arc = sine of the hypotenuse × sine of the angle at the base; cosine of the hypotenuse = cosine of the base × cosine of the perpendicular side»). But in all this Delambre arbitrarily puts the sine of the *adjusted latitude* (sin [β ± β′]) of the preceding problem for the sine of the *latitude of the star* (sin β); and he thinks that what al-Battānī calls *the difference of the passage of the star through the middle of the sky* is the arc BC. Wrongly indeed: the name could not then fit the thing, nor could such an interpretation be reconciled with the rule that follows: «add (or subtract) its difference of passage through the middle of the sky to the times *of the right ascensions that correspond to the degree in which the star is*», where plainly the point I is meant, not the point C, which in no way shows the right ascension of the degree of the

[a] The print has sin ω in the second fraction of the first formula; Nallino's Addenda et emendanda (Part I, p. LXXIII) correct it to sin ε.

star's longitude. Moreover, Delambre's interpretation takes as already known the arc ♈C, which al-Battānī, on the other hand, does not teach how to compute. In my interpretation the quantity D (or the arc IH) is drawn at once from the table of the declinations of the sun (f. 177v–178r) or from the rules of chapter IV. — Schiaparelli.]

On chapter 19

a) — p. 32.

To compute the *half arc of the day* P of a star of known declination δ.

The arc $P = 90° +$ the ascensional difference if the declination is northern; $P = 90° -$ the ascensional difference if the declination is southern.

The ascensional difference the author now calls *the difference* (or *variation*) *of the quadrant of the circle of the star*; for the quadrant (90°) must be increased or diminished by this quantity to give the half arc of the day, that is, the arc of the parallel that the star runs through on the given day from its rising to its passage through the meridian. In chapter XIII, where he had considered not any star but only the points of the ecliptic, he had called the ascensional difference *the argument of the difference of the day in the quadrant.* He repeats formula (β) of chapter XIII, which serves for all stars:

$$(\alpha) \qquad \sin(\text{asc. diff.}) = R\,\frac{\sin\varphi\,\sin\delta}{\cos\varphi\,\cos\delta}.$$

For the rest, he had already taught the computation of the half arc of the day of the sun in chapter IX, whose formula (β) is general and applies to any star.

On chapter 20

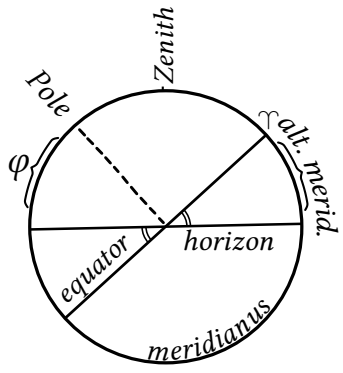
a) — p. 33.

The rules are very easy; they are set out more fully in Ptolemy, *Almag.*, II, 7 (ed. Halma, vol. I, pp. 90–108). In present-day astronomy, which does not use oblique ascensions, they are of no use.

On chapter 21

a) — pp. 33–34.

To compute *the hours of the night that have elapsed*, by any star.



First let the altitude of the star on the meridian circle be found, by adding the declination δ of the star to the [meridian] altitude of the beginning of Aries, or subtracting it, as the case requires. This rule of the author is the same as the rule set out in ch. XV for computing the meridian altitude of the sun:

(α) merid. alt. = complement of the latitude ± declin.;

for, as appears at once from the figure, the meridian altitude of the beginning of Aries (and of any other point of the equinoctial circle) is always the complement of the latitude φ of the given city, that is, the complement of the altitude of the pole of the equator above the horizon of the city[1].

This done, al-Battānī repeats the rule of chapter XVI:

$$(\beta) \qquad \operatorname{vers} t = \operatorname{vers} P - \frac{\sin h \operatorname{vers} P}{\sin(\text{merid. alt.})}.$$

By adding the arc *t* to the half arc of the day of the star, or subtracting it, according as the star is west or east of the meridian circle, the arc of the revolution of the sphere since the rising of the star will be found. Let this arc be increased (or, if the star rises from the horizon before sunset, diminished) by the difference between the oblique ascension of the star and the oblique descension (or the ascension of the nadir) of the sun; there will result the arc of the celestial sphere revolved since sunset, whose degrees are very easily turned into seasonal or equal hours.

If the computation is made by the moon, whose parallax is perceptible, not its true but its apparent coordinates must be used.

On chapter 22

a) — p. 35.

To find the *altitude h* of a star from the hours of the night that have elapsed.

The problem is the converse of the preceding one, whose formula is to be inverted thus:

$$(\alpha) \qquad \sin h = \frac{\sin(\text{merid. alt.}) \, (\operatorname{vers} P - \operatorname{vers} t)}{\operatorname{vers} P}.$$

On chapter 24

a) — pp. 35–36.

To compute the *declination* δ of a star and *the degree of the ecliptic that passes through the meridian circle at the same time as it*, when the degree of the ecliptic that rises at the same time as the star and the rising or setting amplitude of the star are known.

The first rule is

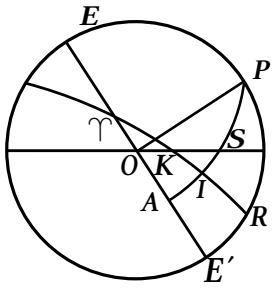
$$(\alpha) \qquad \sin \delta = \frac{\sin(\text{alt. of the beginning of Aries}) \times \sin(\text{ampl.})}{R}.$$

But the [meridian] altitude of the beginning of Aries, as the author has said in several places (see note 1 of p. 196), is the complement of the geographical latitude of the given city; so the formula

[1] See ch. VII (p. 20, ll. 1–2), XIV (p. 29, last line), XV (p. 30, l. 15).

can be written $\sin\delta = \cos\varphi \sin(\text{ampl.})/R$, which follows at once from the other formula of chapter VII, $\sin(\text{ampl.}) = R\sin\delta/\cos\varphi$. Once the declination is known, let the half arc of the day P of the star be computed by the rules of chapter XIX:

$$P = 90^\circ \pm \text{asc. diff.}, \qquad \sin(\text{asc. diff.}) = R\,\frac{\sin\varphi\,\sin\delta}{\cos\varphi\,\cos\delta}.$$



This done, al-Battānī teaches how to find the right ascension of the degree that culminates with the star, which is

obl. asc. of the degree rising with the star + half arc of the day of the star,

or obl. asc. of the degree setting with the star − half arc of the day of the star;

where the right ascension of the degree culminating with the star (which is also the right ascension of the star) is understood to be taken from the **beginning of Capricorn**, as will be clear from the figure drawn here to one who examines the matter more closely. For EE′ is the equator, S the place from which the star rises, ♈ the vernal equinox at that time, ♈R the ecliptic, K the degree of the ecliptic rising at the same time as the star, I the degree culminating with the star; so ♈A will be the right ascension, counted from the beginning of Aries, of the degree I culminating with the star, or the right ascension of the star itself; EA the half arc of the day of the star on the equator; ♈O the oblique ascension of the degree K that rises with the star, or the oblique ascension of the star S itself. Plainly ♈A = ♈O + OA; and since

$$OA = EA - 90^\circ,$$

we shall have ♈A = ♈O + EA − 90°, that is, 90° + ♈A = ♈O + EA. The tables of right ascensions will then give the degree of the ecliptic that corresponds to the right ascension so found.

On chapter 25

a) — pp. 36–37.

To compute the *longitude of a star*, given the declination of the star and the degree of the ecliptic that passes through the meridian circle at the same time as the star.

It is the converse of the problem of chapter XVIII, whose rules it would have been enough to invert. The author, on the other hand, seeks a new solution, and a most unhappy one, as will appear below. Let δ be the declination of the given star, δ′ the declination of the degree culminating with it, ε the obliquity of the ecliptic; the rules are these:

I. When δ and δ′ are on the same side:

$$\delta - \delta' \text{ (or } \delta' - \delta) = \text{adjusted distance.}$$

$$2R - \cos\varepsilon = \text{longest chord.}$$

$$2R - \cos\delta' = \text{exceeding chord.}$$

$$R\,\frac{\sin\varepsilon}{\cos\delta'} = \text{sine of the adjusted inclination.}$$

$$\frac{\sin(\text{adj. incl.}) \times \sin(\text{adj. dist.})}{\cos(\text{adj. dist.})} \times \frac{\text{exceed. chord}}{\text{longest chord}} \times$$

$$\times \frac{\cos\varepsilon}{\cos(\text{adj. incl.})} \times \frac{\sin(\text{solst. dist. of the culm. degree})}{R} =$$

$$= \text{sine of the variation of the star.}$$

It should be noted that the solstitial distance of the degree culminating with the star is counted in right ascensions.

If the degree culminating with the star belongs to the descending signs of the zodiac (from the beginning of Cancer to the end of Sagittarius), the arc of the variation of the star will be subtracted from the right ascension of the degree culminating with the star, or added to it, according as δ is northern or southern. If the culminating degree is in the ascending signs (from the beginning of Capricorn to the end of Gemini), addition will be used when δ is northern, subtraction when it is southern. Let the sum or the difference found be sought in the right ascensions; the degree of the ecliptic corresponding to it will be the longitude of the star.

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II. When δ and δ′ are on opposite sides:

$$\delta + \delta' = \text{adjusted distance.}$$

$$2R - \cos\varepsilon = \text{longest chord.}$$

$$2R - \cos\delta' = \text{exceeding chord.}$$

$$\frac{\sin\varepsilon\,\sin\delta}{\cos(\text{adj. dist.})} \times \frac{\cos\delta'}{R} = \text{sine of the adjusted inclination.}$$

$$\sin(\text{adj. incl.}) \times \frac{\sin\delta}{\cos\delta} \times \frac{\text{exceed. chord}}{\text{longest chord}} = \text{dividend.}$$

$$\cos(\text{adj. incl.}) \times \frac{\text{dividend}}{\cos\varepsilon} \times \frac{\sin(\text{solst. dist. of the culm. degree})}{R} =$$

$$= \text{sine of the variation of the star,}$$

or, putting in the value of the *dividend*,

$$\frac{\cos(\text{adj. incl.}) \times \sin(\text{adj. incl.})}{\cos\varepsilon} \times \frac{\sin\delta}{\cos\delta} \times \frac{\text{exceed. chord}}{\text{longest chord}} \times$$

$$\times \frac{\sin(\text{solst. dist. of the culm. degree})}{R} =$$

$$= \text{sine of the variation of the star.}$$

You use the variation of the star as was said above.

So far the Escorial codex, with which Plato's translation agrees, except for a few words that are plainly missing in Plato through the carelessness of the copyists. But the rules are wholly false; indeed, they can be called true monsters. Delambre, pp. 22–31, gave eight pages to drawing some sense or other out of the author's rule; but such great labour proved entirely useless. Nor does Regiomontanus seem to have been more successful before him; for, as Delambre relates of him elsewhere (p. 353), in a letter to Bianchini he «parle assez brièvement de la méthode d'Albategni

(chap. XXV), et c'est pour la trouver défectueuse. Il ajoute pourtant qu'elle ne serait admissible que pour le colure des solstices où la déclinaison et la latitude sont dans un même plan. Alors en effet $\frac{2-\cos \mathrm{AR}}{2-\cos L'} = \frac{2}{2} = 1$; cependant, à la manière dont il s'exprime, on serait tenté de croire que son manuscrit d'Albategni ne ressemblait guère aux éditions qu'on en a données»[1a] («speaks rather briefly of Albategni's method (ch. XXV), and that in order to find it defective. He adds, however, that it would be admissible only for the colure of the solstices, where the declination and the latitude are in one plane. Then indeed (2 – cos AR)/(2 – cos L′) = 2/2 = 1; yet from the way he expresses himself, one would be tempted to think that his manuscript of Albategni hardly resembled the editions that have been made of it»).

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[Those formulas are not only ridiculously difficult, but manifestly false. Delambre has already shown this in some measure; it will appear more clearly from the following considerations. 1°. The formulas in the two cases set out are different, whereas they ought to have been the same, with δ + δ′ put in the second for δ – δ′ (or δ′ – δ). Not to have understood this shows in the author of the formulas an unskilfulness that can in no way be imputed to al-Battānī, considering what he himself does on similar occasions. — 2°. The formulas appear absurd from the very fact that they cannot fit all the values of the quantities indicated in them. In the second rule the sine and the cosine of the *adjusted inclination* occur; but the sine of this inclination is defined by the formula $\frac{\sin \varepsilon \, \sin \delta}{\cos(\text{adj. dist.})} \times \frac{\cos \delta'}{R}$, which can lead to an absurdity whenever the adjusted distance (that is, δ + δ′) becomes equal to 90°; and no one will deny that this can happen. Then cos (δ + δ′) = nothing, and the sine of the adjusted inclination will be sought in vain in the tables, being infinitely great. Inept, moreover, are the names *longest chord* and *exceeding chord*, given to two quantities that always exceed *R*, and to which no possible geometrical representation in the sphere answers; so that it must be strongly doubted that they could ever be used in solving such problems. The factor determined by their ratio to each other is called «assez étrange», «rather strange», by Delambre, p. 27, and it does indeed look unusual.

It is also very strange that al-Battānī in this chapter did not invert the formulas already found in order to solve the converse problem of chapter XVIII, as he had done elsewhere, and did not see that to draw the latitude and the longitude from the right ascension and the declination is the same as to find the altitude and the azimuth from the declination and the hour angle, since in this last problem it is enough to put the distance between the zenith and the pole = ε, that is, the latitude = 90° – ε.

Lastly, the division of the computation into many small parts designated by special names seems foreign to al-Battānī's usage; he does this very rarely in the other chapters, and only when moved by evident reasons of convenience, as in ch. XXXIX, pp. 78–79[b].

When all this is weighed, I think that what teaches the computation of the *variation of the star*, or of the longitude, does not come from al-Battānī and is a most foolish interpolation. It is an easy conjecture that in the copy from which the Escorial codex and Plato's codex were afterwards derived, a leaf containing al-Battānī's true rules had perished, and that its owner, endowed more with good will than with astronomical knowledge, tried to fill the gap according to his own (alas,

[1] I have not been able to read this letter; but I think it hardly credible that Regiomontanus had before his eyes a text other than the one printed afterwards. The Nuremberg edition seems to have been made from the manuscript to which Regiomontanus himself had added notes here and there in his own hand; and in this chapter it does not differ from the Arabic text except by small gaps, which are manifest mistakes of the copyists. The differences that Delambre noted are probably different interpretations of Regiomontanus, arising from the almost incredible obscurity and ambiguity of Plato's words, or conjectures of Regiomontanus.

[a] Nallino's Addenda et emendanda (Part I, p. LXXIII) correct this passage and the note: the letter is not Regiomontanus' to Bianchini, as Delambre says, but Bianchini's to Regiomontanus, edited by M. Curtze, *Urkunden zur Geschichte der Mathematik im Mittelalter und der Renaissance*, Leipzig 1902, vol. I, pp. 240–41; the Addenda are translated with Part I's front matter.

[b] The words «as in ch. XXXIX, pp. 78–79» are added from Nallino's Addenda et emendanda (Part I, p. LXXIII).

how small!) powers. — Schiaparelli.]

b) — p. 37, ll. 21–25.

To compute the *latitude* β of a star when its declination and longitude are known.

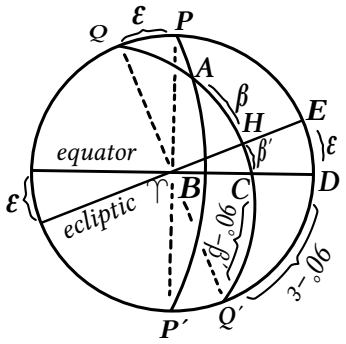
The solution of the problem depends on knowing an arc, which the author defines thus:

(α) $\quad \sin(\text{arc}) = \dfrac{\sin\delta\ \cos\beta'}{\cos\varepsilon},$

where β′ is what al-Battānī calls «the declination of the degree in which you found the star», that is, without doubt, the *second* declination of the degree of the star's longitude, which is the latitude of the degree of the equator that has the same longitude as the star. Then

(β) $\quad \beta = \beta' - \text{arc}, \qquad \text{or} \qquad \beta = \text{arc} - \beta',$

according as β′ is greater or smaller than the arc.



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The way by which the author arrived at his formulas is probably this. Let QPEDP′ be the meridian circle, ϒE the ecliptic, ϒD the equator, PAB the circle of declination passing through the star A, QACQ′ the circle of latitude of the same star A. So AB = δ, AH = β, HC = β′, and AC is the arc sought. From the triangle ABC, right-angled at B, one derives

$$\sin AC = R\,\frac{\sin AB}{\sin(\text{angle } C)} = R\,\frac{\sin\delta}{\sin(\text{angle } C)}.$$

The triangle CDQ′, right-angled at D, gives

$$\sin(\text{angle } C) = R\,\frac{\sin DQ'}{\sin CQ'} = R\,\frac{\cos\varepsilon}{\cos\beta'},$$

whence

(α) $\quad \sin AC = \dfrac{R\sin\delta}{R\,\frac{\cos\varepsilon}{\cos\beta'}} = \dfrac{\sin\delta\ \cos\beta'}{\cos\varepsilon}.$

But neither here nor before, in chapter XVIII (see p. 193), does the author teach how the necessary quantity β′ is computed. It can be drawn from the tables: let the arc of the ecliptic 90° + ϒH be imagined to be an arc of the equator, that is, the right ascension[1] of the point C; the tables will give the corresponding arc (90° + ϒC) of the fictitious ecliptic. When the 90° are thrown away and the tables of the declination of the sun (f. 177v–178r) are entered with the arc ϒC, the fictitious declination β′ will be found.

[1] Those ninety degrees must be added to the arc ϒH because the author counts the right ascensions from the colure of the winter solstice (the beginning of Capricorn), not from the beginning ϒ of Aries. See p. 164.

On chapter 26

a) — p. 38.

To compute *the distance between two stars*, on the great circle that passes through both.

The author puts a geometrical proof before his rules, which are set out on pp. 39–40, as he had already done in chapter XII; and to the proof he also adds an example in numbers, as he does later in chapter XXX, pp. 59–61. Already in note *b* on chapter XII (p. 186) I remarked that it is a probable conjecture that al-Battānī was the first to establish these formulas, and for that reason thought it necessary to prove their truth geometrically.

Regiomontanus strongly disapproves of the author's construction; «verum esset», he says, «si arcus GE esset circuli magni, et ipse primo bene inventus. Sed dico tibi ille processus intricatus est, et modicae reputationis. Utitur enim lineis curvis tanquam rectis, quod tametsi Ptolemaeum fecisse constet, ipse tamen arcus breves loco linearum rectarum accepit, noster vero indifferenter quantoscumque» («it would be true if the arc GE were of a great circle, and had itself first been rightly found. But I tell you that this procedure is intricate and of little worth. For he uses curved lines as if they were straight; and though it is certain that Ptolemy did this, Ptolemy took short arcs in place of straight lines, while our author takes arcs of any size indifferently»). He then proves that the quadrilateral of the four chords AB, BG, GE, EA is symmetric, and can therefore be inscribed in a circle; lastly he proves that chord $GE = R\,\frac{\text{chord of the arc } AB}{\sin(\text{arc } FG)}$, where GE is an arc of a smaller circle.

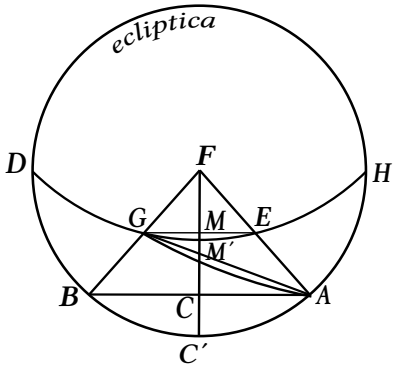
Delambre, pp. 33–34, judges no less harshly that the author's solution is «décidément mauvaise, sans aucune ambiguité» («decidedly bad, without any ambiguity»); he solves the numerical example with logarithmic tables and finds cos AG = cos 64° 20′ 27″; «Albategni la trouve de 54° 19′, trop faible de 10° 1′ 27″» («Albategni finds it to be 54° 19′, too small by 10° 1′ 27″»); and he concludes: «Albategni, en terminant ce chapitre, dit qu'il sert, comme le précédent, pour les nativités..... Pour la gloire d'Albategni, il faut croire ces deux derniers chapitres interpolés par quelque astrologue ignorant et charlatan» («Albategni, in ending this chapter, says that it serves, like the preceding one, for nativities..... For Albategni's glory we must believe these two last chapters to have been interpolated by some ignorant astrologer and charlatan»).

But had Delambre considered the numbers of al-Battānī's computation attentively, he would have found that 54° 19′, which Plato has, is a slip of the pen or of the typesetter for 64° 19′; so, neglecting the seconds, which the author leaves out, the error is only one minute, and it plainly comes from the difference in interpolating the tables of sines. Likewise, if the rules of pp. 39–40 are written in the present trigonometric manner, one finds the fundamental rules by which we solve spherical triangles. It is impossible, then, that the geometrical constructions from which a true computation and sure rules are derived should be false.

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From the numerical example it is very clear that by the name «line *x*» the author denotes not the arc *x* or the chord of the arc *x*, but the straight line that lies under the arc *x*, that is, the sine of that arc. The same expression occurs several times in chapter XXVIII. Nor should it be forgotten that the figure of p. 38, which Plato's translation shows with curved lines, is drawn with straight lines in the Escorial codex, as I have had it printed. Hence it becomes certain that al-Battānī had in mind the orthographic projection of the sphere, which he had already used elsewhere (as I showed in the notes on chapters XI, XII and XVI), and derived the rules from it. The ambiguities would easily be avoided by drawing al-Battānī's figure a little differently, so that the point M of the chord appears at once distinct from M of the arc, and C of the chord from C of the arc; Schiaparelli proposed to me the diagram printed here. Let the sphere be projected orthogonally onto the plane of the ecliptic; AHDB will be the ecliptic, whose pole is F; FG, FE, FB, FA will at one and the same time be the complements of the latitude and the sines of these complements.

Through the two points G, E, of the same latitude, let a *great circle*[1] DH be described, whose arc GE will be the distance between G and E; let the straight lines GE, BA be drawn, bisected by the line FC; lastly let the arc AG of a *great circle* and its chord be marked. Let the midpoints of the chords be distinguished by the letters M and C, the midpoints of the arcs by the letters M′ and C′.



This done, the proportion GM : BC = FG : FB[2] is manifest in the plane triangle FBC; that is, if L and L′ are called the longitudes of the stars and β the latitude,

$$\sin\frac{GE}{2} : \sin\frac{L-L'}{2} = \sin(90^\circ - \beta) : \sin 90^\circ.$$

But $\sin 90^\circ = R$, and the arc GE/2 is half the distance sought; therefore

$$\sin\frac{GE}{2} = \sin\frac{L-L'}{2} \times \frac{\cos\beta}{R},$$

which is al-Battānī's rule, stated on p. 40, ll. 1–2 (and 39, ll. 26–28).

There are also some corruptions or interpolations in the text, which I have pointed out in their places in my translation; they cannot be ascribed to al-Battānī himself, since they are most foolish, wholly superfluous and of no consequence in the author's numerical computation or in the rules of pp. 39–40. The hand of an unskilled reader appears, whose traces we find plainly elsewhere too.

According to the end of the chapter, these rules are most useful for computing the profections in nativities; but that they serve non-astrological questions too appears from al-Battānī's own words where he teaches how to compute the distance of the moon from the sun in a certain and absolute way (ch. XLI, p. 87, ll. 11–13).

b) — p. 39, ll. 1–12.

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Once the distance of the point G from E on a great circle is known by the formula

$$\sin\tfrac{1}{2}\,\text{distance} = \sin\frac{L-L'}{2} \times \frac{\cos\beta}{R},$$

[1]Not a smaller circle, that is, a circle parallel to the great circle BA, as Regiomontanus and Delambre interpreted it.

[2]The codex and Plato add «= FM : FC», which, though true, appears entirely useless, for the knowledge of these lines is neither required nor taught. It must therefore be regarded as an interpolation, which arose because the writer of the archetype from which the Escorial codex and Plato's codex derive wrongly put, in lines 23, 24, 30 and 31, FM ف م for FG ف ط, and FC ف ج for FB ف ب. The corruption pointed out on p. 38, note 6, also comes from this.

the author seeks the distance of the point G from A, also on a great circle. So he observes that in the *rectilinear* quadrilateral GEBA (which is symmetric, and so can be inscribed in a circle) Ptolemy's theorem, stated at the beginning of the chapter (p. 37, ll. 38–40), holds; so that one has

$$\overline{GA}^2 = \overline{GB}^2 + GE \times BA,$$

where the sides GB, BA are chords of known arcs, and GE has just been determined[a]. So the diagonal GA becomes known, which is the chord of the arc GA of a *great* circle, that is, of the distance sought. The solution can rightly be called elegant.

c) — pp. 39–40.

He now states the general rules for computing the *distances between two stars* (in arcs of great circles).

In spherical quadrilaterals, of the two sides whose chords are parallel he calls the one nearer the ecliptic (or the ecliptic itself) *the first side*, the one nearer the pole *the second side*. The other two sides, which are equal and whose chords are not parallel, he calls *the third* and *the fourth side*.

He distinguishes four cases:

I. — When one of the stars is on the ecliptic and the other has the latitude β. — The solution is the one I have already set out in note *b*, after al-Battānī himself; he puts $L - L'$ (that is, the difference of the longitudes) = the first side, $2 \arc\big[\sin\big(\frac{L-L'}{2}\big) \times \frac{\cos\beta}{R}\big]$ = the second side (that is, the arc GE, on which see note *a*), and lastly the chord of β = the third or the fourth side. So he states the rule:

(α) crd distance = $\sqrt{\text{crd first side} \times \text{crd second side} + \text{square of the third side}}$.

From this formula of al-Battānī it is not difficult to derive our cos dist. = cos β cos (L – L′). Since chord $x = 2 \sin \frac{x}{2}$, if we put sines for chords and square both members of equation (α), there results[b]

$$4 \sin^2 \frac{\text{distance}}{2} = 2 \sin\Big(\frac{L-L'}{2}\Big) 2 \sin\Big(\frac{L-L'}{2}\Big) \frac{\cos\beta}{R} + 4 \sin^2 \frac{\beta}{2} =$$
$$= 4 \sin^2\Big(\frac{L-L'}{2}\Big) \frac{\cos\beta}{R} + 4 \sin^2 \frac{\beta}{2}.$$

But $\sin^2 \frac{x}{2} = \frac{R^2 - R\cos x}{2}$; so, dividing everything by 4,

$$\frac{R^2 - R\cos(\text{dist.})}{2} = \frac{R^2 - R\cos(L-L')}{2} \times \frac{\cos\beta}{R} + \frac{R^2 - R\cos\beta}{2},$$

whence, multiplying by 2 and dividing by *R*,

$$R - \cos(\text{dist.}) = \cos\beta - \frac{\cos\beta\ \cos(L-L')}{R} + R - \cos\beta,$$

and lastly $\cos(\text{distance}) = \frac{\cos\beta\ \cos(L-L')}{R}$.

II. — When the stars have the same latitude. — The chord of the distance is then the second side itself; that is, as he taught before and as is derived from the geometrical construction of p. 38 (on which see note *a*):

(β) chord of the distance = $2 \sin\Big(\frac{L-L'}{2}\Big) \frac{\cos\beta}{R}$,

or $\sin \frac{\text{dist.}}{2} = \sin\big(\frac{L-L'}{2}\big) \frac{\cos\beta}{R}$.

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[a] The print has 2 GB². Ptolemy's theorem in the symmetric quadrilateral (GA × EB = GE × BA + GB × EA, with GA = EB and GB = EA) gives GB², as rule (α) below has it; GB² is set here.

[b] The print has cos β / 2R three times in these formulas; Nallino's Addenda et emendanda (Part I, p. LXXIII) correct it to cos β / R.

III. — When the latitudes β and β′ of the stars are different, but on the same side of the ecliptic. Al-Battānī then sets

$$\text{first side} = L - L', \qquad \text{second side} = \text{double arc}\Big[\sin\Big(\frac{L-L'}{2}\Big)\frac{\cos(\beta-\beta')}{R}\Big],$$

$$\text{third side} = \text{fourth side} = \beta - \beta',$$

$$\text{first side of the longitude} = \text{double arc}\Big[\sin\Big(\frac{L-L'}{2}\Big)\frac{\cos\beta}{R}\Big],$$

$$\text{second side of the longitude} = \text{double arc}\Big[\sin\Big(\frac{L-L'}{2}\Big)\frac{\cos\beta'}{R}\Big].$$

Al-Battānī's rule is:

$$(\gamma) \qquad \text{chord of the distance} = \sqrt{\text{crd I side of long.} \times \text{crd II side of long.} + \text{crd}^2 \text{ III side}},$$

which is plainly proved geometrically in the same way as rule (α) (see note *b*). The fundamental formula of our spherical trigonometry is contained in it implicitly. For by squaring both members of the equation and bringing in sines one finds

$$4\sin^2\Big(\frac{\text{distance}}{2}\Big) = 2\sin\Big(\frac{L-L'}{2}\Big)\frac{\cos\beta}{R} \times 2\sin\Big(\frac{L-L'}{2}\Big)\frac{\cos\beta'}{R} + 4\sin^2\Big(\frac{\beta-\beta'}{2}\Big).$$

Dividing everything by 4, and considering that $\sin^2\frac{x}{2} = \frac{R^2 - R\cos x}{2}$,

$$\frac{R^2 - R\cos(\text{distance})}{2} = \sin^2\Big(\frac{L-L'}{2}\Big)\frac{\cos\beta\,\cos\beta'}{R^2} + \frac{R^2 - R\cos(\beta-\beta')}{2} =$$

$$= \frac{R^2 - R\cos(L-L')}{2} \times \frac{\cos\beta\,\cos\beta'}{R^2} + \frac{R^2 - R\cos(\beta-\beta')}{2}.$$

Let everything be multiplied by 2 and divided by *R*; there will result

$$R - \cos(\text{distance}) = \frac{\cos\beta\,\cos\beta'}{R} - \frac{\cos\beta\,\cos\beta'\,\cos(L-L')}{R^2} + R - \cos(\beta-\beta').$$

But $\cos(\beta-\beta') = \frac{\sin\beta\,\sin\beta' + \cos\beta\,\cos\beta'}{R}$; so, taking away the equal terms of opposite sign,

$$\cos(\text{distance}) = \frac{\sin\beta\,\sin\beta'}{R} + \frac{\cos\beta\,\cos\beta'\,\cos(L-L')}{R^2},$$

which we, taking $R = 1$, now write

$$\cos(\text{dist.}) = \sin\beta\,\sin\beta' + \cos\beta\,\cos\beta'\,\cos(L-L').$$

If for β and β′ their complements Δ and Δ′ are taken, that is, the distances from the pole of the ecliptic, the formula becomes

$$\cos(\text{dist.}) = \cos\Delta\,\cos\Delta' + \sin\Delta\,\sin\Delta'\,\cos(L-L'),$$

which is derived at once from the consideration of the spherical triangle FDE, in which the angle DFE = L – L′, and which is always used today in this problem.

IV. — When the latitudes β and β′ are different and on two different sides. It differs from case III only in that β + β′ must be taken for the β – β′ of the third side. Then $\cos(\beta+\beta') = \frac{\cos\beta\,\cos\beta' - \sin\beta\,\sin\beta'}{R}$, so that the formula becomes

$$\cos(\text{dist.}) = -\frac{\sin\beta\,\sin\beta'}{R} + \frac{\cos\beta\,\cos\beta'\,\cos(L-L')}{R^2}.$$

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Al-Battānī's solution is most elegant, and all the more worth mentioning because, after S. Günther, *Studien zur Geschichte der mathematischen und physikalischen Geographie*, Halle 1877–79, part V, pp. 306–308, Georgius Amirucius is said to have been the first to solve this problem without dividing the figure into two right-angled spherical triangles[1]. Amirucius considers the geographical distance of two places on the earth; he uses chords, but follows a way different from al-Battānī's. Our author's method, however, so much older, seems to me also easier and more elegant than that of Amirucius. So this chapter XXVI, which Regiomontanus and Delambre attacked so bitterly, appears to be one of the chief parts of the work, once some mistakes of the copyists are cleared away and certain perhaps more obscure expressions are rightly understood.

On chapter 27

a) — p. 40, ll. 28–29.

Only one known author besides al-Battānī mentions such a *sidereal year of the Egyptians and the Babylonians*; for Ptolemy himself, whom al-Battānī cites afterwards, does not know it, as I shall say in note *b*.

The reading of the Arabic text, 365 1/4 + 1/120 (that is, $365^d\ 6^h\ 12^m$), differs from the printed translation of Plato, «ex 365 diebus et quarta ultraque parte ex 131 diei partibus» («of 365 days and a quarter, and beyond that one part of 131 parts of a day»)[a], which can only mean 365 1/4 + 1/131 (that is, $365^d\ 6^h\ 10^m\ 59.6^s$). So too a glossator (perhaps Regiomontanus) understood it, whose words in the margin of the Nuremberg edition are these: «ex 365 diebus, 15 minut. et 27 secund. et 30 ter. fere» («of 365 days, 15 minutes, 27 seconds and about 30 thirds»), that is, $365^d\ 6^h\ 11^m$[b]; and the Bologna edition took this marginal gloss into the text in place of Plato's true words. Nor, lastly, should it be passed over in silence that the Vatican codex of Plato has «ex 365 dierum [*sic*] et quarta unaque parte ex 130 diei partibus» («of 365 days and a quarter and one part of 130 parts of a day»), that is, $365^d\ 6^h\ 11^m\ 4.6^s$.

It is not easy to judge which is the genuine reading. In chapter LII (p. 127) the astrologers (Greek?) who supposed a trepidation of the fixed stars are credited with a (tropical?) year «exceeding 365 1/4 days by a fifth of an hour, or about a fifth», that is, 365 days $6^h\ 12^m$, or about this length. It is not impossible that those astrologers are the same as the ancient Egyptians and Babylonians of this chapter; but even if this were certain, we should have no criterion for settling the true reading. For «by a fifth of an hour» agrees exactly with the 12 minutes of the Escorial codex, while «or about a fifth» can agree not badly with Plato's $10^m\ 59.6^s$ (or rather $11^m\ 4.6^s$, as in the Vatican codex).

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The reading of the Arabic codex, however, seems more probable to me. Not only does the comparison I shall make below yield, among the Indians and the Persians, a year of 365 days, 6 hours, 12 minutes or a few seconds *more* (not less); but in a late Arab writer I find a passage that agrees excellently with the Escorial codex. Saḥnūn ibn ʿUthmān al-Maydawī al-Wānsharīshī, who wrote a commentary on the astronomical poem that ʿAbd al-Raḥmān al-Akhḍarī finished in the month

[1] R. Wolf, *Handbuch der Astronomie, ihrer Geschichte und Litteratur*, Zürich 1890–93, vol. II, p. 208, § 429, note *a*; Cantor, vol. II, pp. 416–417; von Braunmühl, vol. I, p. 134. — Amirucius, a Greek by nation, born at Trebizond, embraced the Islamic religion after 1461 and held a public office at Constantinople among the Turks; he died about 1490.

[a] Nallino's Addenda et emendanda (Part I, p. LXXIII) show that Plato wrote 1/130, not 1/131 as the editions have: the Vatican, Paris and Venice manuscripts and the book called *Almagesti parvum* agree; Plato seems to have found in his Arabic copy قل, 130, for قك, 120.

[b] The print has 360^d for 365^d.

of Dhū l-Qaʿda 939 H (June 1533), says the following: «What [al-Akhḍarī] relates of the non-Arab year[1] [that it comprises 365 1/4 days] is what is commonly set out. It is also said [by others that this year comprises] a part of 120 parts of a day; such is the opinion of the people of Egypt and Babylon (*qawl ahl miṣr wa bābil*), who used for this the conjunction of the sun with some fixed or wandering star until the time when the sun returned to it. This opinion is manifestly false in the highest degree. ‹For the truth is that the time of the year [is counted] from the conjunction of the sun with some fixed point of the sphere until its return to the same point, namely one of the equinoctial or solstitial points.› So relates al-Jādirī»[2]. If the whole passage of al-Battānī, p. 40, ll. 28–36, is compared with this, it will appear either that al-Battānī and the late al-Jādirī drew from the same source, or that al-Battānī himself is al-Jādirī's first source.

But let us now seek the origin of such a sidereal year. Relying on al-Battānī's authority and following the reading of the Bologna edition, Fréret (*Mémoires de l'Académie des Inscriptions*, old series, vol. XVI, pp. 214–215) ascribes the sidereal year of $365^d\ 6^h\ 11^m$ (which differs little from the value accepted today, $365.256374^d = 365^d\ 6^h\ 9^m\ 10.75^s$) to the Egyptians and the Chaldaeans; Montucla (*Histoire des Mathématiques*, 2nd ed., Paris 1799–1802, vol. I, p. 59), Bailly (*Histoire de l'Astronomie ancienne*, Paris 1775, pp. 149, 165–166, 392, 403), Lepsius (*Die Chronologie der Aegypter*, Berlin 1849, vol. I, p. 197) and others, the most recent of them P. Tannery (*Recherches sur l'histoire de l'astronomie ancienne*, Paris 1893, p. 306), to the Chaldaeans only. This sidereal year served well to prove that the Chaldaeans knew the annual precession of the equinoctial points long before Hipparchus[3].

Letronne, in his *Nouvelles recherches sur le calendrier égyptien*, memoir II, § 4, no. 2, pp. 103 ff. (Mémoires de l'Académie des Inscriptions, vol. XXIV, 2nd part, 1864), though he keeps the 11 minutes of the Bologna edition, thinks that al-Battānī's Egyptians and Babylonians are the very astrologers of chapter LII. Later, on p. 110, he writes: «Maintenant il est clair que la rétrogradation de 1 degré en 80 ans[4] donne un arc annuel de $40''$, parcouru en $16^m\ 19^s$ de temps. Si nous ajoutons 16^m à l'année tropique qu'Hipparque et Ptolémée ont conclu de leurs observations = 365 jours 5 heures 55^m, nous avons la fameuse année sidérale d'Albatégni, de 365 jours 6 heures 11^m. Il est à remarquer que c'est la même opération qui avait donné à Regiomontanus et à Purbach leur année sidérale si exacte de 365 jours, 6 heures $9^m\ 12^s$. Ils l'ont obtenue, disent-ils, dans l'hypothèse de la trépidation de l'astronome arabe Thebit….. Les anciens Égyptiens, comme les Chaldéens, sont tout à fait désintéressés dans la question. Nous n'avons ici qu'une rêverie astrologique postérieure à l'un des deux astronomes grecs, sinon à tous les deux» («Now it is clear that the retrogradation of 1 degree in 80 years gives a yearly arc of $40''$, run through in $16^m\ 19^s$ of time. If we add 16^m to the tropical year that Hipparchus and Ptolemy concluded from their observations, 365 days 5 hours 55^m, we have the famous sidereal year of Albatégni, of 365 days 6 hours 11^m. It should be noted that it is the same operation that had given Regiomontanus and Purbach their very exact sidereal year of 365 days 6 hours $9^m\ 12^s$. They obtained it, they say, on the hypothesis of the trepidation of the Arab astronomer Thebit….. The ancient Egyptians, like

[1] *Sana ʿajamiyya*, that is, the barbarian year, in the same sense in which the Greeks called all foreigners barbarians. From what follows in the text of the poem it is clear that the Roman year is meant.

[2] Saḥnūn al-Wānsharīshī, *Mufīd al-muḥtāj ʿalā l-manẓūma al-musammāh bi-l-Sirāj li-l-Akhḍarī*, Cairo 1314 H (1896–97), p. 33. — ʿAbd al-Raḥmān al-Jādirī (the name is also written الكادري) wrote an astronomical poem in the year 794 H (1391–92); Suter, *Die Mathematiker und Astronomen der Araber*, p. 172. — The words enclosed in ‹ › are also quoted, as taken from al-Jādirī, by Muḥammad ibn Saʿīd al-Marghītī al-Sūsī (who died at the end of the year 1089 H, a little before the beginning of 1679), *al-Mumtaʿ fī sharḥ al-maqnaʿ*, Fez 1313 H (1895–96), p. 3 of the third part. [*Translator:* The print has al-Mumtiʿ and al-muqniʿ; Nallino's Addenda et emendanda (Part I, p. LXXIII) correct them to al-Mumtaʿ and al-maqnaʿ.]

[3] For the Chaldaeans had a civil year of $365^d\ 6^h$ (365 1/4), which some ancients say was tropical. If, then, they had also known a sidereal year 11 or 12 minutes longer than that, the knowledge of precession could rightly have been ascribed to them. — But that precession was unknown before Hipparchus has been sufficiently proved by G. Schiaparelli, *Le sfere omocentriche di Eudosso, di Callippo e di Aristotele*, Milan 1875, pp. 20–23 (= Pubblicazioni del R. Osservatorio di Brera, no. IX), and by Th.-H. Martin in the dissertation that I shall cite below.

[4] It is the annual precession according to the astrologers of whom al-Battānī speaks in chapter LII.

the Chaldaeans, have nothing at all to do with the question. We have here only an astrological fancy later than one of the two Greek astronomers, if not than both»).

Th.-H. Martin thinks no differently, *Mémoire sur cette question: La précession des équinoxes a-t-elle été connue des Égyptiens ou de quelque autre peuple avant Hipparque?* (Mémoires présentés par divers savants à l'Académie des Inscriptions, 1st series: sujets divers, vol. VIII, 1st part, 1869); for in ch. II, § 3, pp. 369–372, he holds all that Letronne wrote to be true, and on p. 372 he ascribes that sidereal year to Alexandrian astrologers who lived between Hipparchus and Ptolemy and professed the trepidation of the fixed stars.

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Letronne's whole reasoning, however, depends on an error of computation that Martin did not notice[1]. For 1° (or 3600″) divided by 80 years gives a yearly arc of 45″, not 40″. The daily motion of the sun in Ptolemy is 0° 59′ 8″, or 3548″; divided by 45″, this shows that the arc of 45″ is run through in the 78.844th part of a day. Let the minutes of time that a day contains, that is, 1440^m, be divided by 78.84; there will result 18.264^m, or about $18^m\ 16^s$, not $16^m\ 19^s$ as Letronne has it. Let this space of time be added to Ptolemy's tropical year; one will find $365^d\ 5^h\ 55^m + 18^m\ 16^s = 365^d\ 6^h\ 13^m\ 16^s$, which differs from the sidereal year mentioned in al-Battānī (whichever reading is accepted as good, the 12^m of the codex or Plato's about 11^m). So I think that all that Letronne and Martin write about the origin of such a year must be rejected. It remains to inquire whether the length of the solar year that al-Battānī and al-Wānsharīshī (after al-Jādirī) ascribe to the Egyptians and the Babylonians (al-Battānī also to the astrologers who followed the theory of trepidation) is mentioned elsewhere too.

The Indian astronomers know a sidereal year that differs from the one just mentioned only by some added seconds of time. For they have cycles, which they call *yuga*, comprising some thousands of years, to which they assign a certain number of days; by dividing the number of days by the years of one *yuga*[a], the length of the year is easily derived. I take the table of the Indian years from J. Burgess, *Notes on Hindu Astronomy and the history of our knowledge of it* (in the Journal of the R. Asiatic Society of Great Britain, new series, vol. XXV, 1893, p. 722); I give the age of the books after G. Thibaut, *Astronomie, Astrologie und Mathematik* [*der Inder*], Strassburg 1899 (= Grundriss der indo-arischen Philologie, begründet von G. Bühler, fortgesetzt von F. Kielhorn, vol. III, part 9):

Name of the book	*Age after Christ*	*Length of the year*
Sūrya-Siddhānta[2]	before the 5th or 6th century	$365^d\ 6^h\ 12^m\ 36.56^s$
Parāśara-Siddhānta	before the 5th or 6th century	365 6 12 31.50
Pauliśa-Siddhānta	before the 5th or 6th century	365 6 12 36
Romaka-Siddhānta	before the 5th or 6th century	365 5 55 12
Prior Ārya-Siddhānta	499	365 6 12 30
Brahma-Siddhānta	628	365 6 12 9
Posterior Ārya-Siddhānta	650–1150	365 6 12 36.84
Siddhānta-Śiromaṇi	1150	365 6 12 9

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Except, then, for the *Romaka-Siddhānta*, which gives Ptolemy's tropical year, all the Indian (sidereal) years, if the seconds are neglected, have the same length that the Escorial codex mentions in this chapter, and the Escorial codex and Plato in chapter LII. Nor should it be forgotten that

[1] Martin, however, knew al-Battānī only from what Letronne reports; both, as we shall see in the notes on chapter LII, ascribe to al-Battānī certain things that he never said.

[2] In the older recension of the book the year seems to have been exactly $365^d\ 6^h\ 12^m\ 36^s$, as appears from *The Pañchasiddhāntikā, the astronomical work of Varāha Mihira, edited with an original commentary in sanskrit and an english translation and introduction by G. Thibaut and Sudhākara Dvivedī*, Benares 1889, p. 5 of the translation, ch. I, stanza 14.

[a] The Latin says that the years of one yuga are divided by the number of days.

Varāhamihira, who died in 587, gave in his *Pañcasiddhāntikā* the computation of the mean and the true longitude of the sun according to the old recension of the book *Pauliśa-Siddhānta*, from which a year of exactly $365^d\ 6^h\ 12^m$ is derived[1].

The Arab al-Bīrūnī, *India*, pp. 177, 178, 185, 222 of the text (= I, 350, 353, 368 and II, 39 of the translation), mentions the Indian year of $365^d\ 6^h\ 12^m\ 9^s$ on the authority of the book entitled *Brahma-Siddhānta*. It is known, moreover, that this *Brahma-Siddhānta*, written by Brahmagupta, was translated into Arabic, or rather reworked, by order of the caliph al-Manṣūr (754–775), and became known among the Arabs under the name *al-Sindhind* (a corruption of the Indian name); of it al-Masʿūdī, *Tanbīh*, p. 221, relates «that the solar year, according to the Indian *al-Sindhind*, consists of 365 days, a quarter of a day and a fifth of an hour, with 1/400 of an hour added», that is, $365^d\ 6^h\ 12^m\ 9^s$. The same appears from an Arab writer (in al-Bīrūnī, *Chron.*, p. 51 of the text, 60–61 of the translation) who added 729^s to the Roman year.

The year of $365^d\ 6^h\ 12^m$ seems to have been known to the ancient Persians too. Almost all the Arab writers who discuss the old chronology of the Persians relate that the Persians had a year of only 365 days, but intercalated one month of thirty days every 120th year, so that those 120 years were equal to 120 years of 365 1/4 days, since 120 : 4 = 30; and this until the Arab rule was set up in Persia (about 650); afterwards, however, they never used the intercalation. The scholars of our age have interpreted this Arab tradition in different ways; compare L. Ideler, *Handbuch der mathematischen und technischen Chronologie*, Berlin 1825–26, vol. II, pp. 540–550; Th. Benfey and M. A. Stern, *Ueber die Monatsnamen einiger alter Völker, insbesondere der Perser, Cappadocier, Juden und Syrer*, Berlin 1836, pp. 140–156; von Gutschmid, *Ueber das iranische Jahr* (in Berichte über die Verhandlungen der k. sächsischen Gesellschaft der Wissenschaften zu Leipzig, phil.-hist. Cl., XIV, 1862, pp. 1–9); lastly, as Schiaparelli points out to me, E. W. West, *Pahlavi texts translated*, vol. V, Oxford 1897 (= Sacred Books of the East, vol. XLVII), pp. XLIV–XLVI of the introduction, who, relying on solid arguments, puts the origin of this Persian calendar in the time of Darius Hystaspis (about 505 BC).

But another Persian year too, which the scholars just cited seem not to know, is mentioned by al-Bīrūnī, *Chron.*, p. 11 of the text, 12–13 of the translation: «The Persians reckoned the year as 365 days, without fractions, until, after 120 years had elapsed, a whole month [of 30 days] had been collected from the quarters of a day, and a whole day from the fifths of an hour that, with them, follow the quarter of a day. So they added a whole month to the year after every 116 years were completed. This usage of the Persians was followed by the ancient inhabitants of [the lands of] Khwārizm[2] and of Sogdiana, and by all who professed the Persian religion»[3]. It is, then, the year of 365 1/4 days + 1/5 hour (= $365^d\ 6^h\ 12^m$), which al-Battānī gives in the same numerical form in ch. LII.

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The same al-Bīrūnī, *Chron.*, p. 52 of the text, 61–62 of the translation, relates the following: «The

[1] *Pañchasiddhāntikā*, p. 14 of the translation, ch. III, stanzas 1–3. Another passage of this book (p. 5, ch. I, stanzas 11–13), where the year of the same *Pauliśa-Siddhānta* is discussed, is so obscure that the English translator has not dared to give any translation.

[2] It is the region south of the Aral lake, whose chief city is now called Khiva. Al-Bīrūnī was born in it, and he readily mentions the institutions of his homeland.

[3] He adds afterwards that this did not happen under the Pēshdādian dynasty, the oldest of all the Persian dynasties. This whole passage is read in the same words, but without al-Bīrūnī's name, in the description of Egypt written by al-Maqrīzī (d. 27 Ramaḍān 845 = 9 February 1442): al-Maqrīzī, *Kitāb al-mawāʿiẓ wa-l-iʿtibār fī dhikr al-khiṭaṭ wa-l-āthār*, Būlāq 1270 H, vol. I, p. 261. — Elsewhere al-Bīrūnī (*Chron.*, pp. 43–44 of the text, 53–55 of the translation, and in many places) mentions the simple intercalation of a month after 120 years, which depends on the year of $365^d\ 6^h$.

like[1] was set out by Ḥamza ibn al-Ḥasan al-Iṣfahānī[2] in his epistle on [the day of] *al-nayrūz*[3], where he defends the Persians as to their use of taking a solar year of 365 days + 6 hours + 1/5 + 1/400 of an hour [= $365^d\ 6^h\ 12^m\ 9^s$], while the Romans (*al-Rūm*), on the other hand, leave out the fraction that follows the six hours. He relies on the argument that the astronomer Muḥammad ibn Mūsā ibn Shākir[4] explained it and diligently investigated it in a book of his on the solar year, giving geometrical proofs and showing the errors of certain ancients. — But we have examined the observations of Muḥammad ibn Mūsā and of his brother Aḥmad, [and have found] that they show only that these fractions are less than six hours. The book that [Ḥamza] cites is the one ascribed to Thābit ibn Qurra; for Thābit was a protégé of this family, and collected what is read in this book from it and from those who polished its sciences [its books] for it[5]. The purpose of [this little book] is to prove the inequality and the difference of the solar years, since the apogee [of the sun] is movable [because of the precession of the equinoxes]; so that, to find the mean course of the sun in them, he would have needed equal revolutions and motions proportional to their times; but the only revolutions that appeared equal to him were those in the eccentric, from a given point of the eccentric to the return to the same point. This revolution, as Ḥamza says, contains fractions [of a day] exceeding six hours; but it is not called the solar year, since the year of the sun, as we have already determined, is the one in which the state, or the conditions, of nature for producing life and death [in all things] become again as they were before».

From what we have said so far it is clear that before the rule of the Arabs the Indians and the Persians knew, and indeed used, the sidereal year of $365^d\ 6^h\ 12^m\ 9^s$, or of only $365^d\ 6^h\ 12^m$, which is the value that al-Battānī ascribes in this chapter (when the reading of the codex and of al-Wānsharīshī, or rather of al-Jādirī, is accepted) to the Egyptians and the Babylonians, and in chapter LII to certain astrologers who flourished before Ptolemy.

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If a conjecture may be put forward in such obscure questions, I suspect that al-Battānī's information is true. Those Babylonians (from whom the astrologers of chapter LII probably do not differ) would then be Chaldaeans, contemporaries of Hipparchus or a little later, who could have derived such a sidereal year from observations carefully made over so many centuries. The most recent investigations show well enough that the astronomical and astrological science of the Chaldaeans flourished in the Hellenistic period too and did not neglect the old observations; so that, besides the Greek astronomy handed down to us by Ptolemy, another existed at the same time in Mesopotamia, fragments of which are preserved on clay tablets with cuneiform inscriptions[6]. A Babylonian, indeed, as Schiaparelli points out to me, was that Seleucus who, on the evidence of Pseudo-Plutarch (*De placitis philosophorum*, II, 1), dared after Aristarchus to teach the heliocentric system and the infinity of the world[7]. Much that is read in the Indian astronomers and does not agree with the doctrines of the Almagest, above all observational data, seems to

[1] He had spoken before of the Indian year.

[2] He flourished in the city of Baghdad in the first half of the fourth century H (the tenth century); he belonged to the party of the men called *shuʿūbiyya*, who tried to show in their historical and philological writings that the Persian race far surpassed the Arab stock in antiquity, nobility, language, manners and institutions. See I. Goldziher, *Muhammedanische Studien*, Halle a. S. 1889–90, vol. I, pp. 209–212.

[3] The Arabic form of the Persian name *nawrōz*, «new day», by which the Persians called the first day of the year, at the vernal equinox.

[4] He died in the month of Rabīʿ I 259 H (January 873).

[5] The text of al-Bīrūnī seems to me slightly corrupt, and not well interpreted by Sachau in his translation; I read: اذ كان صنيعة هولاء القوم ومِن بينهم وممن كان يُهذِّب لهم علومَهم جَمَلَ ما فى هذا الكتاب, with وممن for ومن and جَمَلَ for وجَمَلَ. — «The family» is the family of the sons of Mūsā ibn Shākir, of whom Muḥammad and Aḥmad are mentioned above; most skilled in the mathematical sciences and authors of many books, they spent the greatest diligence and liberality on furthering the disciplines. They greatly favoured Thābit ibn Qurra, who died in the month of Ṣafar 288 (February 901). See Suter, *Die Mathematiker*, no. 43 (pp. 20–21) and 66 (pp. 34–38).

[6] See what F. X. Kugler was able to derive from tablets not later than Hipparchus, *Die babylonische Mondrechnung. Zwei Systeme der Chaldäer über den Lauf des Mondes und der Sonne.* Freiburg i. Br. 1900.

[7] See G. Schiaparelli, *I precursori di Copernico nell'antichità*, Milan 1873, pp. 35–36 and 51 (= Pubblicazioni del R. Osservatorio di Brera, no. III). Seleucus lived before Hipparchus.

come from Chaldaeo-Hellenistic science. So it would be no wonder if the Chaldaeans had taught that sidereal year at the beginning of the Christian era to the Indians and the Persians, who seem to have given little attention to exact celestial observations[1]. And in fact Kugler, p. 91, shows that a sidereal year of $365.25953^d = 365^d\ 6^h\ 13^m\ 43.4^s$ was known to the Babylonians in the fourth and third centuries BC (what he says of al-Battānī on p. 92 should now be corrected).

b) — p. 40, l. 29 – p. 41, l. 2.

«*But Ptolemy says that they*» etc. — Ptolemy nowhere mentions that year of the Egyptians and the Babylonians; but in *Almag.* III, 2 (ed. Halma, vol. I, pp. 151–152) one reads the considerations, which al-Battānī reports, on the solar year that cannot be derived from the returns of the sun to the same fixed, non-wandering star.

The words «*so that he did not doubt the motion of the sun*» etc. seem to me to mean that Hipparchus nevertheless did not find such inequalities in the motion of the sun as to think it necessary, in order to represent them, to give the sun a third circle besides the ecliptic and the eccentric (or the epicycle). Al-Battānī had in mind, I think, this passage of the *Almagest*, where the precession of the fixed stars observed by Hipparchus is discussed: «What chiefly drives him [that is, Hipparchus] into that doubt is that from the returns of the sun to the solstitial and the equinoctial points the time of the year comes out less than 365 1/4 days, but from the returns to the fixed stars greater. Hence he conjectures that the sphere of the fixed stars too advances very slowly towards the following signs [that is, eastwards], as the planets also move in the direction contrary to the first motion [of the heavens], perpendicularly to the circle that passes through both poles, of the equator and of the ecliptic» (*Almag.* III, 2, ed. Halma, vol. I, p. 151).

c) — p. 42, ll. 6–8.

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The observation of the autumnal equinox. — Since al-Battānī follows the usage of the Byzantine indictions, putting the beginning of the Roman year on the first day of September, 19 Aylūl 1194 after Dhū l-Qarnayn will be 19 September of the year 882, not of 883. With this agrees 8 Pachōn, if it is referred to the vague years (without intercalation) of the era of Nabonassar[2]. From the epoch of the era of Nabonassar (26 February 748 BC) to the beginning of the year 882, 748 + 881 = 1629 Julian or Roman years elapsed, that is, 1629 vague years + 1629/4 days = 1630 vague years + 42 days. The first day of the month Thoth has therefore gone back 42 days (besides a whole vague year) with respect to 26 February; so 1 Thoth 1631 of Nabonassar = 15 January 882, and 8 Pachōn (which is the 248th day of the Egyptian year) = 19 September 882[a].

It remains to inquire how the words «4 3/4 hours before the rising of the sun on the 19th day of Aylūl» are to be understood, which, by the astronomical way of reckoning the day (that is, from noon), would mean 19 Sept., $13^h\ 15^m$. Had al-Battānī meant the astronomical day here, he

[1] I know of only two Persian observations between Ptolemy and the Hijra, which Ibn Yūnus reports, ch. VIII (in Caussin, p. 234), one about the year 470, the other about 630, for determining the longitude of the solar apogee. Whether they were true observations I hold uncertain.

[2] Not to the fixed years (of 365 days, with the Julian intercalation) of the Egypto-Coptic era mentioned in chapter XXXII and elsewhere, or of the era of Diocletian or of the Martyrs; both begin on 29 August, so that 8 Pachōn becomes 3 May. That al-Battānī meant the era of Nabonassar is clear from the very fact that in the Arabic text he calls the month *bākhūn*, after the Greek παχών, not after the Arabic *bashans*. Al-Battānī and the other Arabs give the Coptic months in the Greek form (which arose from the Memphitic Coptic) when they use the era of Nabonassar, in the Arabic form (from the Theban Coptic) when they use the eras of Augustus or of Diocletian. Likewise a little before, where he reports Ptolemy's observations of the equinoxes, *atūr* (ἀθύρ) and *fārmūth* (φαρμουθί) for the *hatūr* and *barmūdha* that are read in chapter XXXII (p. 67).

[a] By the usual reckoning the era of Nabonassar begins on 26 February 747 BC, and the vague year that begins on 15 January 882 is its 1630th; the dates 15 January and 19 September 882 are right.

would have referred the hour to noon, not to sunrise; it is probable, then, that the author had in mind the Muslim civil day, which begins at sunset. The observation was therefore in the morning hours of the civil 19th day, from whose noon only the astronomical 19th day begins; and since at the time of the equinox days and nights are equal, the time given will be 18 Sept. 882, $13^h\ 15^m$ mean time of al-Raqqa, in the astronomical usage. — This interpretation becomes quite certain from the interval of 743 vague years, 178 days and about $17^h\ 36^m$ that al-Battānī states to have elapsed between his observation and Ptolemy's, the difference of longitude between Alexandria and al-Raqqa being put at 2/3 of an hour. For the observations were:

Ptolemy: 25 Sept. 139, 19^h mean time of Alexandria $= 19^h\ 40^m$ al-Raqqa

al-Battānī: 18 Sept. 882, $13^h\ 15^m$ al-Raqqa;

so the interval is: (882 – 139) Julian years – (25 – 18) days – ($19^h\ 40^m$ – $13^h\ 15^m$) hours = 743 Julian years – 7 days – $6^h\ 25^m$ = 742 Julian years + 358 days – $6^h\ 25^m$.

From the year 139 to 882 there were 186 leap years; so the interval is:

742 vague years + (358 + 186) days – $6^h\ 25^m$ = 743 vague years + 179 days – $6^h\ 25^m$ = 743 vague years + 178 days + $17^h\ 35^m$,

as al-Battānī has it. Al-Battānī's observation was therefore 882 Sept. 18, $13^h\ 15^m$ mean time of al-Raqqa.

From the tables of R. Schram (*Hilfstafeln für Chronologie*, in the Denkschriften d. k. k. Akademie der Wissenschaften zu Wien, mathem. Classe, 1882, vol. XLV) one learns that in the year 882 the autumnal equinox fell on 18 September, $11^h\ 49^m$ mean time of Greenwich. According to the astronomical observations made by Chesney in 1835, the longitude of al-Raqqa is 39° 3′ 57″ E of Greenwich, $2^h\ 36^m$ in time; so the hour of al-Battānī's observation corresponds to $13^h\ 15^m$ – $2^h\ 36^m$ = $10^h\ 39^m$ mean time of Greenwich. Al-Battānī therefore observed the equinox $1^h\ 10^m$ before the true moment of time derived from Schram's tables.

[If we consider the astronomical instruments of that age, al-Battānī's observation must be judged excellent. Without doubt al-Battānī used the long alidade or the brick (the triquetrum and the mural quadrant), which are described in the last chapter of the book, or perhaps a very tall and most accurately made gnomon, to measure the meridian altitude of the sun on the days before and after the equinox; by making a proportion between the two measurements he could compute the moment of time between those two noons at which the meridian altitude of the sun would have been equal to the altitude of the celestial equator (or, as al-Battānī often says, to the altitude of the beginning of Aries) in the city of al-Raqqa (= 90° – the latitude of the city). Astronomers today, too, work in a similar way. — Schiaparelli.]

d) — p. 42, ll. 9–21.

The Arab astronomers mostly determined the *length of the year* by comparing their observations with Ptolemy's. Ibn Yūnus, ch. V (in Caussin, p. 158), however, says that it is better to divide the difference of the equinoxes found by the interval since the observations of Hipparchus.

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Ibn Yūnus, in the introduction and in ch. V (in Caussin, pp. 56 and 150 ff.), reports the mean motion of the sun in 365 days according to the Arab astronomers older than al-Battānī or of the same age as follows:

Aḥmad al-Nahāwandī, the calculator, in the city of Jundīsābūr in Susiana (today Shāhābād), by observations made at the beginning of the ninth century and by dividing the interval of time since Hipparchus (ch. V, Caussin 156–158)	359° 45′ 40″ 40‴.
The astronomers of the caliph al-Maʾmūn (Yaḥyā ibn Abī Manṣūr, al-ʿAbbās ibn Saʿīd al-Jawharī, Sanad ibn ʿAlī and others), at the observatory of al-Shammāsiyya[1] in the city of Baghdad, in the year 214 H (10 March 829 – 26 Febr. 830), 198 of Yazdajird (began 28 Apr. 829), by dividing the interval since Ptolemy[2]	359° 45′ 45″ 14‴ 24^{IV},
The astronomers of the same caliph (Sanad ibn ʿAlī, Khālid ibn ʿAbd al-Malik al-Marwarrūdhī, ʿAlī ibn ʿĪsā and others), in the city of Damascus, in the year 217 H (6 Febr. 832 – 25 Jan. 833), 201 of Yazdajird (began 27 Apr. 832)	359° 45′ 46″ 33‴ 50^{IV} 43^{V}.
The sons of Mūsā ibn Shākir, in the first half of the ninth century (ch. V, Caussin p. 150)	359° 45′ 39″ 58‴ 2^{IV}.
Aḥmad, son of Mūsā ibn Shākir, in the tables of the year 220 of Yazdajird (began 23 Apr. 851)	359° 45′ 40″.
Muḥammad ibn Aḥmad al-Samarqandī, in the year 234 of Yazdajird (began 19 Apr. 865), in the city of Samarqand (Maracanda)	359° 45′ 39″ 58‴ (var. 48‴).
The Banū Amājūr, who flourished in the same age as al-Battānī (ch. V, Caussin, p. 152)	359° 45′ 39″ 45‴.

Ḥabash, in his book of astronomy (fol. 157r), relates that the exceeding part of the revolution (*faḍl al-dawr*), that is, the amount (expressed in degrees) by which the whole circuit of the sun exceeds 365 days, was found by the astronomers of the caliph al-Maʾmūn:

at the observatory of al-Shammāsiyya	86° 43′ 39″ 36‴ 47^{IV}
in the city of Damascus	86° 41′ 25″ 14‴ 9^{IV}
by Yaḥyā ibn Abī Manṣūr	86° 35′ 13″ 46‴

In his tables Ḥabash adopted the first value, writing it in round figures 86° 43′ 39″ 37‴, or, with the degrees turned into time, 5^h 46^m 54^s 36^t. If the exceeding parts of the revolution are turned into hours and added to 365 days, the length of the tropical year comes out, according to

the observation of al-Shammāsiyya	365^d	5^h	46^m	54.640870^s
the observation of Damascus	365	5	46	45.682388
Yaḥyā ibn Abī Manṣūr	365	5	46	20.917777
the tables of Ḥabash	365	5	46	54.6

The mean motion in 365 days, according to the tables of Ḥabash (fol. 29v), is 359° 45′ 45″ 15‴ 27^{IV} 5^{V}; in a day 0° 59′ 8″ 20‴ 35^{IV} 25^{V} 0^{VI}. Astronomers today make the length of the tropical year of the sun 365^d 5^h 48^m 47.5^s (= 365.2422166 days); the mean motion in 365 days 359° 45′ 40.523″, in a day 0° 59′ 8.3302″.

On chapter 28

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a) — p. 43, ll. 11–20.

[1] The name of a gate and of a quarter of the city near the left bank of the Tigris, at the far northern end of the city; M. Streck, *Die alte Landschaft Babylonien nach den Arabischen Geographen*, Leiden 1900 ff., pp. 133–35; G. Le Strange, *Baghdad during the Abbasid Caliphate*, Oxford 1900, pp. 169–176.

[2] In one passage of the introduction (p. 56 Caussin) the seconds are 44″ (مد), but elsewhere (pp. 74 and 232) 45″ (مه).

The intervals of time between the equinoxes and the solstices. From many observations of his own he derived:

from the vernal equinox to the summer solstice:	a little less than $93^d\ 14^h$[1]
from the autumnal equinox to the vernal equinox:	about $178^d\ 14\ 1/2^h$
from the vernal equinox to the autumnal equinox:	about $186^d\ 14\ 1/2 + 1/4^h$
their sum:	about $365^d\ 5\ 1/4^h$.

In ch. XXVII, p. 42, he had established that the tropical year comprises $365^d\ 5^h\ 46^m\ 24^s$; so the difference is about $31^m\ 24^s$.

[It is strange that al-Battānī found so small a difference in the sum of two intervals, each of which is determined by observing two equinoctial times; for in these observations an uncertainty of several hours could very easily arise. Much more difficult indeed is the observation of the solstice, in which an error of half a day would not have been thought too great in al-Battānī's time. In his computation, then, he was helped by the happiest fortune, or rather he was able to use many very good observations of his own and of others. Near the solstices the declination of the sun changes by only a few seconds of arc in a day; and since the moment of time at which the declination becomes greatest has to be determined by observations, it is easily understood how hard this task was for the ancient astronomers, even though they used the shadows of long gnomons. Twelve hours before and after the solstice the declination of the sun is only 3 1/2 seconds less than at the time of the solstice; so that the Greeks not seldom erred by a whole day or more in determining these solstitial times. — For the rest, it should be noted that the error of 31 minutes that appeared in al-Battānī's computation is of very little consequence in this problem; for in that space of time the sun runs a little more than 1′ on the ecliptic, which was then the limit of the quantities that could be neglected. — Schiaparelli.]

To show the care of the observations, I shall now give the true intervals computed for the years 882[2] and 883 for mean Greenwich time, with the help of the tables of R. Schram, *Hilfstafeln für Chronologie* (Denkschriften der kais. Akademie der Wissenschaften zu Wien, mathem. Cl., 1882, vol. XLV):

	In the year 882	*In the year 883*
vernal equinox	74.9046^d = 15 March, $21^h\ 42.6^m$	75.1470^d = 16 March, $3^h\ 31.7^m$
summer solstice	168.4195 = 17 June, 10 4.1	168.6580 = 17 June, 15 46.5
autumnal equinox	261.4923 = 18 Sept., 11 48.9	261.7360 = 18 Sept., 17 39.8
winter solstice	350.6147 = 16 Dec., 14 45.2	350.8579 = 16 Dec., 20 35.4

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	In the year 882	*In the year 883*
From the vernal equinox to the summer solstice	$93^d\ 12^h\ 21.5^m$	$93^d\ 12^h\ 15.8^m$
From the vernal equinox to the autumnal equinox	186 14 6.3	186 14 8.0
From the autumnal equinox to the vernal equinox	178 15 42.5	178 15 40.6

Other equinoctial and solstitial observations of the Arabs, made at Damascus and Baghdad between the years 830 and 851, are handed down to us by Ibn Yūnus, ch. IV, in Caussin pp. 146–148, and by Delambre, pp. 85–86 (where 851 should be read for 831 on p. 86, l. 1).

b) — p. 44, l. 15.

[1] In the computation of p. 44, l. 5, he plainly chose $93^d\ 13^h\ 53.5^m$; for he states that in this interval of time the sun ran 92° 14′ 10″ instead of 92° 14′ 26″, which by al-Battānī's tables correspond to $93^d\ 14^h$. [*Translator:* The print has $93^d\ 14^m$ for $93^d\ 14^h$.]

[2] The year in which, as he himself relates in ch. XXVII, p. 42, he observed the autumnal equinox.

Anomaly, ἀνωμαλία (*ikhtilāf al-ḥaraka*, «variation, inequality of motion»), properly denoted the difference between the true longitude and the mean longitude of the sun or of a planet. Geometrically, on the hypothesis of the eccentric, it was the angle contained by the two lines drawn from the centre of the sun to the centre of the ecliptic, or of the world, and to the centre of the eccentric; on the hypothesis of the epicycle, the angle made by the two lines drawn from the centre of the ecliptic to the centre of the epicycle and to the centre of the sun. Very often, however, al-Battānī (as presently in what follows) and the other Arabs call this angle or this difference *taʿdīl*, «equation»[1], whence the present astronomical name equation of the centre. By the name anomaly[2], on the other hand, they denote, as the Greeks often did and the moderns always do, what would more rightly be called the *argument of the anomaly* or the *anomalistic motion*, that is, the arc of the eccentric or of the epicycle that the sun or the planet has run from the apogee to the given moment of time[3].

The greatest anomaly, or the greatest equation of the centre of the sun, al-Battānī found to be 1° 59′ 10″[4]; the eccentricity 2^p 4 3/4′, the radius of the eccentric being taken as 60 parts[5]. According to Leverrier, who put the greatest equation of the centre for the beginning of the year 1850 at 1° 55′ 18.31″ and the eccentricity of the solar (or rather the terrestrial) orbit at 0.016771063, the equation in the year 880 must have been 1° 58′ 2.85″, the eccentricity 0.017182700. — Other values of the equation adopted by authors older than al-Battānī or contemporary with him are these: I 214

Hipparchus and Ptolemy (*Almag.* III, 4, ed. Halma vol. I, pp. 188–190)	2° 23′
The Indian book entitled *Sūrya-siddhānta*, transl. Burgess and Whitney, p. 220, note (before the 5th or 6th century)	2° 10′ 31″
The astronomers of the caliph al-Maʾmūn, in the city of Baghdad, in the year 829 (Ibn Yūnus, introd., in Caussin p. 56)[6]	1° 59′
The same, in the year 832, at Damascus (Ibn Yūnus, loc. cit.)	1° 59′ 51″
The three brothers Banū Mūsā ibn Shākir (Ibn Yūnus, ch. V, in Caussin pp. 150 and 152)	2° 0′ 50″
Aḥmad, one of the three, for the year 220 of Yazdajird, which began on 23 April 851 (Ibn Yūnus, loc. cit.)	2° 0′ 8″
Ḥabash, Berlin codex ff. 30r–32r	1° 59′ 0″
The family of the Banū Amājūr (Ibn Yūnus, ch. V, in Caussin p. 152)	2° 0′ 50″

In the second half of the eleventh century al-Zarqālī, or Arzachel, was the first, it seems, who dared to profess a secular variation of the eccentricity and of the equation of the centre; Purbach (1423–1461) followed him in Europe. To the astronomers of the 16th and 17th centuries the decrease was not certain; Kepler, Longomontanus, Bullialdus and Riccioli denied it; Magini, Tycho Brahe, Snellius and Lansbergius affirmed it. Compare I. B. Riccioli, *Almagestum novum, astronomiam veterem novamque complectens*, Bologna 1651, vol. I, pp. 158–159.

[1] A word that Rudloff and Hochheim, al-Jaghmīnī pp. 238, 240, 248 and in many places, translate «Rectification».

[2] But when the moon and the other planets are concerned, al-Battānī almost always uses the word الحاصة *al-ḥāṣṣa*, «the [motion] falling to [the planet]», as does al-Bīrūnī, *India* (see Sachau's note in the translation, II, 363); others the word الخاصة *al-khāṣṣa*, «the proper or peculiar [motion]».

[3] In present-day astronomy the angle that the radius vector makes with the major axis of the elliptical orbit is called the *true anomaly*.

[4] The number 1° 58′ that is commonly given (Delambre p. 36, R. Wolf, *Handbuch*, I, 447, § 206) is an error of the printed translation of Plato; see p. 44, note 3.

[5] If the radius is taken as one part, one finds $\frac{2^p\ 4'\ 45''}{60^p} = 0.034652766\ldots$; Ptolemy had found $\frac{2^p\ 29'\ 30''}{60^p} = 0.0415277\ldots$, but, to make the computation more convenient, he chose the fraction 1/24 = 0.04166…. It should be noted that the eccentricity on the hypothesis of the eccentric is nearly the same as *twice* the eccentricity of Kepler's ellipse; so, if the numbers of the ancients are compared with the modern ones, they must be halved, and al-Battānī must be given the eccentricity 0.017326, Ptolemy 0.0207638… (or 0.02083….). For the ancients the eccentricity was the sine of the greatest equation of the centre.

[6] This value was adopted in the *Tested Tables*. Elsewhere Ibn Yūnus (ch. VI, in Caussin p. 232, and Delambre p. 93) wrongly puts this observation in the following year, that is, 199 of Yazdajird (which began on 28 Apr. 830), 215 H (which began on 27 Febr. 830).

c) — p. 44, ll. 29–37.

I. — *The observation of the apogee of the sun* in the year 1194 of Dhū l-Qarnayn (which began on 1 Sept. 882).

He relates that he found the longitude of the apogee to be 82° 17′. In the computation he puts the arc 7° 43′ as corresponding to the sine 8^p 4′; so 90° – 7° 43′ = 82° 17′. But in al-Battānī's tables, ff. 176v–177r, the arc corresponding to the sine 8^p 4′ is about 7° 44′, so that the longitude of the apogee would be 82° 16′. Since the author, as we shall see elsewhere, gives all the apogees the motion of the precession of the equinoxes, or of the fixed stars, that is, 54″ 33‴ a year, those two apogees will be 82° 15′ 16″ and 82° 14′ 16″ for the year 1191, to which al-Battānī's tables are mostly referred.

I cannot settle the doubt. The number 82° 15′ is confirmed by the passage of ch. XXXIII, p. 72: «on the first day of the month Ādhār of the year 1191 of Dhū l-Qarnayn [= 1 March 880] the apogee of the sun was at about 22° 15′ of Gemini, that is, 82° 15′ from the beginning of Aries»; and in the following lines the same number is repeated twice, and, as I shall show in note *d* on ch. XXXI, it is also drawn from Ibn al-Samḥ and al-Jaghmīnī. But at the end of ch. XLV, p. 114, he says that in the same year 1191 the apogee of Venus was «at 82° 14′, the apogee of the sun being as that of Venus»; and he gives this longitude, 82° 14′, to the apogee of Venus in the tables too, f. 217v. Likewise Ibn Yūnus (ch. V, Caussin p. 155) testifies that al-Battānī found the apogee of the sun at 22° 14′ of Gemini (82° 14′ from the beginning of Aries); and the same number, as regards Venus, is drawn from al-Bīrūnī, as I shall show in note *d* on ch. XXXI. — But in this computation a difference of 1 minute was of very little consequence in those times[1].

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[Al-Battānī's observation must be judged excellent, since the problem was a very difficult one at that time. By the formula $\omega = 280° 21′ 21.5″ - 61.6995″ \cdot t + 0.0001823″ \cdot t^2$, with which Leverrier determines the longitude ω of the *perigee* for *t* Julian years before 1850[2], the longitude of the *apogee* of the sun in the year 884 must have been 83° 50′ 51″; al-Battānī found 82° 17′, that is, a number 1 1/2° smaller than the true one. But since the problem depends on very uncertain elements, it would be unjust to demand more subtlety in these matters from the ancient astronomers. — Schiaparelli.]

II. — *The method of computing the longitude of the apogee of the sun.*

As he himself admits on p. 43, l. 11, al-Battānī follows the method of Ptolemy, which depends on knowing the equinoxes and the summer solstice, and also the length of the tropical year, from which the mean daily motion of the sun is derived. Let the mean motion of the sun in the spring of a given year be *v*, in the following summer *a*; the eccentricity of the orbit of the sun, or the greatest anomaly, is $FE = \sqrt{FS^2 + ES^2}$, where from al-Battānī's proof, with the radius of the eccentric = 60^p = *R*, it is clear that

$$FS = \sin\Big[v - \Big(90° + \frac{(v+a) - 180°}{2}\Big)\Big] = \sin\frac{v-a}{2},$$

$$ES = \sin\frac{(v+a) - 180°}{2} = -\cos\frac{v+a}{2}.$$

[1]Nor of any consequence in the computation of eclipses, in which the true longitude of the sun must be known. True longitude = mean longitude ± equation of the centre; the equation of the centre is found by entering the tables with the mean anomaly, which is the difference between the place of the apogee and the mean longitude. A change of 1 minute in the apogee (and so in the mean anomaly) brings into al-Battānī's equation of the centre a change that is at most 2 seconds (when the mean anomaly is about 1° or 360°), and that becomes 0″ when the mean anomaly comprises 110° or 250°. [*Translator:* With al-Battānī's eccentricity the change vanishes where the equation is greatest, at a mean anomaly of about 92° or 268°; at 110° and 250° it is about 0.65″.]

[2]But in his formula the hundredths of the seconds are not absolutely certain.

We can therefore write:

$$\text{eccentricity} = FE = \sqrt{\sin^2 \frac{v-a}{2} + \cos^2 \frac{v+a}{2}}.$$

The longitude of the apogee T is the arc AH, whose complement is the arc HB, or the angle FES. But plainly

$$\sin(90^\circ - \text{long. of the apogee}) = \sin FES = R \frac{ES}{FE},$$

where the quantities ES, FE have just been computed.

This method of Ptolemy is also set out at length by al-Bīrūnī, *Chron.*, pp. 183–85 of the text, 164–67 of the translation; afterwards he adds: «But when the *moderns*[1] had recognized that it is very difficult and almost impossible to determine the times of the solstices exactly, they preferred in their observations the points that mark the middles of the four seasons of the year, that is, the middles of the fixed signs [that is, 15° of Taurus, 15° of Leo, 15° of Scorpio, 15° of Aquarius]. The method, however, that my master Abū Naṣr Manṣūr ibn ʿIrāq[2], a client of the caliph, found for solving this problem rests on knowing any three points of the ecliptic and on knowing the length of the solar year exactly. In my book entitled *The request for testimony on the discrepancies in observations* I have shown that this method is as much better than the method of the modern astronomers as theirs is better than the method of the ancients»[3]. And in fact in this way points of the ecliptic can be chosen whose determination is easier than that of the solstices; it seems to be the same solution that Delambre proposed, *Hist. de l'astron. ancienne*, vol. II, pp. 121–133. — Of the difficulty of measuring the time of the solstice I have already spoken on p. 212[a]. I 216

III. — *The motion of the apogee of the sun.*

Delambre, p. 36, rightly denies that al-Battānī knew the *proper* motion of the solar apogee, as Bailly and many others wrongly say; in fact, at the end of chapters XXXIII and XLIV, he gives the apogee of the sun only the motion of the other apogees and of the fixed stars, that is, the motion of precession.

Delambre, however, is wrong to wonder that al-Battānī did not compare his solar apogee with that of Hipparchus and Ptolemy (who have 65° 30′), and so derive a peculiar annual motion of the apogee of 25 seconds (by comparison with Ptolemy's apogee) or of 21 (with Hipparchus's). Al-Battānī understood very well that the Greek longitude of the solar apogee was of no use; for Ptolemy[4], though he gives the motion of precession to the fixed stars and to the apogees of the planets, thinks the apogee of the sun immovable, affirming that in the year 463 after the death of Alexander (139) he found it at the same longitude, 65° 30′, at which Hipparchus had found it about 265 years before! Theon himself, lastly (*Tables manuelles*, ed. Halma, vol. I, p. 54), in the

[1] *Al-muḥdathūn*, that is, the astronomers who flourished in the time of al-Maʾmūn (813–833) and after.

[2] On him see Suter, *Die Mathematiker*, pp. 81–2, no. 186. — I think it very probable that the first part of this new solution is contained in the little work entitled «*The computation of the eccentricity, from the royal Almagest of Abū Naṣr ibn ʿIrāq*», which is kept in London, in the library called the «Library of the India Office».

[3] This whole passage of al-Bīrūnī on the apogee is also found edited, translated and explained in Sachau and Holetschek, *Eine Berechnung der Entfernung des Sonnen-Apogäums von dem Frühlingspunkte bei Albîrûnî* (Sitzungsberichte der philos.-histor. Classe der k. k. Akad. der Wissenschaften zu Wien, 1876, vol. LXXXII, pp. 243–266).

[4] *Almag.* III, 4 (ed. Halma vol. I, p. 184); likewise in the *Hypotheses*, ed. Halma, Paris 1820, pp. 44–45. — Proclus, *Hypotyposes*, ed. Halma, pp. 89–90 and 120, says that the astronomers found the apsides of the sun immovable, the apogee at 5 1/2° of Gemini (65° 30′) and the perigee at 5 1/2° of Sagittarius (245° 30′). But Proclus (5th century) also denies the precession of the equinoxes.

[a] Nallino's Addenda et emendanda (Part I, p. LXXIV) add that, on Snellius's authority, Montucla says that al-Zarqālī, about 1070–80, used observations of the sun at both equinoxes and at any third point, or at any three points, to compute the longitude of the solar apogee: the method of which al-Bīrūnī speaks; the Addenda are translated with Part I's front matter.

fourth century, always puts the apogee of the sun at 65° 30′[a].

How, then, could al-Battānī derive from the statements of the Greeks the small proper motion of the solar apogee, which is 11.46″ a year and in the interval of 745 years from Ptolemy to al-Battānī reaches only 2° 22′ 18″? It is already of the greatest consequence that the Arabs declared false the opinion of the Greeks on the immobility of the solar apogee, and recognized that it shares in the motion of the precession of the equinoxes, by which the longitudes of the fixed stars increase. Al-Masʿūdī writes, *al-Tanbīh*, p. 222: «The astrologers and astronomers of our time do not disagree with the sages of the Indians, the Chaldaeans, the Egyptians, the Greeks, the Romans and others in putting the apogee of the sun in the sign of Gemini; the disagreement among them concerns only its immobility and its motion, as we have said. Thābit ibn Qurra al-Ṣābiʾ al-Ḥarrānī[1] wrote an epistle to defend Hipparchus's opinion that the apogee of the sun moves, against what Ptolemy had said»[2]. — Caussin, p. 234, relates the following after Ibn Yūnus: «Dans ce même chap. VIII, Ebn Iounis rapporte deux observations faites par les Perses, postérieurement à Ptolémée, qui ont servi à reconnoître le mouvement de l'apogée du soleil que Ptolémée croyoit immobile. Par la première de ces observations, qui remonte à l'an 470 environ, ère vulgaire, l'apogée du Soleil fut trouvé dans 17° 55′ des gémeaux; et par la seconde, 160 ans environ après, 630 ère vulgaire, dans 20° des gémeaux» («In this same chapter VIII, Ibn Yūnus reports two observations made by the Persians after Ptolemy, which served to recognize the motion of the apogee of the sun, which Ptolemy believed immovable. By the first of these observations, which goes back to about the year 470 of the common era, the apogee of the sun was found at 17° 55′ of Gemini; and by the second, about 160 years later, 630 of the common era, at 20° of Gemini»). — Ibn Ezra[3], who died in the second half of the twelfth century, says in his preface to the Hebrew translation of the *Foundations of the tables of al-Khwārizmī*, written by a certain Ibn al-Muthannā, that Ptolemy wrongly believed the apogee of the sun to be immovable, whose motion, equal to the motion of the fixed stars, «the learned men of Ishmael», that is, the Arabs, were the first to recognize[4].

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Al-Masʿūdī, loc. cit., relates «that the Indians and others thought that the apogee of the sun moves 1° in every 100 years, so that it stays 3000 years in each sign of the zodiac and goes round the ecliptic in 36000 years»[5]. But al-Masʿūdī is gravely mistaken when he says that the Indians too ascribe to the apogee of the sun this motion of 1 degree in 100 years, or 36″ a year (which is the motion of precession according to Ptolemy). For the Indians do hold that the apogees of the sun and the planets move, but very slowly. In the book *Sūrya-siddhānta*, Burgess's translation, ch. I, st. 41–44, pp. 170–172, the apogee of the sun is said to complete its revolution in 11,162,790.7 years, that is, to run 1 minute in 516.8 years; Brahmagupta, on the other hand, and the Arabs who translated his *Brahmasphuṭa-siddhānta* in the eighth century, say that the revolution is completed in 9,000,000 years, so that 1′ is run in 416.66… years[6]. When they imagined these things is not clear[1]; but without doubt they are dreams for which astronomical observa-

[1]He died in the month of Ṣafar 288 H (February 901); Suter, *Die Mathematiker und Astronomen der Araber*, no. 66, pp. 34–38; M. Steinschneider, *Thabit ben Korra, bibliographische Notiz* (in the Zeitschrift für Mathematik und Physik, vol. XVIII, 1873, pp. 331–338).

[2]See also the passage of al-Bīrūnī that I quoted on p. 208, ll. 11 ff.

[3]The passage is given in Hebrew and in German in Steinschneider, *Zur Geschichte der Uebersetzungen aus dem Indischen ins Arabische* (in the Zeitschr. der deutsch. morgenländischen Gesellschaft, vol. XXIV, 1870, p. 356); but there the Hebrew *gabhūt* is wrongly translated «Höhe», height, instead of «Apogäum», apogee.

[4]See also the *Astronomy* of Barhebraeus, pp. 25–26 of the text.

[5]Al-Bīrūnī, *Chron.*, p. 9 of the text, 11 of the translation: «These returns [of the sun to some point of the ecliptic] are equal according to Ptolemy, since he found no motion in the apogee of the sun. But according to the authors of [the Indian book] *al-Sindhind* [= *siddhānta*] and other moderns they are unequal, for their observations show that the apogee has a motion». What motion the Indians gave the apogee we shall see a little below.

[6]Al-Bīrūnī, *India*, ch. L, p. 209 (transl. vol. II, p. 16).

[1]Thibaut says in his preface to the book *The Pañchasiddhântikâ, the astronomical work of Varâha Mihira, edited*

[a]Nallino's Addenda et emendanda (Part I, p. LXXIV) add here that Theon of Smyrna, in the first half of the second century, gives an anomalistic year of 365 1/2 days, so that the solar apogee would run round the whole zodiac in 1461 years; the Addenda are translated with Part I's front matter.

tions gave no foundation, as Burgess (and Whitney), loc. cit., and Thibaut, *Astronomie, Astrologie und Mathematik* [*der Inder*], Strassburg 1899, p. 33 (= Grundriss der indo-arischen Philologie, vol. III, part 9), rightly recognized[2]. — The Arabs, then, were the *first* to establish that *the motion of precession belongs to the apogee of the sun too.*

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The *proper* motion of the apogee, on the other hand, being very small, can be derived only from observations carried on over a very long space of time and with the help of the best instruments. So it is no wonder that the Arabs mostly did not know it; for the discrepancies, and very small ones, that appeared to them in the motion of the longitude of the apogee with respect to the motion of precession they necessarily put down to the fault of the observations, knowing that it is very difficult to measure the place of the solar apogee. Al-Zarqālī[3], however, on the evidence of Aboul Hhassan, ch. XII, p. 132, by comparing the observations of Ptolemy with al-Battānī's and his own, established that the apogee of the sun completes 1° in every 299 Roman years, that is, $12\frac{12}{299}''$ a year[a], which agrees wonderfully with the value 11.46″ accepted today; but he held that this motion takes place forward and backward alternately on the circumference of a small circle, because, as appears from his *Toledan Tables*, the apogee seemed to have advanced 16° 47′ in the 740 years from Ptolemy to al-Battānī, but to have gone back 4° 27′ in the 193 years from al-Battānī to al-Zarqālī. — The Jew Levi ben Gerson, or Leo de Balneolis (1288–1370), in his astronomy written in Hebrew, derived the motion of the apogee of the sun by diligently comparing his own observations with older ones as 1° in 43 2/3 years, that is, 1′ 19″ 39‴ 53.6^{IV} a year[4]. And since he had accepted al-Battānī's annual precession of 54″ 33‴, he put the *proper* motion of the apogee of the sun at 25″ 3‴. — Of the European astronomers, Kepler and Longomontanus were the first to distinguish the motion of the apogee of the sun from the motion of the fixed stars, that is, of precession; see I. B. Riccioli, *Almagestum novum astronomiam veterem novamque complectens*, Bologna 1651, vol. I, pp. 157–58.

IV. — *The longitude of the apogee of the sun* according to the astronomers older than al-Battānī or contemporary with him:

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with an english translation and introduction by G. Thibaut and M. Sudhâkara Dvivedî, Benares 1889, p. XVIII: «The Pañchasiddhântikâ says nothing about the revolutions of the apogees of the sun and planets, and it hence is possible that the old Sûrya Siddhânta was not yet acquainted with the theory held, on entirely insufficient grounds, by the modern treatise, and modern Hindû astronomers in general, that the apogees of the sun and the planets perform a certain number of revolutions within a mahâyuga or kalpa. On the other hand it might be supposed that Varâha Mihira, although acquainted with that doctrine, yet confined himself to stating the places which the apogees occupied at his time, since so much is sufficient for the purposes of a karaṇa-writer». — From what al-Bīrūnī relates in the *India*, ch. L, pp. 211–212 (transl. II, 18–19), he likewise seems to have found no motion of the apogees of the sun and the planets (the moon excepted) in the Indian book *Pulisa-siddhānta*, written before the fifth century, nor in the Arabic work that Abū l-Ḥasan al-Ahwāzī drew from Indian sources.

[2] [They made these revolutions extremely slow so that they could imagine all the celestial longitudes = 0° at a most remote epoch, which they took as the beginning of all times. — Schiaparelli.]

[3] The Arzachel of the Latin writers of the Middle Ages; he lived in the eleventh century (Suter, *Die Mathematiker und Astronomen der Araber*, no. 255, pp. 109–111). It is strange that so notable a passage of Aboul Hhassan (who took the proper motion of the apogee from al-Zarqālī) escaped those who have written the history of astronomy. [*Translator:* The print has «the tenth century»; Nallino's Addenda et emendanda (Part II, p. XXIV) correct it to the eleventh.]

[4] On the evidence of Augustinus Ricius, *De motu octavae sphaerae*, Casale 1513; see also Delambre 381. — 43 2/3 years are 43 years $243^d\ 8^h$; more precisely in the astronomy written (or translated?) in Latin by the same Leo de Balneolis, tract. I, ch. 58 (fol. 46v, col. *a* of the Vatican Latin codex 3098): «Nos vero invenimus eam [that is, the aux or apogee of the sun] moveri per 1 g. 43 annis, 232 diebus, 6 horis et circa 46 minutis» («but we have found that it moves 1 degree in 43 years, 232 days, 6 hours and about 46 minutes»).

[a] The print has 11″ 12/299; Nallino's Addenda et emendanda (Part II, p. XXIV) correct it to 12″ 12/299, as 3600″ : 299 requires.

The Babylonians in the third century BC[1]	76° 35′
All the Greeks from Hipparchus (about 125 BC) to Theon (4th century) and Proclus (5th century)	65° 30′
The first recension of the book *Sūrya-siddhānta*, before the 5th century, according to the *Pañcasiddhāntikā*, Thibaut's translation, ch. IX, st. 7, p. 57	80°
The later *Sūrya-siddhānta*, transl. Burgess and Whitney, notes, pp. 171 and 487	77° 15′
The book *Romaka-siddhānta*, before the 5th century, in the *Pañcasiddhāntikā*, ch. VIII, st. 2, p. 48 of the translation	75°
Āryabhaṭa, writing in 499, according to Thibaut's preface to the *Pañcasiddhāntikā*, p. XVIII	78°
The Arabic book *al-Sindhind*[2], on the evidence of Ibn Hibintā, ff. 1v and 2r	77° 15′
The tables called *al-Shāh*[3], on the evidence of Ibn Hibintā, f. 2r, and of al-Masʿūdī, *al-Tanbīh*, p. 222	80°
The astronomers of al-Maʾmūn, in the city of Baghdad, in the year 198 of Yazdajird, 214 H (829), on the evidence of Ibn Yūnus, introd., in Caussin p. 56[4]	82° 39′
The same, at Damascus, in the year 201 of Yazdajird, 217 H (832), on the evidence of Ibn Yūnus, loc. cit.	82° 1′ 37″
The three brothers Banū Mūsā ibn Shākir computed that the apogee was in the first year of Yazdajird (which began on 16 June 632), as Ibn Yūnus relates, ch. V, Caussin 150	80° 44′ 19″
Aḥmad, one of the three, in the year 220 of Yazdajird (which began on 23 Apr. 851), on the evidence of Ibn Yūnus, loc. cit., and ch. VIII (in Delambre, p. 99)	84° 33′
The family of the Banū Amājūr (Ibn Yūnus ch. V, Caussin p. 152)	83° 35′
Mufliḥ ibn Yūsuf, a client of that family (Ibn Yūnus, loc. cit.)	84° 5′

To these may be added the two *Persian* observations (?) which, taken from the work of Ibn Yūnus, I reported on p. 209, note 2; the later, of about the year 630, gave 80°, like the tables *al-Shāh* (translated into Arabic from Pehlevi, or middle Persian) and the first *Sūrya-siddhānta*.

d) — p. 46, l. 7.

I have kept the length of the line EF as it is in the Escorial codex and in Plato's translation, namely $61^p\ 48'\ 35''$. More exactly it was[a]

$$EF = \sqrt{EM^2 + FM^2} = \sqrt{(61^p\,48'\,2'')^2 + (1^p\,2'\,22\tfrac{1}{2}'')^2} =$$

[1] According to F. X. Kugler, *Die babylonische Mondrechnung*, Freiburg i. B. 1900, p. 90. The numbers have been reduced to our movable ecliptic; on the fixed ecliptic of the Babylonians they were 79° 49′.

[2] It was the translation, made in the eighth century, of the work that the Indian Brahmagupta wrote in 628 and called *Brahma-sphuṭa-siddhānta*. The longitude of the apogee is as in the later Sūrya-siddhānta. — Al-Masʿūdī, *al-Tanbīh* pp. 221–222, says that those who followed the method of the Sindhind put the apogee of the sun at 77° 55′ 14″ in the first year of the Hijra.

[3] Ibn Rusta also mentions them, p. 162, ll. 3–17, from which it is clear that, at least in the Arabic recension, they referred the celestial motions to the first year of Yazdajird (16 June 632). Without doubt they are what other Arabs call the *tables of al-Shahriyār*, for *shāh* and *shahriyār* mean «king» in Persian; at the end of the eighth century, or the beginning of the ninth, Abū l-Ḥasan ʿAlī ibn Ziyād al-Tamīmī had translated them from Persian; see *Kitāb al-Fihrist*, p. 241, l. 5, and 244, l. 29; *Hamzae Ispahanensis Annalium libri X*, edited by J. M. E. Gottwaldt, St Petersburg–Leipzig 1844–1848, vol. I, pp. 200 and 201; Haji Khalfa, *Lexicon bibliographicum et encyclopaedicum*, edited and translated by G. Flügel, Leipzig–London 1835–58, vol. III, p. 566, no. 6952. In al-Bīrūnī, *Chron.*, p. 6 of the text and of the translation, the name is *Tables of the Shahriyārān al-Shāh*. In the time when Abū Maʿshar (d. 886) wrote, they were very widespread among the Arabs, as appears from Ibn Rusta, loc. cit.; the astrologer Māshāʾallāh (d. about 815) relied on them, on the evidence of Ibn Hibintā, f. 224r. [Perhaps the Arabic translation of the Pehlevi book «Tables of Shatro-ayār», which Mānūshchihar mentions in his second epistle (about 880), in E. W. West, *Pahlavi texts translated*, vol. IV, Oxford 1892, p. XLVII = The sacred books of the East, vol. XXXVII. — Schiaparelli.]

[4] So for the year of the Hijra (which began on 15 July 622), having computed a precession of 3° 9′ 1″ in 214 Arab years, they put the longitude of the apogee at 79° 30′. This is the value adopted in their *Tested Tables* and by Ḥabash, f. 28r. — Al-Masʿūdī, *al-Tanbīh* p. 222, wrongly writes that this observation was in the year 217 H.

[a] The print has $(3472.5'')^2$ for $(3742.5'')^2$, whose square follows.

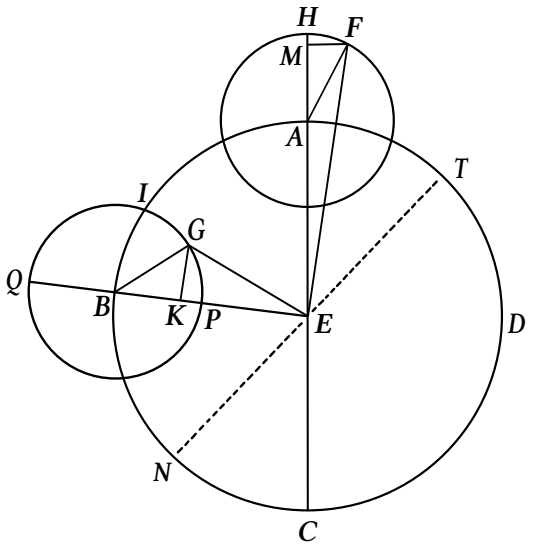
$$= \sqrt{(222\,482'')^2 + (3742.5'')^2} = \sqrt{49\,498\,240\,324^{\mathrm{IV}} + 14\,006\,306.25^{\mathrm{IV}}} =$$
$$= \sqrt{49\,512\,246\,630.25^{\mathrm{IV}}} = 222\,513.47\ldots'' = 61^p\,48'\,33.47''.$$

e) — p. 46, ll. 25–26.

The length of the line EG is given as in the codex and in Plato, namely $58^p\ 12'\ 34''$; but on p. 47, l. 32, the codex more rightly has $32''$ for the seconds. For[a]

$$EG = \sqrt{KG^2 + EK^2} = \sqrt{(1^p\,2'\,22\tfrac{1}{2}'')^2 + (58^p\,11'\,58'')^2} =$$
$$= \sqrt{(3742.5'')^2 + (209\,518'')^2} = \sqrt{14\,006\,306.25^{\mathrm{IV}} + 43\,897\,792\,324^{\mathrm{IV}}} =$$
$$= \sqrt{43\,911\,798\,630.25^{\mathrm{IV}}} = 209\,551.4\ldots'' = 58^p\,12'\,31.4''.$$

f) — p. 48, ll. 2–16.



To compute the *equation of the centre*.

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The geometrical constructions of pp. 46–47 are taken from *Almag.* III, 5 (ed. Halma vol. I, pp. 190–199); al-Battānī's formula, on the other hand, is not in Ptolemy:

$$\sin(\text{eq. of the centre}) = \frac{R\,\frac{e\sin m}{R}}{\sqrt{\left(\frac{e\sin m}{R}\right)^2 + \left(\frac{e\cos m}{R} + 60^p\right)^2}},$$

where *e* is the eccentricity or, on the hypothesis of the epicycle, the radius of the epicycle[1]; *m* the arc of the anomaly, that is, the arc of the eccentric or of the epicycle that the star has run from the apogee to the given moment of time. The rule is very easily derived from the constructions of Ptolemy and al-Battānī. On the hypothesis of the epicycle, where the equation of the centre = the angle HEF,

$$\sin HEF : MF = \sin FME : EF, \quad \text{that is,} \quad \sin(\text{eq. of the centre}) : MF = \sin 90^\circ : EF,$$

[1] In parts of the radius of the eccentric or of the orbit (= 60^p).

[a] The print has 43 911 798 625.25 for the sum, 43 911 798 630.25.

and since sin 90° = *R*,

$$\sin(\text{eq. of the centre}) = R\,\frac{MF}{EF}.$$

1) $$EF = \sqrt{\overline{MF}^2 + (AE + AM)^2} = \sqrt{\overline{MF}^2 + (60^p + AM)^2}.$$

$$AM : AF = \sin AFM : \sin AMF = \cos HAF : R,$$

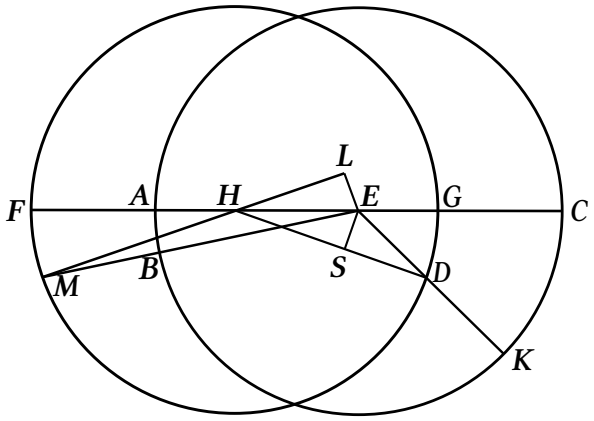
whence $AM = \frac{AF \times \cos HAF}{R} = \frac{e \cos m}{R}$.

$$MF : AF = \sin HAF : \sin AMF,$$

whence $MF = \frac{AF \times \sin HAF}{R} = \frac{e \sin m}{R}$. Let these values of the lines AM, MF be put into 1); there will result

$$EF = \sqrt{\left(\frac{e \sin m}{R}\right)^2 + \left(\frac{e \cos m}{R} + 60^p\right)^2},$$

which is the denominator of al-Battānī's formula.



Similarly on the hypothesis of the eccentric:

$$\sin(\text{eq. of the centre}) = \sin LME = R\,\frac{LE}{ME},$$

$$LE = HE\,\frac{\sin LHE}{R} = HE\,\frac{\sin FHM}{R} = \frac{e \sin m}{R},$$

$$LH = HE\,\frac{\cos LHE}{R} = HE\,\frac{\cos FHM}{R} = \frac{e \cos m}{R},$$

$$ME = \sqrt{\overline{LE}^2 + (HL + 60^p)^2} \quad \text{etc.}$$

Had he known the use of tangents, he would have found more simply, on the hypothesis of the epicycle,

$$\tan(\text{eq.}) = \tan HEF = R\,\frac{MF}{60^p + AM} = R\,\frac{e \sin m}{60^p R + e \cos m},$$

and on the hypothesis of the eccentric

$$\tan(\text{eq.}) = \tan LME = R\,\frac{LE}{60^p + HL} = R\,\frac{e \sin m}{60^p R + e \cos m}.$$

In all these formulas the eccentricity or the radius of the epicycle, that is, *e*, is in *sixtieth* parts of the radius of the eccentric or of the orbit, which is supposed to comprise 60^p. But if the radius of the eccentric or of the orbit is taken as 1^p, and *e* is expressed in its *decimal* parts, as is our custom (that is, if e/60 is put for the *e* of the Greeks and of al-Battānī), plainly 1 must be written everywhere for 60^p.

g) — p. 48, ll. 17–20.

The radii of the epicycles of the moon and the five planets. The correction of the numbers is easy, for these radii are the sines of the greatest equation of the centre (at the mean distance from the earth, if the five planets are concerned). Let the greatest equations (the mean ones for the planets) and their sines be taken from al-Battānī's tables; there will result: I 221

Planet	*Greatest equation (mean for the planets)*	*Its sine (radius of the epicycle)*	*Errors of the codex*	*Errors of Plato*	*Ptolemy*	*al-Farghānī*
Moon	5° 1′ 0″	5^p 15′	6^p	—	5^p 15′	6 1/3^p
Saturn	6° 13′	6^p 29′ 50″	—	7″	6^p 30′	6 1/2^p
Jupiter	11° 3′	11^p 30′	—	add. 5″	11^p 30′	11 1/2^p
Mars	41° 9′	39^p 27′ 22″	19^p 25′	55′	39^p 30′	39 1/2^p
Venus	45° 59′	43^p 9′	—	44^p 9′ 5″	43^p 10′	43 1/6^p
Mercury	22° 2′	22^p 30′ 30″	29^p	—	22^p 30′	22 1/2^p

From the reading of the codex and of Plato it seems that al-Battānī wrongly wrote the radius of the epicycle of Mars as 39^p 25′ 22″ for 39^p 27′ 22″; or perhaps ٥, 25, is a copyists' error for ٧, 27. — In the other planets he kept Ptolemy's numbers[1].

The astronomers of the caliph al-Maʾmūn in the *Tested Tables* (on the evidence of Ibn Yūnus, ch. VI, Caussin pp. 234–236) and Ḥabash, ff. 35r ff., have the mean equations of the planets as al-Battānī does, and the greatest equation of the moon as 5°, leaving out the minute. Ibn Yūnus, loc. cit., changed only a little the mean equation of Venus (46° 25′) and of Mercury (22° 24′).

On chapter 29

a) — p. 48, ll. 33–37.

Like Ptolemy, all the Arab astronomers known to me count the day from noon. Sédillot, however (from chapter I or III of Ibn Yūnus, which are unpublished?), p. 228 of the translation of Oloug Beg, says that Yaḥyā ibn Abī Manṣūr, the chief author of the *Tested Tables*, began the day at midnight, like Hipparchus; on the evidence of al-Bīrūnī, *Chron.* p. 6 of the text and of the translation, the author of the tables of the *Shahriyārān al-Shāh*, of which I have already spoken on p. 218, note 4, did the same.

b) — p. 49, ll. 8–13.

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[*The equation of time.* — In al-Battānī's theory of this equation some things are worth noting. In those times (and still today, nearly) the greatest retardation of the sun with respect to its mean motion came about 20° of Aquarius, the greatest advance a little after the beginning of Scorpio. This is what al-Battānī means by the words «*the place of the diminution is at about 2/3 of Aquarius*» etc. Astronomers today have defined mean time so that it holds the middle place between the oscillations of the solar motion, and so that the correction to be applied to true time to find mean time (that is, the equation of time) is sometimes additive and sometimes subtractive. Ptolemy did otherwise; like Theon after him in the *Handy Tables* (see Delambre, *Histoire de l'astronomie*

[1] In al-Farghānī the radius of the lunar epicycle is 6 1/3^p, both in Golius's edition (ch. XVI, p. 65) and in the Latin translation printed at Nuremberg (1537, fol. 16v). I have followed this edition in setting down the radius of the epicycle of Mars; Golius's edition gives 39 1/6^p.

ancienne, Paris 1817, vol. II, pp. 630–632), he determined the course of mean time so that it agreed with true time when true time had advanced more than in all the other parts of the year, that is, at the beginning of Scorpio. Then the two times agree; in all the other parts of the year true time will lag behind mean time, and the correction to be applied to it to find the true time, that is, the *equation of time*, will always be additive. Its greatest amount therefore fell at 20° of Aquarius, as is gathered from the table of this equation that Delambre gives after Ptolemy and Theon, *Astr. anc.*, II, 632.

Al-Battānī, on the other hand, for what reason I do not well know, supposes that mean and true time agree when true time is at its greatest retardation, that is, at 20° of Aquarius; for this is what the words mean, «*by this day-and-night* [20° of Aquarius] *compute the other days-and-nights of the year in this book*». Plainly, then, in all the other parts of the year true time will *precede* mean time; the correction, or equation, will always be subtractive, and will reach its greatest amount near the beginning of Scorpio, as appears from the table, vol. II, p. 64. Both ways lead to the same result; but Ptolemy's mean time *precedes* al-Battānī's mean time by the whole greatest equation, which Ptolemy makes $33^m\ 30^s$, al-Battānī $31^m\ 12^s$. For Ptolemy's mean time agrees with true time when true time is at its greatest advance, al-Battānī's when true time is at its greatest retardation. So al-Battānī, to make his astronomical tables comparable with Ptolemy's when both are entered with the same argument of time, had to increase all his epochs of the celestial motions by as much as those motions amount to in the interval of $31^m\ 12^s$. Such a correction is negligible in all the motions of the planets; for the sun it is a little more than 1′; for the moon, however, it reaches 18′. In this way at least it can be understood to some extent why he says below (ll. 28–29) *that he added 18′ to the mean longitude of the moon in his tables*. When al-Battānī's mean time is $0^h\ 0^m\ 0^s$, Ptolemy's reaches $0^h\ 32^m\ 12^s$. It also follows that the tables of Ptolemy and of al-Battānī, as regards the equation of time, cannot be compared directly with each other; for in the one the equation is nothing when in the other it is greatest. Had al-Battānī built his table on the elements that Ptolemy used, the sum of the two equations would always be $33^m\ 30^s$; but this sum is always slightly different, since al-Battānī's equation rests on somewhat different elements (another obliquity of the ecliptic, another eccentricity of the solar orbit, another length of the tropical year). — Schiaparelli.]

It is worth reporting here what Delambre, p. 98, relates from ch. III of Ibn Yūnus: «Il [Ibn Yūnus] compte cinq méthodes différentes pour calculer l'équation du tems. La première est celle de Ptolémée. Théon nous l'a expliquée de façon à ne rien laisser à désirer. La seconde est celle des auteurs des Tables qui commencent par le point de l'écliptique où l'équation est nulle. Suivant la troisième, on commence par le point où l'équation est soustractive (et sans doute la plus grande); dans la quatrième, on commence la Table au point où l'équation est additive; enfin, dans la cinquième, on commence au point équinoxial du printemps. Toutes ces méthodes au fond n'en sont qu'une» («He [Ibn Yūnus] counts five different methods of computing the equation of time. The first is Ptolemy's; Theon has explained it to us so as to leave nothing to be desired. The second is that of the authors of the tables that begin with the point of the ecliptic where the equation is nothing. By the third, one begins with the point where the equation is subtractive (and doubtless greatest); in the fourth, the table is begun at the point where the equation is additive; lastly, in the fifth, one begins at the vernal equinoctial point. All these methods are at bottom only one»)[a].

Ḥabash, f. 16v, follows Ptolemy's usage.

c) — p. 49, ll. 15 ff.

[a]Nallino's Addenda et emendanda (Part II, pp. XXIV–XXV) show from Muḥammad ibn Abī Bakr al-Fārisī that the fifth method is that of Yaḥyā ibn Abī Manṣūr and the authors of the *Tested Tables*, who began from the beginning of Aries; they are translated with Part II.

[«*The given mean place of the sun*» here means the initial mean place; «*the other mean place*», the mean place that the sun has reached at that time. Plainly, as in the *Almagest*, the problem is to correct a given interval of true time so as to know the interval of mean time corresponding to it. — Schiaparelli.]

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The rules are set out not very clearly both by Ptolemy, *Almag.* III, 8 (ed. Halma vol. I, pp. 207–210), and by al-Battānī; far better in the commentary of Theon, which Delambre reports in his notes on that passage of the Almagest (ed. Halma, vol. I, p. 28 of the notes). They can be described briefly in this way. Let a given interval of true time between t and t' be converted into an interval of mean time. For each of the times t and t' let the mean and the true longitudes be derived from the tables of the sun; the difference between the two mean longitudes will be the arc of the ecliptic run in mean motion, the difference between the two true longitudes the arc described in true motion. Let the intervals of right ascension that correspond to those two arcs of the ecliptic be taken; there will come out the arcs of the equator that measure the mean time and the true time between t and t'. The difference of these two arcs of the equator, turned into parts of an equal hour, will be the correction of true time by which mean time is found, that is, the equation of time; it will be added to the unequal, or true, time when the equatorial arc of the latter is shorter than the mean one, or subtracted from the unequal time when its arc is longer than the mean one. The given interval will come out expressed in mean, or equinoctial, days-and-nights.

«Suivant les modernes», says Delambre, loc. cit., «l'*équation* du temps est l'excès de la longitude moyenne sur *l'ascension droite vraie*, et cet excès converti en temps à raison de 15° par heure, est l'*équation* du temps, qui sert à convertir le temps vrai en temps moyen. Ce procédé est au fond le même que celui de Ptolémée, à la réserve que le précepte de Ptolémée donne la différence de deux *équations* du temps, au lieu que la règle moderne ne donne qu'une seule *équation.* Avec nos tables d'*équation du temps* nous corrigerions séparément les deux temps donnés, et nous prendrions la différence des deux temps corrigés, qui seroit l'intervalle exprimé en temps moyens» («According to the moderns, the equation of time is the excess of the mean longitude over the true right ascension, and this excess, turned into time at 15° to the hour, is the equation of time, which serves to convert true time into mean time. This procedure is at bottom the same as Ptolemy's, except that Ptolemy's rule gives the difference of two equations of time, whereas the modern rule gives only one equation. With our tables of the equation of time we would correct the two given times separately and take the difference of the two corrected times, which would be the interval expressed in mean time»).

Tannery, *Recherches sur l'astronomie ancienne*, Paris 1893, p. 178, says that the Ptolemaic method prevailed until Tycho Brahe. See Riccioli, *Alm. novum*, I, 177–78.

d) — p. 49, ll. 25–34.

[The wording of this passage is very obscure. First he means that *in the method he used the mean time will always be less than the time that the place of the sun indicates*, that is, than true time; this is clear from what I said in note *b*, pp. 221–22. — He then adds that *this will be so for a very long time*, until *the motion of the solar apogee* arising from precession *has* perceptibly *changed the course of the part of that equation that comes from the anomaly of the solar motion.* — Then he speaks of the 18 minutes by which he increased the longitude of the moon at the epoch of the tables, whose cause I explained in note *b*, p. 222. — Lastly, confirming what precedes, he indicates that the equation of his table must *always be subtracted* from true time to give mean time, and *added* to mean time to find true time.

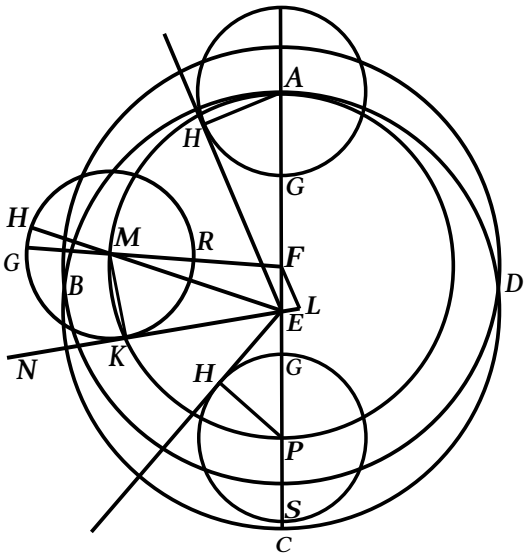
The words «*at the end of a long space of time, in which the variation of the apogee of the sun, which we have found on the ecliptic, will grow; so that what comes from the inequality of the motion of the sun will change*» point to two discoveries of great importance. The one, which I have already

mentioned, is that *the equation of time changes slowly over the years with the secular motion of the apogee*; the other, discussed in note *c* on ch. XXVIII (pp. 216–219), is that *Ptolemy's opinion of the unchanging distance of the solar apogee from the equinox is false.* — Schiaparelli.]

On chapter 30

a) — p. 50 – p. 51, l. 27.

Everything so far is taken from *Almag.* V, 2 (ed. Halma vol. I, pp. 288–292). The figure and the proof of p. 51, however, which are corrupt in the codex and in Plato, I have easily healed in the translation with very small changes, under Schiaparelli's guidance. The figure of the codex and of Plato is this:



[While what is read on pp. 52–54 agrees well enough with the doctrine of the *Almagest*, the figure and the proof of p. 51 (as the codex and Plato have them) are false and contrary to what is set out afterwards and to the figure of p. 53. Plainly the writer of the archetype from which the Escorial codex and Plato's codex arose, having drawn the figure carelessly, foolishly tried to fit the proof to it. — What the circle may be that has its centre at F and passes through B and D cannot be understood; it is wholly useless. But a point of the greatest consequence in Ptolemy's theory is missing, which is marked in the corrected figure with the letter F′; it is as far from the earth E as the centre F of the eccentric is on the opposite side, and it shows the true direction of the apogee H of the epicycle, from which the computation of the arc GH is derived. Instead of F′ the centre of the eccentric itself is considered, and by the two lines FM, EM an arc GH is defined which is falsely supposed to be the arc GH of the figure of p. 53. There is a similar error in the proof. When small corrections are made on p. 51, ll. 13–18, al-Battānī's text is well restored. — Schiaparelli.]

b) — p. 51, l. 36 – p. 52, l. 10.

All the numbers are taken from the *Almagest*, as I have noted in their place. For the 5° 1′ of the simple inequality the authors of the *Tested Tables* (see Ibn Yūnus ch. VI, Caussin p. 232) and Ḥabash, ff. 35r ff., adopted only 5°.

[The simple inequality at its greatest amount (5° 1′) is the *least* value of the greatest equation on the epicycle; the greatest is, according to Ptolemy, 7° 40′. The mean comes out 6° 20 1/2′; such would be the greatest equation of the centre of the moon if the second inequality, which we call the *evection*, did not exist. Today the mean value is put at 6° 18′, the second inequality at ± 1° 16′; so according to Ptolemy 5° 1′, 6° 20 1/2′, 7° 40′; according to the moderns 5° 2′, 6° 18′, 7° 34′. — Schiaparelli.]

The third inequality, which we call the *variation*, was unknown before Tycho Brahe; some wrongly said that Abū l-Wafāʾ (d. 388 H, which began on 2 Jan. 998) had already found it, as Carra de Vaux has sufficiently shown, *L'Almageste d'Aboû 'l-Wéfa al-Bûzdjâni*, pp. 434–471 (in the Journal Asiatique, 8th series, vol. 19, 1892). But in the tables of Ptolemy (and of the Arabs), as Tannery first noticed, *Rech. sur l'astr. ancienne*, p. 214, besides the term 1° 19′ 30″ sin (2 [☾ − ☉] − *m*), that is, besides the evection[1], there is also a term 18′ sin 2 (☾ − ☉), which is nearly half the variation. See also Carra de Vaux, p. 470.

c) — p. 53, ll. 18–20.

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[The words printed in italics seem to me interpolated. Al-Battānī had made the computation before, $42'\,38'' = 60' \frac{1°\,53'}{2°\,40'}$; now, as if it were something new and wonderful, we are taught that $2°\,40' = 1°\,53' \frac{60'}{42'\,38''}$. Having made this computation (which gives 2° 39′ for 2° 40′), our interpolator, not knowing what this number is, looks for it in the sixth column of the table and finds it under the argument 110°, which has nothing to do with this problem; entangled in his own error, he changes 110° into 120°, to get back to Ptolemy's number, which measures the angle AEM, that is, the mean anomaly of the eccentric. But in the table something entirely different corresponds to the number 120°. Plainly the author of these lines understood nothing of this computation, and he cannot possibly be al-Battānī. — Schiaparelli.]

d) — p. 54, ll. 19–30.

[*The lunar tables.* — When al-Battānī's tables are compared with Ptolemy's, what the author relates appears true. To the motion of the moon in longitude he added the correction that he had already applied before (ch. XXVII, p. 42) to the motion of the sun, so that the mean elongation (☾ − ☉) remains as in Ptolemy. The anomalistic motion too is unchanged.

Ptolemy's motion of the node, or of the latitude[2], he says he found diminished by 27′ in the interval of about 750 years between him and Ptolemy. Comparing the tables of the two, I find al-Battānī's motion in latitude smaller than Ptolemy's indeed, but not by exactly that amount; it is quite certainly gathered that al-Battānī diminished the motion in latitude by exactly three minutes in a century, which in 750 years makes 22 1/2′ instead of 27′. — The number 27′ of the codex and of Plato is, however, certain, for it is confirmed by what is related in ll. 28–30. If a correction of 27′ appeared in 750 years, in the roughly 1200 years that had elapsed since the era of Dhū l-Qarnayn the correction will be $27' \times \frac{1200}{750} = 43.2'$. These are the 3/4 of a degree by which, as al-Battānī says, the motion in latitude had to be diminished from the epoch of the era to «*this our time*», before the epochs of that motion, noted in the last column of vol. II, p. 72, were computed.

It may seem strange to some that al-Battānī found a correction of the motion in latitude (or of the motion of the node) of only 27 minutes (22 1/2′ in the tables) in 750 years, when in this space of time the correction of the motion in longitude had come to no less than 4° 31′. The matter

[1]By the sign ☾ I denote the mean longitude of the moon, by ☉ the mean longitude of the sun, by *m* the mean anomaly of the moon.

[2]The motion in latitude in al-Battānī is the sum of the motion in longitude and the motion of the node.

is explained thus. The ancient astronomers referred the motion of the moon in longitude to the motion of the sun, using the epochs of observed eclipses; so whatever error affected the motion of the sun affected the motion of the moon in the same interval of time. But the motion of the sun (or the length of the year) depended on observations of the equinoxes, each of which could give an error *of several hours*, indeed *of days*. — The motion of the nodes of the moon, on the other hand, was found from the position of the moon with respect to the nodes in the same eclipses; and this position could easily be determined within *a few minutes of arc* from the magnitude of the eclipses observed.

I add a comparison of the lunar tables of Ptolemy, al-Battānī and Hansen[1]:

	Inclination of the lunar orbit	*Mean motion in longitude in 100 Roman years*	*Mean motion in anomaly in 100 Roman years*	*Mean motion in latitude in 100 Roman years*
Ptolemy	5° 0′	307° 21′ 37″	198° 29′ 56″	82° 2′ 43″
al-Battānī	5 0	307 57 46	198 29 56	81 59 43
Hansen	5 8 2/3	307 53 40	198 50 38	82 2 40

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Al-Battānī, then, determined the motion in longitude much better than Ptolemy; he kept the inclination of the orbit and the motion in anomaly unchanged; he corrected the motion in latitude less well. But in this motion Hansen's number is not so certain that the comparison can be made safely. — Schiaparelli.]

The astronomers of the caliph al-Ma'mūn had already found, in the *Tested Tables* (see Ibn Yūnus ch. VI, Caussin p. 232), a motion in longitude better than Ptolemy's, but they had put the motion of the node too small. The lunar motions in a Persian year of 365 days (without intercalation) are these:

	Mean motion in longitude	*Mean motion in anomaly*	*Mean motion of the node*
Ptolemy[2]	129° 22′ 46″ 13‴ 50^{IV} 32^{V} 30^{VI}	88° 43′ 7″ 28‴ 41^{IV} 13^{V} 55^{VI}	19° 20′ 0″ 58‴ 53^{IV} 52^{V} 25^{VI}
Tested Tables	129 23 5 51	88 43 7 28 41	19 19 33 40
Ḥabash	129 23 6	88 43 7	19 19 39
al-Battānī	129 23 8	88 43 7	19 20 37[a]

e) — p. 54, l. 33.

«*And it ends beyond the sphere of Mercury*». — In this very chapter, p. 61, he computes that the shadow of the earth, when the sun is at its greatest geocentric distance, extends 254 2/3 earth radii beyond the centre of the earth. In chapter L, pp. 120–121, he finds

the nearest distance of Mercury	64 1/6 earth radii
the farthest distance of Mercury and the nearest distance of Venus	166 earth radii
the farthest distance of Venus	1070 earth radii

[1] [Whose numbers are taken from the *Monthly Notices of the Royal Astronomical Society*, vol. XV, p. 7. — Schiaparelli.]

[2] These numbers, computed to the sixths, are in *Almag.* IV, 3 (ed. Halma vol. I, pp. 222–25, and the tables, pp. 226–237). The motion of the node is not given there, but it is drawn at once by taking the mean motion in longitude from the mean motion in latitude.

[a] Al-Battānī's motion of the node in 365 days is 19° 19′ 38″ in Schiaparelli's computation of the fourth eclipse (p. 231), which takes it from his tables, and the motions in 100 Roman years in the table above likewise make it smaller than Ptolemy's; the 19° 20′ 37″ printed here is 59″ greater.

Hence it follows that in al-Battānī's system of the heavens the shadow of the earth passes beyond the sphere of Mercury and enters the sphere of Venus itself.

f) — p. 55, l. 15.

[Mercury and Venus are never seen in opposition to the sun; hence it is understood at once why neither planet can fall into the shadow of the earth. — Schiaparelli.]

g) — p. 56, ll. 9–21.

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The first eclipse (of the sun), observed on 8 Āb (August) 1202 of Dhū l-Qarnayn, 891, at $1^h\ 7\ 1/2^m$, in the city of al-Raqqa[1]. It bears the number 4995 in Oppolzer, *Canon der Finsternisse* (Denkschriften der k. Akad. der Wissenschaften zu Wien, Mathem. Cl., vol. LII, 1887). Halley, in the dissertation cited on p. 56, note 2, tried to clear the elements of all these eclipses of the countless errors of the editions of Plato, having collected the roots of the mean motions, corrected as far as was possible, from chapters XXVII, XXVIII and XXX. He restored much successfully; but he indicated only the emendations, leaving out the computations by which he had found them.

[To correct the numbers of the codex and of Plato properly, the author's computations must be repeated from his own tables and hypotheses. First, al-Battānī says that the time of the true conjunction preceded the apparent conjunction, that is, the middle of the eclipse, by about an eighth of an hour (see p. 56, ll. 16–17); it follows that the time of the true conjunction was $1^h\ 0^m$ after noon. Further, there is no doubt that al-Battānī's time, given in seasonal hours in the observation, is *true* time, not *mean*. With the sun at 19° of Leo, al-Battānī's table (vol. II, p. 63) gives the equation of days-and-nights $3°\ 16' = 13^m\ 4^s$, to be subtracted; hence the mean time of the true, or geocentric, conjunction is $0^h\ 47^m$ after noon, leaving out a few seconds of time, which are wholly negligible in this matter. The computation, then, taken from the volume of tables, pp. 72–83, is as follows:

Mean motions	*Mean long.* ☉	*Mean long.* ☾	*Mean anom.* ☾	*Asc. node* ☾
Dhū l-Qarnayn 1191	342° 48′ 50″	165° 40′ 17″	108° 20′ 0″	176° 23′ 0″
+ 11 years	359 21 47	9 35 37	282 2 11	212 42 13
Tammūz (July)	150 48 17	215 59 20	198 56 33	8 6 6
+ 8 days	7 53 7	105 24 40	104 31 12	0 25 25
+ 30 minutes	0 1 14	0 16 28	0 16 20	0 0 4
+ 15 minutes	0 0 37	0 8 14	0 8 10	0 0 2
+ 2 minutes	0 0 5	0 1 6	0 1 5	0 0 0
	140° 53′ 57″	137° 5′ 42″	334° 15′ 31″	37° 36′ 50″
				137° 5′ 42″
mean motion in lat. ☾				174° 42′ 32″

[1]From astronomical observations made in 1836 Chesney established the longitude of the city as 39° 3′ 57″ E of Greenwich and the latitude as 35° 55′ 35″ N; the place of observation was the north-west corner of the ruins of the palace once built by the caliph Hārūn al-Rashīd. [*Translator:* The print has 1835; Nallino's Addenda et emendanda (Part I, p. LXXIV) correct it to 1836.]

a) **Sun**[1]:				
mean long.	140° 53′ 57″	mean long. ☉	140° 53′ 57″	
apogee 1202	82 25 0	equation ☉	− 1 39 37	
mean anomaly ☉	58° 28′ 57″	true long. ☉	139° 14′ 20″	
b) **Moon**:				
mean long.	137° 5′ 42″	mean long. ☾	137° 5′ 42″	
mean long. ☉	140 53 57	simple equation	+ 2 5 48	
elongation ☾ − ☉	3 48 15	true long. ☾	139° 11′ 30″	
double elong.: 2 (☾ − ☉)	7° 36′ 30″			
mean anomaly ☾	334° 15′ 31″	mean motion in lat. ☾	174° 42′ 32″	
equation of the anomaly	− 1 5 24	simple equation	+ 2 5 48	
true anomaly ☾	333° 10′ 7″	true motion in lat.	176° 48′ 20″	
		dist. of ☾ from the node	3° 11′ 40″	at the true conj.
		subtract for 1/8 hour	4 8	
		dist. of ☾ from the node	3° 7′ 32″	at the app. conj.
		true lat. ☾	+ 0° 16′ 18″	at the app. conj.

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I have gathered the elements of the eclipse derived in this way in column I of the small table set out below; but we could not use them for our purpose. For al-Battānī did not take the longitude of the moon from his tables; rather, he supposed, as was certain from the observation, that the true longitudes of the sun and the moon were equal. So for the moon he took 139° 14′ instead of the 139° 11′ 30″ that the tables gave. Column II of the following table shows the numbers that al-Battānī must have found from this hypothesis of his; it will now be proper to compare them with the numbers of the codex and of Plato, set out in columns III and IV. Halley's emendations, mentioned above, have also been added in column V.

	Col. I	*Col. II*	*Esc. codex*	*Plato*	*Halley's corr.*
Mean longitude of the sun	140° 53′ 57″	140° 53′ 57″	140° 54′	140° 54′	…
True longitude of the sun	139 14 20	139 14 20	139 14	139 14	…
Mean longitude of the moon	137 5 42	137 8 32	*134* 7	137 *50*	137° 6′
True longitude of the moon	139 11 30	139 14 20	139 14	139 14	…
True anomaly of the moon	333 10 7	333 12 57	*332 57*	*332 57*	333 7
Mean motion in lat. of the moon	174 42 32	174 45 22	174 43	174 43	…
True motion in lat. of the moon	176 48 20	176 51 10	176 51	176 *11*	176 51
True latitude of ☾ at the middle of the eclipse	0 16 18	0 16 12	0 16	0 16	…
True motion in lat. of ☾ at the middle of the eclipse	176 52 25	176 55 15	176 55	*177 11*	176 55

Leaving aside small differences of one or two minutes, the comparison of column II with the numbers of the Escorial codex proves beyond all doubt that two corrections must be made in the codex, namely قلز 137° for قلد 134° in the mean longitude of the moon, and سلج يج 333° 13′ for سلب نز 332° 57′ in the true anomaly of the moon. — The corrupt numbers of the codex and of Plato are printed in italic, or sloping, type. — Schiaparelli.]

h) — p. 56, l. 22 – p. 57, l. 9.

The second eclipse (of the sun), observed on 22 Kānūn II 1212 of Dhū l-Qarnayn, that is, 22 January 901, at about 20 1/3^{h}[a] in the city of Antioch[2]. — In Oppolzer's canon it bears the number 5016.

[1] With the longitude of the apogee computed as 82° 15′ in the year 1191 of Dhū l-Qarnayn; see note *c* on ch. XXVIII (p. 214).

[2] According to M. Hartmann's map, *Karte des Liwa Haleb*, 1 : 220 000 (in the Zeitschrift der Gesellschaft für Erd-

[a] The print has 20^{h} 1/8. Al-Battānī puts the middle of the eclipse at Antioch about 3 2/3 equal hours before noon of 23 Kānūn II (p. 56), that is, at 20^{h} 20^{m} after noon of the 22nd, the time taken below.

[The noon of which al-Battānī speaks is certainly *true* noon, that is, the culmination of the sun[1]. At Antioch, then, the *true* time of the observation was $20^h\ 20^m$; in the city of al-Raqqa it was $20^h\ 30^m$, or a little more, as al-Battānī has it. Hence the *mean* time at al-Raqqa is found to be $20^h\ 29^m$, since the equation of days-and-nights for the eighth degree of Aquarius is found to be $15' = 1^m$ (see the tables in vol. II, p. 61). He says, moreover, that the true conjunction took place 50^m later; so the mean time of that conjunction was $21^h\ 19^m$ for the meridian of al-Raqqa, or a little more. For this reason the computations will have to be made for $21^h\ 20^m$.

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Mean motions	*Mean long.* ☉	*Mean long.* ☾	*Mean anom.* ☾	*Asc. node* ☾
Dhū l-Qarnayn 1211	343° 0′ 0″	299° 15′ 50″[a]	148° 1′ 59″	203° 11′ 23″
Kānūn I	301 36 34	71 58 41	37 53 5	16 12 10
+ 22 days	21 41 3	289 52 51	287 25 47	1 9 55
+ 20 hours	0 49 17	10 58 49	10 53 15	0 2 39
+ 1 hour	0 2 28	0 32 56	0 32 40	0 0 8
+ 1/3 hour	0 0 49	0 10 59	0 10 53	0 0 3
	307° 10′ 11″	312° 50′ 6″	124° 57′ 39″	220° 36′ 18″[b]
				312 50 6
mean motion in lat. ☾				173° 26′ 24″

a) **Sun**:

mean long.	307° 10′ 11″	mean long. ☉	307° 10′ 11″	
apogee	82 34 0	equation ☉	+ 1 25 40	
mean anom. ☉	224° 36′ 11″	true long. ☉	308° 35′ 51″	

b) **Moon**:

mean long. ☾	312° 50′ 6″	mean long. ☾	312° 50′ 6″	
mean long. ☉	307 10 11	simple equation	− 4 14 0	
elongation ☾ − ☉	5 39 55	true long. ☾	308° 36′ 6″	
double elongation	11° 19′ 50″			
mean anomaly ☾	124° 57′ 39″	mean motion in lat. ☾	173° 26′ 24″	
equation of the anomaly	+ 1 41 0	simple equation	− 4 14 0	
	126° 38′ 39″	true motion in lat. ☾	169° 12′ 24″	
		dist. of ☾ from the node	10° 47′ 36″	at the true conj.
		add for 50^m	27 33	
		dist. of ☾ from the node	11° 15′ 9″	at the app. conj.
		true lat. ☾	0° 58′ 25″	at the app. conj.

The elements of the eclipse found by this computation are entered in column I of the following table. The true longitudes of the sun and the moon differ from each other by only a few seconds; it is therefore plain that the time of the true conjunction derived from the observation agrees very well with al-Battānī's tables. Al-Battānī, however, put the true longitude of both bodies at 308° 35′, that is, about one minute less; accepting this hypothesis, I have tried to restore his computations in column II. Lastly, in the three following columns I have added the numbers of the codex, of Plato and of Halley for comparison.

kunde zu Berlin, vol. XXIV, 1894), the longitude of the city is 36° 8 2/3′ E of Greenwich, the latitude 36° 11′ 52″ N.

[1] [The ancient astronomers knew only one *noon*, namely the one that divides the natural day in the *middle*, and that was therefore called μεσημβρία, or *midday*. — Schiaparelli.]

[a] The print has 299° 15′ 30″; the computation of the fourth eclipse below has 299° 15′ 50″ for the same year, and the sum 312° 50′ 6″ is made with it.

[b] The print has 220° 36′ 10″; the motions of the node add up to 220° 36′ 18″, with which the motion in latitude, 173° 26′ 24″, is made.

	Col. I	*Col. II*	*Esc. codex*	*Plato*	*Halley*
Mean longitude of the sun	307° 10′ 11″	307° 9′ 20″	307° 9′	307° 9′	…
Mean longitude of the moon	312 50 6	312 49 0	312 49	312 49	…
True longitude of the sun	308 35 51	308 35 0	308 35	308 35	…
True longitude of the moon	308 36 6	308 35 0	308 35	308 35	…
True anomaly of the moon	126 38 39	126 37 33	126 *32*	*156 55*	126° 35′
Mean motion in lat. of the moon	173 26 24	173 25 18	173 25	173 *55*	173 25
True motion in lat. of the moon	169 12 24	169 11 18	*199* 11	169 *41*	169 11
The same at the middle of the eclipse	168 44 51	168 43 45	*198* 45	168 45	…
True latitude of the moon	0 58 25	0 58 29	0 59	0 59	…

The errors of the codex, then, are: in the anomaly of the moon لب 32′ for لز 37′; in the true motion in latitude of the moon قضط 199° for قصط 169°; in the true motion in latitude of the moon at the middle of the eclipse قضح 198° for قصح 168°. — Schiaparelli.]

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i) — p. 57, ll. 10–20.

The third eclipse (of the moon), observed on 23 Tammūz 1194 of Dhū l-Qarnayn, that is, 23 July 883, very little more than 8 equinoctial hours after noon, in the city of al-Raqqa. It bears the number 3232 in Oppolzer's canon.

[With the sun at 4° 2′ of Leo, the equation of days-and-nights is found to be 3° 4′, or $12^m\ 16^s$, to be subtracted; hence the true time[a] of the middle of the eclipse, which is also that of the true opposition, is very little more than $7^h\ 48^m$. Entering al-Battānī's tables with this time, I obtained numbers everywhere smaller than al-Battānī's. But weighing the matter more attentively, I saw that the word *very little* can be interpreted so that $7^h\ 55^m$ may rather be taken instead of $7^h\ 48^m$.

Mean motions	*Mean long.* ☉	*Mean long.* ☾	*Mean anom.* ☾	*Asc. node* ☾
Dhū l-Qarnayn 1191	342° 48′ 50″	165° 40′ 17″	108° 20′ 0″	176° 23′ 0″
+ 3 years	359 17 19	28 9 24	266 9 22	57 58 53
Ḥazīrān (June)	120 14 58	167 31 14	153 55 40	6 27 37
+ 23 days	22 40 11	303 3 26	300 29 41	1 13 5
+ 7 hours	0 17 15	3 50 35	3 48 38	0 0 56
+ 55 minutes	0 2 16	0 30 11	0 30 0	0 0 7
	125° 20′ 49″	308° 45′ 7″	113° 13′ 21″	242° 3′ 38″
mean long. ☾				308° 45′ 7″
mean motion in lat. ☾				190° 48′ 45″

a) *Sun*:			
mean long.	125° 20′ 49″	mean long. ☉	125° 20′ 49″
apogee	82 17 42	equation	− 1 19 17
mean anom.	43° 3′ 7″	true long. ☉	124° 1′ 32″
b) *Moon*:			
mean long. ☾	308° 45′ 7″	mean long. ☾	308° 45′ 7″
mean long. ☉	125 20 49	simple equation	− 4 43 43
elongation ☾ − ☉	183 24 18		304 1 24
double elongation	6° 48′ 36″		180
		true long. of the *shadow*	124° 1′ 24″
mean anom. ☾	113° 13′ 21″	mean motion in lat. ☾	190° 48′ 45″
equation of the anom.	+ 1 0 17	simple equation	− 4 43 43
true anom. ☾	114° 13′ 38″	true motion in lat. ☾	186° 5′ 2″
		latitude ☾	− 0° 31′ 52″

[a] So in the print. Subtracting the equation from the observed true time gives the mean time, as for the other three eclipses; the table on p. 232 gives $7^h\ 48^m$ as the mean time of this eclipse.

From this is derived the computation of the first column of the following table; there follow the numbers of the codex, of Plato and of Halley:

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	Col. I	*Esc. codex*	*Plato*	*Halley*
Mean longitude of the sun	125° 20′ 49″	125° 21′	125° *51′*	125° 21′
True longitude of the sun	124 1 32	124 2	124° *5*	124 1
Mean longitude of the moon	308 45 7	308 45	308 45	…
True longitude of the *shadow*	124 1 24	124 2	124 *5*	…
Mean anomaly of the moon	113 13 21	113 *0*	*93 0*	113 8
True anomaly of the moon	114 13 38	114 *9*	*94 10*	114 10
Mean motion in lat. of the moon	190 48 45	190 49	190 49	…
True motion in lat. of the moon	186 5 2	186 5	186 5	…
Southern latitude of the moon	0 31 52	0 32	0 32	…

In the mean anomaly of the moon I think the minutes have dropped out, for in place of ه 0′ one must read يج 13′. In the true anomaly one must read قيد يج 114° 13′, or قيد يد 114° 14′, instead of قيد ط 114° 9′. — Schiaparelli.]

k) — p. 57, ll. 21–32.

I. — *The fourth eclipse* (of the moon), observed at Antioch on 2 Āb 1212 of Dhū l-Qarnayn, that is, 2 August 901, 15 1/3^h after noon (in true time), or 15 1/3^h + 1/4^h if the time is reduced to the meridian of the city of al-Raqqa. It is number 3261 in Oppolzer's canon.

[With the sun at 14° 36′ of Leo, the equation of days-and-nights comes to 3° 10′, to be subtracted, or 13^m. Hence follows the mean time in the city of al-Raqqa, namely 15^h 22^m; the computations may be made for this moment of time.

Mean motions	*Mean long.* ☉	*Mean long.* ☾	*Mean anom.* ☾	*Asc. node* ☾
Dhū l-Qarnayn 1211	343° 0′ 0″	299° 15′ 50″	148° 1′ 59″	203° 11′ 23″
+ 1 year	359 45 46	129 23 8	88 43 7	19 19 38
Tammūz (July)	150 48 17	215 59 20	198 56 33	8 6 6
+ 2 days	1 58 17	26 21 10	26 7 48	0 6 21
+ 15 hours	0 36 58	8 14 7	8 9 56	0 1 59
+ 22 minutes	0 0 54	0 12 5	0 11 59	0 0 3
	136° 10′ 12″	319° 25′ 40″	110° 11′ 22″	230° 45′ 30″
				319 25 40
mean motion in lat. ☾				190° 11′ 10″

a) Sun:			
mean long.	136° 10′ 12″	mean long.	136° 10′ 12″
apogee	82 33 12	equation	− 1 33 59
mean anom.	53° 37′ 0″	true long.	134° 36′ 13″
b) Moon:			
mean long. ☾	319° 25′ 40″	mean long. ☾	319° 25′ 40″
mean long. ☉	136 10 12	simple equation	− 4 48 57
elongation	183 15 28	true long. ☾	314° 36′ 43″
double elongation	6° 30′ 56″		
mean anom. ☾	110° 11′ 22″	mean motion in lat. ☾	190° 11′ 10″
equation of the anomaly	+ 0 57 39	simple equation	− 4 48 57
true anom. ☾	111° 9′ 1″	true motion in lat.	185° 22′ 13″
		lat. ☾	− 0° 28′ 9″

This computation agrees well enough with al-Battānī's numbers, as appears from the following table:

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	Col. I	*Esc. codex*	*Plato*	*Halley*
Mean longitude of the sun	136° 10′ 12″	*134*° 10′	136° 10′	…
True longitude of the sun	134 36 13	134 36	134 36	…
Mean longitude of the moon	319 25 40	319 24	319 *54*	319° 24′
True longitude of the moon	314 36 43	314 36	314 36	…
Mean anomaly of the moon	110 11 22	110 *4*	110 7	…
True anomaly of the moon	111 9 1	111 *5*	*91 5*	111 5
Mean motion in lat. of the moon	190 11 10	190 10	*109* 10	190 10
True motion in lat. of the moon	185 22 13	185 21	185 *51*	185 21
True southern lat. of the moon	0 28 9	0 28	0 28	…

In the Escorial codex, then, the degrees of the mean longitude of the sun (قلو 136° for قلد 134°), the minutes of the mean anomaly (يا 11′ for د 4′) and the minutes of the true anomaly (ط 9′ for ه 5′, an easy error in Maghribi script) must be corrected. — Schiaparelli.]

II. — *General remarks on these four eclipses.*

[On these four eclipses, observed by al-Battānī and brought into computation, it is fitting to add a few remarks; and first on their excellent agreement with al-Battānī's tables. The following table shows, in the first place, for each eclipse the mean time of the true conjunction from the observation, for the meridian of the city of al-Raqqa; in the second and third places, the true longitudes of the sun and the moon according to the tables for the same moment of the true conjunction (or opposition), which would certainly have to be equal (or to differ from each other by 180°) if no error had crept into the observation and the tables. In the fourth place it shows the difference ☾ − ☉, which gives the error of the elongation of the moon arising from the tables; lastly, the difference between the time of the true conjunction (or opposition) taken from the observation (which I denote by the letter O) and the time of the true conjunction (or opposition) derived from the tables (which I mark with the letter T).

Eclipses	*Mean time from observation*	*True longitude from the tables*		*Difference in elongation* ☾ − ☉	*Difference of time at the true syzygy* O − T
		of the sun	*of the moon or shadow*		
I. 891, 8 Aug.[a]	$0^h\ 47^m$	139° 14′ 20″	139° 11′ 30″	− 2′ 50″	$-\ 5.7^m$
II. 901, 22 Jan.	21 20	308 35 51	308 36 6	+ 0 15	+ 0.5
III. 883, 23 Jul.	7 48	124 1 14	123 57 34	− 3 40	− 7.3
IV. 901, 2 Aug.	15 22	134 36 13	134 36 43	+ 0 30	+ 1.0[b]

The errors ☾ − ☉ and O − T must certainly be regarded as very small; indeed, they seem much smaller than should be expected in observations made without a telescope, at a time when the means of determining time accurately were also lacking. For this reason the suspicion arises that al-Battānī corrected his own tables to fit these eclipses.

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The matter stands otherwise if we compare the four eclipses with the more recent tables of the moon. The theory of the moon, although cultivated with the greatest labours by the astronomers

[a] The print has 7 Aug.; the eclipse is that of 8 August 891 (p. 226), $0^h\ 47^m$ after noon in mean time.

[b] The print has + 0.1^m. The other rows turn the difference in elongation into time at 2 minutes of time to 1′ (2′ 50″ into 5.7^m, 15″ into 0.5^m, 3′ 40″ into 7.3^m); at that rate 30″ give 1.0^m.

of our age, has not yet reached such certainty that the ancient observations of the Babylonians, Greeks and Arabs may safely be judged by it. On the contrary, a few years ago a man above all praise, Theodor Oppolzer, strove hard, in his tables of the syzygies and above all in the great *Canon of Eclipses*, to make those observations in some way help the theory. F. K. Ginzel afterwards took the same road: using the ancient observations of total eclipses of the sun, he very diligently sought to work out new corrections of Oppolzer's Canon[1], which R. Schram finally expressed in small tables admirably suited to the use of astronomers[2]. Thus, through the joint labours of three astronomers, tables have come into being, resting partly on a theoretical and partly on an empirical foundation, which must be considered to represent the course of the moon for ancient times more accurately than any others.

To make the comparison of our eclipses with these tables properly, one must first remember that al-Battānī's mean time is not the same as the mean time we use. It was shown earlier, in note *b* on ch. XXIX (pp. 221–22), that for al-Battānī the equation of days-and-nights is always subtractive and for Ptolemy additive, so that about 32^m must be added to al-Battānī's mean time to bring it to Ptolemy's usage. But the way in which our astronomers usually compute that equation is so arranged that it holds very nearly the middle between al-Battānī's method and Ptolemy's. Therefore, so as not to be entangled in overly intricate and unnecessary computations, it will be enough for our purpose to add to al-Battānī's mean time half the interval just mentioned, namely 16^m, to bring the author's numbers to our usage, which the authors of Oppolzer's Canon also followed.

The other reduction is that of longitude, from the meridians of the cities of Antioch and al-Raqqa, where the observations were made, to the meridian of Greenwich, which Oppolzer and his associates used. For Antioch (where the second and fourth eclipses were observed) I have taken the longitude east of Greenwich as $36^\circ\ 9' = 2^h\ 24^m\ 36^s$, on Hartmann's authority; for al-Raqqa (where the first and third eclipses were observed) $39^\circ\ 4' = 2^h\ 36^m\ 16^s$, according to Chesney's observations. With these things weighed, I have gathered the elements of the comparison in the following table.

Eclipses	*True time of the geocentric syzygy from al-Battānī*	*Equation of days + 16^m*	*Difference in longitude from Greenwich*	*Mean time of the geocentric syzygy at the meridian of Greenwich*		*Difference between al-Battānī and Ginzel*
				al-Battānī	*Oppolzer–Ginzel*	
I. 891, 7/8 Aug.	$1^h\ 0^m\ 0^s$	$+\ 2^m\ 56^s$	$2^h\ 36^m\ 16^s$	$22^h\ 26^m\ 40^s$	$22^h\ 13^m\ 12^s$	$+\ 13^m\ 28^s$
II. 901, 22 Jan.	21 10 0	+ 15 0	2 24 36	19 0 24	18 46 42	+ 13 42
III. 883, 23 Jul.	8 0 0	+ 3 44	2 36 16	5 27 28	5 9 24	+ 18 4
IV. 901, 2 Aug.	15 20 0	+ 3 20	2 24 36	12 58 44	12 31 18	+ 17 26[a]

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It is gathered at once that in all the eclipses the observation puts the syzygy about a quarter of an hour later than the tables do. Some part of this difference must, I have no doubt, be ascribed to the observations themselves. For the beginning of the obscuration cannot be perceived by the naked eye until some minutes of time have elapsed after the geometrical contact; and this must have happened above all to the ancient astronomers, who could not predict the beginning

[1] *Astronomische Untersuchungen über Finsternisse*, in the Sitzungsberichte der mathem.-naturwissenschaftliche Classe der k. Akad. der WW. zu Wien, vol. LXXXV, section II, 1882, pp. 663–747; vol. LXXXVIII, sect. II, 1883, pp. 629–755; vol. LXXXIX, sect. II, 1884, pp. 491–559.

[2] *Reductionstafeln für den Oppolzer'sche Finsterniss-Canon, zum Uebergang auf die Ginzel'schen empirischen Correctionen*, in the Denkschriften der k. Akad. der WW. zu Wien, mathem. Classe, vol. LVI, 1889.

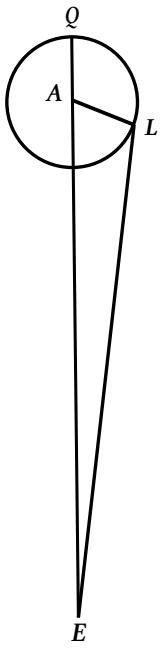
[a] So in the print: $12^h\ 58^m\ 44^s$ less $12^h\ 31^m\ 18^s$ is $27^m\ 26^s$, while the next paragraph speaks of about a quarter of an hour in all four eclipses, so that one of the three numbers of this row is misprinted.

of an eclipse accurately enough from tables. This uncertainty had no place in noting the end of the obscuration; so in the estimate of the middle of the eclipse there must necessarily have been some retardation. But whether the whole error of a full quarter of an hour must be ascribed to this cause I dare not affirm; for al-Battānī himself, in a rather rough manner, seems to have noted his times no more accurately than in intervals of ten minutes.

Lastly, it will be useful to compare the *magnitudes* of the lunar eclipses observed by al-Battānī with Oppolzer's computations. The magnitude of eclipse III, which al-Battānī estimated at 10 digits, was 11.2 digits according to Oppolzer. The magnitude of eclipse IV, which al-Battānī saw to be 12 digits or a little less, is found from Oppolzer's computations to be somewhat greater, namely 12.6 digits. — Al-Battānī did not report the magnitudes of the two solar eclipses, I and II. — Schiaparelli.]

l) — p. 58, ll. 1–4.

[«*The sun was towards the apogee*». — For in eclipse III the anomaly of the sun was 42°, in IV 52°.



«*The moon occupied the same place, about 1 1/2^p less than the mean distance*». — That is, about 1 1/2^p of the 60 parts into which the mean distance (AE in the figure of p. 51) of the moon in conjunction with the sun and in opposition to it is taken to be divided. In fact, in eclipse III the true anomaly of the moon on the epicycle was 114° 16′, in IV 111° 4′; and these are the amounts of the angle QAL in the figure drawn here, where E is the earth, A the centre of the epicycle, AE = 60^p, AL = 5.247^p. The computation of the triangle AEL gives EL = 58.04^p for eclipse III and EL = 58.32^p for IV. So what al-Battānī calls «about 1 1/2^p» was exactly 1.96^p and 1.68^p; plainly he made the computation approximately.

«*Between the first and the second latitude there were 0° 3′ 50″*». — According to al-Battānī's numbers the first latitude (eclipse III) «was about 0° 32′ towards the south», the second (eclipse IV) «about 0° 28′» (he omitted to indicate the side, but from the motion in latitude, 185° 21′, it is plain that it too was southern). The difference, then, is 4′ for 3′ 50″. But a more exact computation that I made gave 0° 31′ 52″ and 0° 28′ 3″[a] for those two latitudes; difference 3′ 49″, agreeing well with

[a]The computation of the fourth eclipse on p. 231 and the table on p. 232 have 0° 28′ 9″, with which the difference would be 3′ 43″.

al-Battānī's 3′ 50″. Plainly al-Battānī had made the computations of the eclipses more precisely, and afterwards gave round numbers in the book. — Schiaparelli.]

m) — p. 58, ll. 4–5.

«*The difference of the two eclipses is contained 8 3/4 times in the lunar diameter*». — This does not agree with the author's observations. In eclipse III the part obscured was 5/6 of the diameter; in IV almost the whole moon. The difference is 1 − 5/6 = 1/6; so he should have written 6 for 8 3/4. [I think that al-Battānī, so as not to be led to an absurd lunar diameter barely reaching 23 minutes (as would necessarily arise from the factor 6), wanted somehow to bring to a numerical ratio the «a little more than 5/6» of the third eclipse and the «very little less than its diameter» of the fourth; but since it was difficult to express these by numbers, he chose the factor 8 3/4, estimating the lunar diameter at 33 1/2′. Al-Battānī only wanted to show (not very clearly, it seems) that the diameter of the moon he proposed could be reconciled well enough with these two eclipses. So what al-Battānī says must not be considered a true determination of the lunar diameter, but only a proof that the observations do not conflict with the diameter he had adopted. — Schiaparelli.]

n) — p. 58, ll. 9–11.

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[«*The hourly motion of the moon at the times of the syzygies …*». — These hourly motions, the greatest 36′ 10″ and the least 30′ 12″ (Plato 15″, for in him 2 very often becomes 5), do not agree with the table of vol. II, p. 88, from which 36′ 4″ and 30′ 18″ are derived. I do not know the cause of the difference; perhaps al-Battānī's computation was made only approximately. — In the computations that follow he supposes that the greatest diameter is to the least as the greatest hourly motion to the least. This would be true if the hourly motion in *anomaly* were equal to the motion in *longitude*, as happens with the sun; but even in the diameter of the moon the error seems negligible. — Schiaparelli.]

o) — p. 58, ll. 16–18.

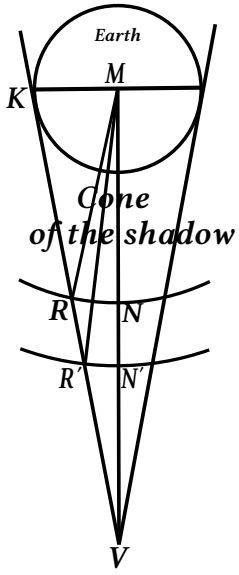
He puts the *semidiameter of the shadow* at about 0° 38 1/2′ when the moon is at its apogee, and at about 0° 46′ when the moon is at its perigee; the difference, then, is about 0° 7 1/2′.

Ibn Yūnus (introduction, Caussin pp. 70–72) makes these observations: «Abū l-ʿAbbās al-Faḍl ibn Ḥātim al-Nayrīzī[1] and Muḥammad ibn Jābir al-Battānī also err when they put the variation of the semidiameter of the shadow between the greatest and the least distance [of the moon] at 7′ 12″. Anyone even less versed in the studies of geometry than those two men will recognize that this opinion is false, if he considers Ptolemy's demonstration [*Almag.* V, 15] of the distance of the sun from the centre of the earth. It is impossible for the variation of the semidiameter of the shadow between the apogee distance of the moon and the perigee distance to be as small as they make it, or even 8′ or 9′; it must come out much greater if in determining it you follow the way of geometrical demonstration. I once made the computation, and found 0° 10′ 17″, putting the sun and the moon[2] at the greatest distance from the earth. The variation that affects the semidiameter of the shadow on account of the nearness of the sun to the earth cannot exceed 1 minute»[3].

[1] Anaritius in the Latin writers of the Middle Ages. He died about 922; Suter, *Die Mathematiker und Astronomen*, p. 45, no. 88.

[2] [Better: the sun at its apogee, the moon at the apogee of the eccentric. — Schiaparelli.]

[3] Al-Battānī about 0′ 50″.



[What Ibn Yūnus says is demonstrated in this way. Let RN be the radius of the shadow at the least distance, measured in minutes and seconds by the angle RMN; R′N′ the radius of the shadow at the greatest distance, measured by the angle R′MN′. KRM is the greatest parallax of the moon, amounting according to al-Battānī to 63′ 51″[a] at the syzygies; KR′M is the least parallax, according to al-Battānī 53′ 34″. Let φ be the angle KVM at the vertex of the cone of the shadow. In the triangle MRV we have

$$KRM = 63'\,51'' = \varphi + RMV,$$

$$KR'M = 53'\,34'' = \varphi + R'MV,$$

$$\text{difference} \quad \overline{10'\,17''} = RMV - R'MV = RMR',$$

and RMR′ is, as an angle, the difference between the greatest and the least shadow. Plainly the angle 10′ 17″ is always the same, although the distance of the sun and the angle of the cone of the shadow vary. — Schiaparelli.]

Al-Battānī plainly found his numbers in this way. He supposed (see p. 58, ll. 7–8) that the ratio of the diameter of the shadow to the diameter of the moon is the one that Ptolemy has, *Almag.* V, 14 (ed. Halma vol. I, p. 343), namely 2 3/5 to 1; with this ratio the diameters of the shadow 91.858′ and 76.7′ are derived from al-Battānī's greatest and least diameters of the moon (35 1/3′ and 29 1/2′), and so the semidiameters 45′ 54″ and 38′ 42″[b]. The difference is 7′ 12″, which Ibn Yūnus mentions; al-Battānī writes about 7 1/2′, since he puts the semidiameters in round numbers as 46′ and 38 1/2′.

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p) — p. 58, ll. 27–31.

From his observations of eclipses al-Battānī determined the *angular diameters of the moon* and *of the sun* better than Ptolemy. The diameter of the *sun* was always 31′ 20″ in the Greek author, who did not know that it varies. Ours recognized the variation of the diameter and found for the different distances from the earth:

[a] The print has 64′ 51″ here and in the first line below; 63′ 51″ less 53′ 34″ gives the difference 10′ 17″ that follows, which Ibn Yūnus found, and 53′ 34″ is the angle whose sine is 1/64;10, the parallax at al-Battānī's greatest distance of the moon, 64;10 earth radii.

[b] The halves of 91.858′ and 76.7′ are 45′ 56″ and 38′ 21″, whose difference is 7′ 35″, al-Battānī's «about 7 1/2′»; 38′ 42″ is not half of 76.7′, and the computation does not give the 7′ 12″ of Ibn Yūnus.

31′ 20″ at the greatest distance, 32′ 30″ at the mean, 33′ 40″ at the least.

The astronomers of our age[1] have established:

31′ 31.98″ at the greatest distance, 32′ 3.64″ at the mean, 32′ 36.48″ at the least.

The diameter of the *moon* according to

	at the greatest distance	*at the mean*	*at the least*
Ptolemy	31′ 20″	33′ 20″	35′ 20″
al-Battānī	29′ 30″	32′ 25″	35′ 20″
the moderns[2]	29′ 26″	31′ 8″	33′ 34″

This is a matter of no small moment, for from his measurements al-Battānī (as Delambre, p. 39, already remarked) infers in ll. 29–30 *that annular eclipses of the sun sometimes occur*; these, on the contrary, could not have happened according to Ptolemy's measurements, which make the diameter of the sun equal to the diameter of the moon at apogee; see Delambre, *Histoire de l'astronomie ancienne*, Paris 1817, vol. II, pp. 234–235, and P. Tannery, *Recherches sur l'astronomie ancienne*, Paris 1893, pp. 224 and 233. Among the Greeks, however, in the first century BC, the Peripatetic philosopher Sosigenes had already recognized the variation of the solar diameter and, on the testimony of Proclus Diadochus[3], had said that in perigee eclipses the sun «is not seen wholly covered, but exceeds the lunar disc with the rim of its circumference and gives light unhindered», ὁρᾶσθαι μὴ ὅλον ἐπιπροσθούμενον, ἀλλὰ τοῖς ἄκροις τῆς ἑαυτοῦ περιφερείας ὑπερβάλλειν τὸν κύκλον τῆς σελήνης καὶ φωτίζειν οὐκ ἐμποδιζόμενον. Proclus wants to prove this opinion false by the Ptolemaic measurements, saying that if it were true it would follow that the apparent diameter of the moon at the apogee points differs from the diameter of the sun; and this is contrary to Ptolemy's doctrine.

The Arabs after al-Battānī did not call annular eclipses of the sun into doubt. In 1279, in an astronomy written in Syriac, Barhebraeus relates, p. 88 of the text: «They found that the diameter of the sun varies from the apogee to the perigee from 31′ to 34′[4], the diameter of the moon from 29′ to 36′. When the two apparent diameters are equal, the whole sun is eclipsed without any shadow being found to exceed; when the diameter of the sun is greater than that of the moon, not only is there no exceeding shadow, but on the sun a luminous crown is seen around the moon that hides it».

Even at the beginning of the seventeenth century some Europeans doubted that a total eclipse of the sun could occur; see G. Schiaparelli, *Le sfere omocentriche di Eudosso, di Callippo e di Aristotele*, Milan 1875, p. 62, note 31 (= Pubblicazioni del R. Osservatorio di Brera, no. IX).

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[Perhaps already through the first eclipse, of 8 August 891, al-Battānī could have been confirmed in his opinion that *annular* eclipses sometimes occur; for according to Oppolzer's *Canon* it was annular along an oblique line crossing Asia Minor, Cilicia and Mesopotamia. News of the marvellous spectacle that had appeared in places not very far away could have reached him as he observed at al-Raqqa. — Two other annular eclipses occurred not far from al-Raqqa, on 26 April 906 and 5 April 916, as appears from Oppolzer's *Canon*, chart no. 101. — Schiaparelli.]

[1] I follow those who publish, at the Paris observatory, the very famous ephemerides *La Connaissance des Temps* and the *Annuaire du Bureau des Longitudes*.

[2] According to the Babylonians of the fourth and third centuries BC, 29′ 26.9″, 31′ 51.5″, 34′ 16.2″, which are better even than the Ptolemaic and Albatenian quantities. See Kugler, *Die babylonische Mondrechnung*, Freiburg i. Br. 1900, p. 126.

[3] *Hypothèses et époques des planètes de C. Ptolémée, et Hypotyposes de Proclus Diadochus publiées et traduites par Halma*, Paris 1820, p. 111. — See also Delambre, *Hist. de l'astr. anc.*, vol. I, p. 314.

[4] Barhebraeus almost always uses round numbers; if a given quantity consists of degrees, he leaves out the minutes; if of minutes, the seconds.